census_income

June 30, 2025

1 Census Income Dataset Analysis

In this project we will cover indepth analysis of the Census data where extraction was done by Barry Becker from the 1994 Census database. A set of reasonably clean records was extracted from the US census data and in order to add anonimity among many things a lot of data has been masked. Even then its still one of the most extenstive Census Income dataset avaialable as generally only tables statistics are available for Census data across multiple countries.

Main predictive goal for the dataset was to predict for the proportion of people having income greater than 50K USD, adjusting for the inflation this amount would be 102K USD as of date of the making this report. This is the individual income, not the household income, and following that 50K was indeed a representation of high income group on 1994.

During this project we will cover Explanatory Data Analysis, Hypothesis Tests, Linear Models such as Generalized Linear Models, Generalized Additive Models, Bayesian Hierarchical Model, and a simple Neurel Network. Goal is to first clean the dataset, perform the EDA, and then check some planned hypothesis. After that we will fit various models and check for both inference and predictive capabilities of them. We will also analyze the use of Bayesian Mixture Model using MCMC algorithms for the dataset and whether they add to the value.

1. Explanatory Data Analysis
2. Hypothesis Tests
3. Linear Model
4. Bayesian Hierarchical Model
5. Neural Network
6. Conclusion
7. Bibliography

1.1 Explanatory Data Analysis

Data used for this analysis is based on the US Census data for 1996 available at UC Irvine ML Repositor[1]. Data card for the dataset is as follows:

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
age	Feature	Integer	Age	N/A		no
workclass	Feature	Categorical	Income	Private, Self-emp-not-inc, Self-emp-inc, Federal-gov, Local-gov, State-gov, Without-pay, Never-worked.		yes

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
fnlwgt	Feature	Integer		N/A		no
education	Feature	Categorical	Education Level	Bachelors, Some-college, 11th, HS-grad, Prof-school, Assoc-acdm, Assoc-voc, 9th, 7th-8th, 12th, Masters, 1st-4th, 10th, Doctorate, 5th-6th, Preschool.		no
education-num	Feature	Integer	Education Level	N/A		no
marital-status	Feature	Categorical	Other	Married-civ-spouse, Divorced, Never-married, Separated, Widowed, Married-spouse-absent, Married-AF-spouse.		no
occupation	Feature	Categorical	Other	Tech-support, Craft-repair, Other-service, Sales, Exec-managerial, Prof-specialty, Handlers-cleaners, Machine-op-inspct, Adm-clerical, Farming-fishing, Transport-moving, Priv-house-serv, Protective-serv, Armed-Forces.		yes
relationship	Feature	Categorical	Other	Wife, Own-child, Husband, Not-in-family, Other-relative, Unmarried.		no
race	Feature	Categorical	Race	White, Asian-Pac-Islander, Amer-Indian-Eskimo, Other, Black.		no
sex	Feature	Binary	Sex	Female, Male.		no
capital-gain	Feature	Integer		N/A		no
capital-loss	Feature	Integer		N/A		no
hours-per-week	Feature	Integer		N/A		no
native-country	Feature	Categorical	Other	United-States, Cambodia, England, Puerto-Rico, Canada, Germany, Outlying-US(Guam-USVI-etc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica, Vietnam, Mexico, Portugal, Ireland, France, Dominican-Republic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala, Nicaragua, Scotland, Thailand, Yugoslavia, El-Salvador, Trinadad&Tobago, Peru, Hong, Holand-Netherlands.		yes
income	Target	Binary	Income	>50K, <=50K.		no

There is no specific information about the variables besides the table above in official source. While most of the variables are self explanatory, fnlwgt is an important weight parameter here used in survey designs of cases like this where authors of the data converted a bigger dataset into smaller one. This variable tells us how much weight the observation in the sample weight in the population. We can ignore it and perform sample based inference and predictions or we include the weights as parameter as Additive Models and Bayesian models do allow us to take them into consideration, but as a negative it increases model complexity by a lot and model training time increases by a lot, in bayesian model we moved from rjags to rstanarm model which uses Hamiltonian Monte Carlo

(HMC) instead of Metropolitis Hastings or Gibbs Sampling in rjags, we were able to reduce the training time from 2-3 hours to few minutes in non weighted but weighted model still took 2+ hours to train on a competent machine.

EDA will be broken into 3 parts: 1. Initial data checking and hypothesis formation 2. Data Cleaning and further EDA 3. Hypothesis Testing

We have broken EDA into smaller parts in order to form planned hypothesis. Generally such hypothesis should have been made before the survey itself and we could have used post hoc hypothesis tests but by just observing the data initially without going for in depth analysis so that we could know what type of data we have. In this way we could avoid data dredging.

[3]:
```
#loading the dataset

census <- read.csv('adult.data.csv', header = FALSE)
headers <- c("age", "workclass", "fnlwgt", "education", "education_num",
               "marital_status", "occupation", "relationship", "race", "sex",
               "capital_gain", "capital_loss", "hours_per_week",␣
 ↪"native_country",
               "income_group")

colnames(census) <- headers
head(census)
```

A data.frame: 6 × 15

	age <int>	workclass <chr>	fnlwgt <int>	education <chr>	education_num <int>	marital_status <chr>
1	39	State-gov	77516	Bachelors	13	Never-married
2	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse
3	38	Private	215646	HS-grad	9	Divorced
4	53	Private	234721	11th	7	Married-civ-spouse
5	28	Private	338409	Bachelors	13	Married-civ-spouse
6	37	Private	284582	Masters	14	Married-civ-spouse

[3]:
```
sapply(census, class)
```

age 'integer' **workclass** 'character' **fnlwgt** 'integer' **education** 'character' **education_num** 'integer' **marital_status** 'character' **occupation** 'character' **relationship** 'character' **race** 'character' **sex** 'character' **capital_gain** 'integer' **capital_loss** 'integer' **hours_per_week** 'integer' **native_country** 'character' **income_group** 'character'

[4]:
```
str(census)
```

```
'data.frame':   32561 obs. of  15 variables:
 $ age           : int  39 50 38 53 28 37 49 52 31 42 …
 $ workclass     : chr  " State-gov" " Self-emp-not-inc" " Private" " Private"
…
 $ fnlwgt        : int  77516 83311 215646 234721 338409 284582 160187 209642
45781 159449 …
 $ education     : chr  " Bachelors" " Bachelors" " HS-grad" " 11th" …
```

```
 $ education_num : int  13 13 9 7 13 14 5 9 14 13 …
 $ marital_status: chr  " Never-married" " Married-civ-spouse" " Divorced" "
Married-civ-spouse" …
 $ occupation    : chr  " Adm-clerical" " Exec-managerial" " Handlers-cleaners"
" Handlers-cleaners" …
 $ relationship  : chr  " Not-in-family" " Husband" " Not-in-family" " Husband"
…
 $ race          : chr  " White" " White" " White" " Black" …
 $ sex           : chr  " Male" " Male" " Male" " Male" …
 $ capital_gain  : int  2174 0 0 0 0 0 0 0 14084 5178 …
 $ capital_loss  : int  0 0 0 0 0 0 0 0 0 0 …
 $ hours_per_week: int  40 13 40 40 40 40 16 45 50 40 …
 $ native_country: chr  " United-States" " United-States" " United-States" "
United-States" …
 $ income_group  : chr  " <=50K" " <=50K" " <=50K" " <=50K" …
```

[5]:
```
summary(census)
```

```
      age         workclass              fnlwgt          education
 Min.   :17.00   Length:32561       Min.   :  12285   Length:32561
 1st Qu.:28.00   Class :character   1st Qu.: 117827   Class :character
 Median :37.00   Mode  :character   Median : 178356   Mode  :character
 Mean   :38.58                      Mean   : 189778
 3rd Qu.:48.00                      3rd Qu.: 237051
 Max.   :90.00                      Max.   :1484705
 education_num   marital_status      occupation         relationship
 Min.   : 1.00   Length:32561       Length:32561       Length:32561
 1st Qu.: 9.00   Class :character   Class :character   Class :character
 Median :10.00   Mode  :character   Mode  :character   Mode  :character
 Mean   :10.08
 3rd Qu.:12.00
 Max.   :16.00
     race               sex             capital_gain    capital_loss
 Length:32561       Length:32561       Min.   :    0   Min.   :   0.0
 Class :character   Class :character   1st Qu.:    0   1st Qu.:   0.0
 Mode  :character   Mode  :character   Median :    0   Median :   0.0
                                       Mean   : 1078   Mean   :  87.3
                                       3rd Qu.:    0   3rd Qu.:   0.0
                                       Max.   :99999   Max.   :4356.0
 hours_per_week  native_country     income_group
 Min.   : 1.00   Length:32561       Length:32561
 1st Qu.:40.00   Class :character   Class :character
 Median :40.00   Mode  :character   Mode  :character
 Mean   :40.44
 3rd Qu.:45.00
 Max.   :99.00
```

```
[5]: table(census[['income_group']])
```

```
 <=50K   >50K
 24720   7841
```

```
[6]: #sample proprotion based on the sex, while weights may change the proportions␣
     ↪they would not vary by much
     prop.table(table(census[['sex']]))
```

```
   Female      Male
0.3307945 0.6692055
```

```
[7]: table(census[['sex']])
```

```
 Female   Male
  10771  21790
```

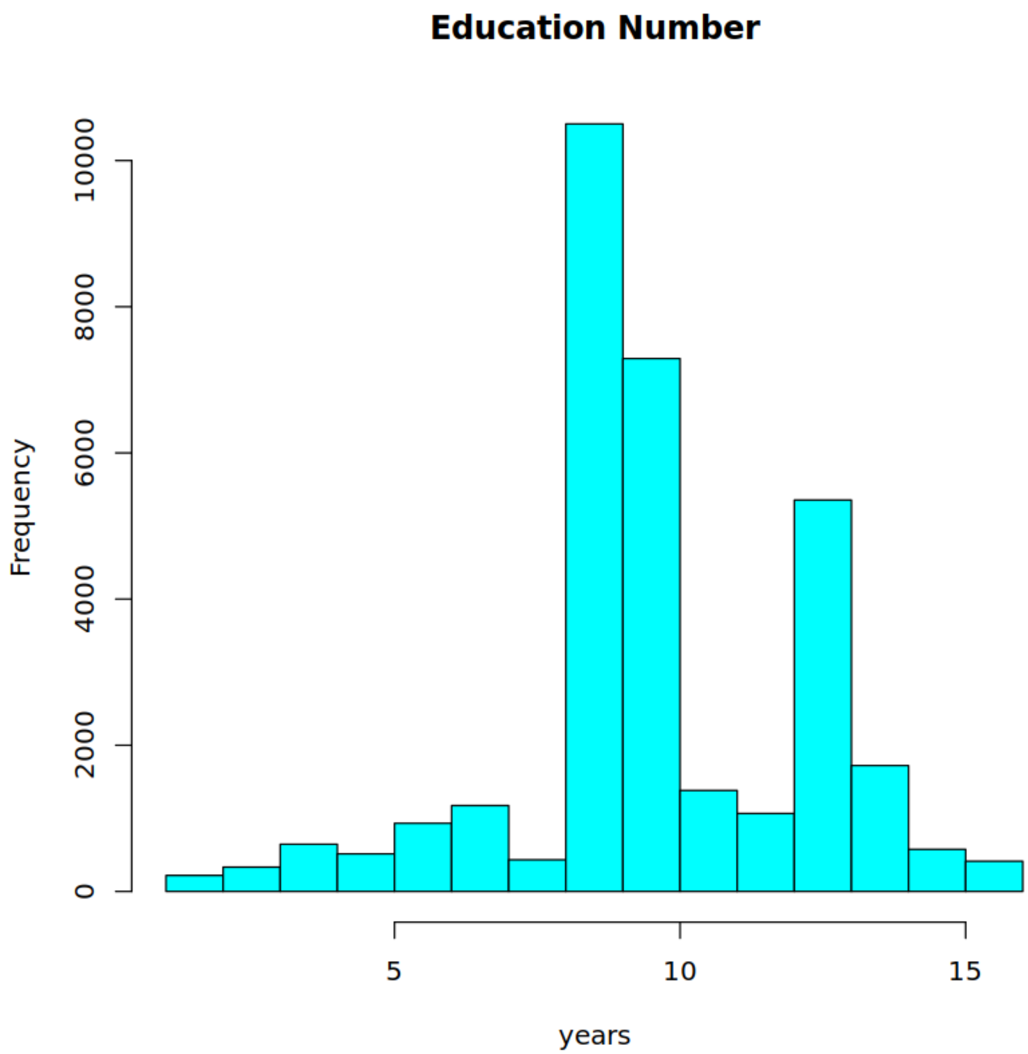
```
[8]: summary(census[['education_num']])
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.00    9.00   10.00   10.08   12.00   16.00
```

```
[10]: var(census[['education_num']])
```

6.61888990703293

```
[9]: hist(census[['education_num']], main = 'Education Number', xlab = 'years', col␣
     ↪= 'cyan')
```



At this stage we are have done a preliminary data checking in order to observe the dataset in hand. We will use this analysis to desing conservative planned hypothesis which could have been naturally during the period of the data with which we can avoid data dredging.

Nevertheless, we can observe some NA values present in the dataset in the form of values like 99 and 99999. We will analyze them further.

After giving a quick look and checking other sources relevant in 1990s we will test following planned hypothesis in EDA, this does not include the post hoc hypothesis we may perform at that stage: 1. Mean Education Years for American working class is 11.5. We are forming a conservative planned hypothesis here and using data for 1990 specifically as it would be a natural data for this dataset. This Null Hypothesis is formed on the basis of NCES Compiled data in 1990 [2], if we were comparing it across different countries HDI [3] based hypothesis of 13.3 would be a better fit for the same period. 2. Proportion of Women in income group of >50k is 0.441. In 1990 as per

World Bank data [3] percentage of women in american workforce was 44.1%, we will use it as null hypothesis and test the hypothesis that even in high income group this overall proportion holds true. 3. Proportion of workforce having salary more than 50k is 0.15. This null hypthesis has been formed based on the general trend and proportions at 1990 as per data from BLS and Census while following the wage stagnation during that period [4].

For both planned hypothesis above we will use 95% confidence value, and with sufficient power as we can see below. We will perform the test after other steps for EDA, we formed them early so that we can avoid data dredging situation

[11]:
```
# Post hoc power analysis for both planned hypothesis
n_sample <- dim(census)[1]
achieved_power <- pwr.t.test(n = n_sample, d = 1/2, sig.level = 0.05, #mean␣
 ↪difference assumed is 1 and estimated variance as 4
                              type = "one.sample", alternative = "two.sided")
print(achieved_power)

women_sample = 10771
achieved_power_h2 <- power.prop.test(n = 10771, p1 = 0.441, p2 = 0.35, sig.
 ↪level = 0.05, alternative = "two.sided") #estimated real prop for h2 is 0.35
print(achieved_power_h2)
```

```
     One-sample t test power calculation

              n = 32561
              d = 0.5
      sig.level = 0.05
          power = 1
    alternative = two.sided


     Two-sample comparison of proportions power calculation

              n = 10771
             p1 = 0.441
             p2 = 0.35
      sig.level = 0.05
          power = 1
    alternative = two.sided

NOTE: n is number in *each* group
```

Now we have formed our planned hypothesis and performed post hoc power analysis we are ready to move on to Data Cleaning and further EDA. On a side note, power is enough for our hypothesis because of large number of observations even if we ignore the sample weights and consider only the sample obseervations.

[13]:
```
#checking NA Values
print(sum(is.na(census)))
sapply(census, function(x) sum(is.na(x)))
```

```
[1] 0
```

age 0 **workclass** 0 **fnlwgt** 0 **education** 0 **education_num** 0 **marital_status** 0 **occupation** 0 **relationship** 0 **race** 0 **sex** 0 **capital_gain** 0 **capital_loss** 0 **hours_per_week** 0 **native_country** 0 **income_group** 0

[14]:
```
dim(census)
```

1. 32561 2. 15

[15]:
```
summary(census)
```

```
     age         workclass              fnlwgt          education
 Min.   :17.00   Length:32561       Min.   :  12285   Length:32561
 1st Qu.:28.00   Class :character   1st Qu.: 117827   Class :character
 Median :37.00   Mode  :character   Median : 178356   Mode  :character
 Mean   :38.58                      Mean   : 189778
 3rd Qu.:48.00                      3rd Qu.: 237051
 Max.   :90.00                      Max.   :1484705
 education_num   marital_status      occupation         relationship
 Min.   : 1.00   Length:32561       Length:32561       Length:32561
 1st Qu.: 9.00   Class :character   Class :character   Class :character
 Median :10.00   Mode  :character   Mode  :character   Mode  :character
 Mean   :10.08
 3rd Qu.:12.00
 Max.   :16.00
     race                sex             capital_gain    capital_loss
 Length:32561       Length:32561       Min.   :    0   Min.   :   0.0
 Class :character   Class :character   1st Qu.:    0   1st Qu.:   0.0
 Mode  :character   Mode  :character   Median :    0   Median :   0.0
                                       Mean   : 1078   Mean   :  87.3
                                       3rd Qu.:    0   3rd Qu.:   0.0
                                       Max.   :99999   Max.   :4356.0
 hours_per_week  native_country     income_group
 Min.   : 1.00   Length:32561       Length:32561
 1st Qu.:40.00   Class :character   Class :character
 Median :40.00   Mode  :character   Mode  :character
 Mean   :40.44
 3rd Qu.:45.00
 Max.   :99.00
```

[16]:
```
#checking 99.0 hours per week data
print(length(census[census$hours_per_week  == 99.0,]))
head(census[census$hours_per_week  == 99.0,])
```

```
[1] 15
```

A data.frame: 6 × 15

	age <int>	workclass <chr>	fnlwgt <int>	education <chr>	education_num <int>	marital_statu <chr>
936	37	Private	176900	HS-grad	9	Married-civ-s
1173	25	Private	404616	Masters	14	Married-civ-s
1888	55	Self-emp-not-inc	184425	Some-college	10	Married-civ-s
3579	37	Self-emp-inc	382802	Doctorate	16	Married-civ-s
4087	50	?	174964	10th	6	Married-civ-s
4309	35	Self-emp-not-inc	166416	HS-grad	9	Married-civ-s

[17]:
```
#checking 99.0 hours per week data
print(length(census[census$capital_gain  == 99999,]))
head(census[census$capital_gain  == 99999,])
```

```
[1] 15
```

A data.frame: 6 × 15

	age <int>	workclass <chr>	fnlwgt <int>	education <chr>	education_num <int>	marital_status <chr>
1247	54	Self-emp-inc	166459	Prof-school	15	Married-civ-spouse
1369	52	Private	152234	HS-grad	9	Married-civ-spouse
1483	53	Self-emp-inc	263925	HS-grad	9	Married-civ-spouse
1529	52	Private	118025	Bachelors	13	Married-civ-spouse
1617	46	Private	370119	Prof-school	15	Married-civ-spouse
1683	43	Private	176270	Bachelors	13	Married-civ-spouse

While outliers are common in both of the variables above, specfic values such 99 or 999999 or so are used as reprentative of NA values. Of course, we may be wrong and they may actually contain important data however due to low number of such observations and the fact that data set itself does not provide any information about the na values we will remove them from the dataset for our analysis

[18]:
```
data <- census[!(census$capital_gain  == 99999 | census$hours_per_week  == 99.
 ↪0),]
summary(data)
```

```
      age          workclass              fnlwgt          education
 Min.   :17.00   Length:32317       Min.   :  12285   Length:32317
 1st Qu.:28.00   Class :character   1st Qu.: 117849   Class :character
 Median :37.00   Mode  :character   Median : 178470   Mode  :character
 Mean   :38.53                      Mean   : 189837
 3rd Qu.:48.00                      3rd Qu.: 237065
 Max.   :90.00                      Max.   :1484705
 education_num   marital_status      occupation         relationship
 Min.   : 1.00   Length:32317       Length:32317       Length:32317
 1st Qu.: 9.00   Class :character   Class :character   Class :character
 Median :10.00   Mode  :character   Mode  :character   Mode  :character
 Mean   :10.07
 3rd Qu.:12.00
```

```
 Max.   :16.00
     race                sex             capital_gain      capital_loss
 Length:32317       Length:32317       Min.   :    0.0   Min.   :   0.00
 Class :character   Class :character   1st Qu.:    0.0   1st Qu.:   0.00
 Mode  :character   Mode  :character   Median :    0.0   Median :   0.00
                                       Mean   :  591.7   Mean   :  87.63
                                       3rd Qu.:    0.0   3rd Qu.:   0.00
                                       Max.   :41310.0   Max.   :4356.00
 hours_per_week  native_country     income_group
 Min.   : 1.00   Length:32317       Length:32317
 1st Qu.:40.00   Class :character   Class :character
 Median :40.00   Mode  :character   Mode  :character
 Mean   :40.24
 3rd Qu.:45.00
 Max.   :98.00
```

[19]:
```r
#checking categorial variables specifically as there can be some na values
 ↪there too
cat_cols <- sapply(data, function(x) !is.numeric(x))
cat_data <- data[, cat_cols]
head(cat_data)
```

A data.frame: 6 × 9

	workclass <chr>	education <chr>	marital_status <chr>	occupation <chr>	relationship <chr>
1	State-gov	Bachelors	Never-married	Adm-clerical	Not-in-family
2	Self-emp-not-inc	Bachelors	Married-civ-spouse	Exec-managerial	Husband
3	Private	HS-grad	Divorced	Handlers-cleaners	Not-in-family
4	Private	11th	Married-civ-spouse	Handlers-cleaners	Husband
5	Private	Bachelors	Married-civ-spouse	Prof-specialty	Wife
6	Private	Masters	Married-civ-spouse	Exec-managerial	Wife

[20]:
```r
for (col in names(cat_data)) {
  print(paste("Table for", col))
  print(table(data[[col]]))
}
```

```
[1] "Table for workclass"

               ?      Federal-gov        Local-gov     Never-worked
            1827              958             2083                7
         Private     Self-emp-inc Self-emp-not-inc        State-gov
           22570             1071             2491             1296
     Without-pay
              14
[1] "Table for education"

        10th         11th         12th      1st-4th      5th-6th
         926         1174          432          168          333
```

```
     7th-8th          9th   Assoc-acdm    Assoc-voc    Bachelors
         639          512         1062         1377         5304
   Doctorate      HS-grad      Masters    Preschool  Prof-school
         394        10460         1698           51          524
Some-college
        7263
[1] "Table for marital_status"

             Divorced     Married-AF-spouse    Married-civ-spouse
                 4421                    23                 14791
Married-spouse-absent         Never-married             Separated
                  416                 10655                  1022
              Widowed
                  989
[1] "Table for occupation"

                ?      Adm-clerical      Armed-Forces      Craft-repair
             1834              3764                 9              4084
  Exec-managerial   Farming-fishing Handlers-cleaners Machine-op-inspct
             4012               982              1369              2001
    Other-service   Priv-house-serv    Prof-specialty   Protective-serv
             3282               147              4055               645
            Sales      Tech-support  Transport-moving
             3622               925              1586
[1] "Table for relationship"

       Husband  Not-in-family Other-relative      Own-child      Unmarried
         13024           8263            979           5063           3434
          Wife
          1554
[1] "Table for race"

Amer-Indian-Eskimo Asian-Pac-Islander              Black              Other
               311               1024               3110                269
             White
             27603
[1] "Table for sex"

Female   Male
 10730  21587
[1] "Table for native_country"

                         ?                   Cambodia
                       574                         19
                    Canada                      China
                       120                         75
                  Columbia                       Cuba
                        59                         95
```

```
     Dominican-Republic                      Ecuador
                     69                           28
            El-Salvador                      England
                    106                           89
                 France                      Germany
                     28                          137
                 Greece                    Guatemala
                     29                           64
                  Haiti           Holand-Netherlands
                     44                            1
               Honduras                         Hong
                     13                           20
                Hungary                        India
                     13                           97
                   Iran                      Ireland
                     43                           24
                  Italy                      Jamaica
                     73                           81
                  Japan                         Laos
                     60                           18
                 Mexico                    Nicaragua
                    641                           34
Outlying-US(Guam-USVI-etc)                      Peru
                     14                           31
            Philippines                       Poland
                    196                           60
               Portugal                  Puerto-Rico
                     37                          114
               Scotland                        South
                     12                           78
                 Taiwan                     Thailand
                     50                           18
        Trinadad&Tobago                United-States
                     19                        28951
                Vietnam                   Yugoslavia
                     67                           16
[1] "Table for income_group"

 <=50K   >50K
24660   7657
```

From the above analysis we can see that while categorial data does not have specific NA values there are observations associated with ? value which might indicate NA values

[21]:
```
#brief look at ? related data and we can observe that it may have arose during␣
↪to data collection itself
head(cat_data[cat_data$workclass == ' ?',])
```

A data.frame: 6 × 9

	workclass <chr>	education <chr>	marital_status <chr>	occupation <chr>	relationship <chr>	race <chr
28	?	Some-college	Married-civ-spouse	?	Husband	Asia
62	?	7th-8th	Married-spouse-absent	?	Not-in-family	Whi
70	?	Some-college	Never-married	?	Own-child	Whi
78	?	10th	Married-civ-spouse	?	Husband	Whi
107	?	10th	Never-married	?	Own-child	Whi
129	?	HS-grad	Married-civ-spouse	?	Husband	Whi

[22]:
```
data[, cat_cols] <- lapply(data[, cat_cols], function(x) {
  x[x == " ?"] <- NA
  return(x)
})

data = na.omit(data)
```

[23]:
```
#after removing NA values dataset has 29936 rows
print(sum(is.na(data)))
print(dim(data))
```

```
[1] 0
[1] 29936    15
```

[24]:
```
head(data)
```

A data.frame: 6 × 15

	age <int>	workclass <chr>	fnlwgt <int>	education <chr>	education_num <int>	marital_status <chr>
1	39	State-gov	77516	Bachelors	13	Never-married
2	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse
3	38	Private	215646	HS-grad	9	Divorced
4	53	Private	234721	11th	7	Married-civ-spouse
5	28	Private	338409	Bachelors	13	Married-civ-spouse
6	37	Private	284582	Masters	14	Married-civ-spouse

[17]:
```
summary(data)
```

```
      age          workclass              fnlwgt          education
 Min.   :17.00   Length:29936       Min.   :  13769   Length:29936
 1st Qu.:28.00   Class :character   1st Qu.: 117679   Class :character
 Median :37.00   Mode  :character   Median : 178514   Mode  :character
 Mean   :38.39                      Mean   : 189846
 3rd Qu.:47.00                      3rd Qu.: 237647
 Max.   :90.00                      Max.   :1484705
 education_num   marital_status      occupation         relationship
 Min.   : 1.00   Length:29936       Length:29936       Length:29936
 1st Qu.: 9.00   Class :character   Class :character   Class :character
 Median :10.00   Mode  :character   Mode  :character   Mode  :character
 Mean   :10.11
```

```
 3rd Qu.:12.00
 Max.   :16.00
     race                sex             capital_gain     capital_loss
 Length:29936       Length:29936       Min.   :    0.0   Min.   :   0.00
 Class :character   Class :character   1st Qu.:    0.0   1st Qu.:   0.00
 Mode  :character   Mode  :character   Median :    0.0   Median :   0.00
                                       Mean   :  603.6   Mean   :  88.68
                                       3rd Qu.:    0.0   3rd Qu.:   0.00
                                       Max.   :41310.0   Max.   :4356.00
 hours_per_week  native_country     income_group
 Min.   : 1.00   Length:29936       Length:29936
 1st Qu.:40.00   Class :character   Class :character
 Median :40.00   Mode  :character   Mode  :character
 Mean   :40.73
 3rd Qu.:45.00
 Max.   :98.00
```

[25]:
```r
#reinitiating cols data
cat_cols <- sapply(data, function(x) !is.numeric(x))
cat_data <- data[, cat_cols]

for (col in names(cat_data)) {
  print(paste("Table for", col))
  print(table(data[[col]]))
}
```

```
[1] "Table for workclass"

      Federal-gov        Local-gov          Private     Self-emp-inc
              941             2058            22165             1032
 Self-emp-not-inc        State-gov      Without-pay
             2449             1277               14
[1] "Table for education"

        10th         11th         12th      1st-4th      5th-6th
         815         1047          376          151          288
     7th-8th          9th   Assoc-acdm    Assoc-voc    Bachelors
         552          453         1004         1302         4998
   Doctorate      HS-grad      Masters    Preschool  Prof-school
         357         9801         1603           45          491
Some-college
        6653
[1] "Table for marital_status"

             Divorced     Married-AF-spouse    Married-civ-spouse
                 4194                    21                 13892
Married-spouse-absent         Never-married             Separated
                  368                  9701                   936
```

```
             Widowed
                 824
[1] "Table for occupation"

     Adm-clerical       Armed-Forces       Craft-repair    Exec-managerial
             3715                  9               4015               3942
  Farming-fishing  Handlers-cleaners  Machine-op-inspct      Other-service
              977               1349               1965               3200
  Priv-house-serv     Prof-specialty    Protective-serv              Sales
              141               3956                640               3557
     Tech-support   Transport-moving
              909               1561
[1] "Table for relationship"

       Husband   Not-in-family  Other-relative       Own-child       Unmarried
         12304            7690             887            4461            3200
          Wife
          1394
[1] "Table for race"

 Amer-Indian-Eskimo  Asian-Pac-Islander               Black               Other
                286                 884                2804                 229
              White
              25733
[1] "Table for sex"

 Female   Male
   9747  20189
[1] "Table for native_country"

                  Cambodia                     Canada
                        18                        106
                     China                   Columbia
                        68                         56
                      Cuba         Dominican-Republic
                        92                         66
                   Ecuador                El-Salvador
                        27                        100
                   England                     France
                        85                         26
                   Germany                     Greece
                       128                         29
                 Guatemala                      Haiti
                        63                         42
        Holand-Netherlands                   Honduras
                         1                         12
                      Hong                    Hungary
                        19                         13
```

```
                   India                       Iran
                      97                         42
                 Ireland                      Italy
                      24                         68
                 Jamaica                      Japan
                      80                         57
                    Laos                     Mexico
                      17                        608
               Nicaragua  Outlying-US(Guam-USVI-etc)
                      33                         14
                    Peru                Philippines
                      30                        186
                  Poland                   Portugal
                      56                         34
             Puerto-Rico                   Scotland
                     109                         11
                   South                     Taiwan
                      70                         41
                Thailand          Trinadad&Tobago
                      17                         18
           United-States                    Vietnam
                   27293                         64
              Yugoslavia
                      16
[1] "Table for income_group"
```

<=50K	>50K
22601	7335

We will fit three type of models in this notebook: Linear Models, Bayesian Heirarchical model, and a Logistic Regression based Deep Learning Model. And for them we need different type of datasets

During our project we tested multiple versions of this dataset and found that only by reducing the levels across the categorial variables we would be able to reduce multicollinearity for linear and bayesian models. For the basic neurel network there is no need to do so.

[26]:
```
#IN GAM or GLM number of tests can lead to faulty tests so we will try to␣
↪reduce the number of groups across the dataset logically

df_gm <- subset(data, select = -c(native_country)) #first we will remove native␣
↪country altogether
head(df_gm)
```

A data.frame: 6 × 14

	age	workclass	fnlwgt	education	education_num	marital_status
	<int>	<chr>	<int>	<chr>	<int>	<chr>
1	39	State-gov	77516	Bachelors	13	Never-married
2	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse
3	38	Private	215646	HS-grad	9	Divorced
4	53	Private	234721	11th	7	Married-civ-spouse
5	28	Private	338409	Bachelors	13	Married-civ-spouse
6	37	Private	284582	Masters	14	Married-civ-spouse

[28]:

```
#reducing the factor levels across the categories by recoding their classes␣
↪logically

df_gm <- df_gm %>%
  mutate(
    # Recode workclass into four groups
    workclass = recode(workclass,
      " Private" = "Private",
      " Private" = "Private",
      " State-gov" = "Government",
      " Federal-gov" = "Government",
      " Local-gov" = "Government",
      " Self-emp-inc" = "Self-employed",
      " Self-emp-not-inc" = "Self-employed",
      " Without-pay" = "Unpaid/Other"
    ),
    # Recode education:
    education = case_when(
      education %in% c(" Preschool", " 1st-4th", " 5th-6th", " 7th-8th") ~␣
↪"Less than High School",
      education %in% c(" 9th", " 10th", " 11th", " 12th", " HS-grad") ~ "High␣
↪School",
      education %in% c(" Some-college", " Assoc-acdm", ' Assoc-voc') ~ "Some␣
↪College/Associate",
      education == " Bachelors" ~ "Bachelors",
      education == " Masters" ~ "Masters",
      education == " Doctorate" ~ "Doctorate",
      TRUE ~ "Other"
    ),
    # Recode marital status into two groups
    marital_status = ifelse(marital_status %in% c(" Married-civ-spouse", "␣
↪Married-AF-spouse", " Married-spouse-absent"),
                            " Married", " Not Married"),
    # Recode relationship into three groups
    relationship = case_when(
      relationship %in% c(" Husband", " Wife") ~ "Spouse/Partner",
      relationship == " Own-child" ~ "Child",
```

```r
    relationship %in% c(" Other relative", " Not in family", " Unmarried") ~
↪"Other/Nonfamily",
    TRUE ~ "Other/Nonfamily"
  ),
  occupation = case_when(
  occupation %in% c(" Exec-managerial", " Prof-specialty", " Tech-support") ~
↪"Management/Professional",
  occupation %in% c(" Adm-clerical", " Sales") ~ "Office/Clerical & Sales",
  occupation %in% c(" Handlers-cleaners", " Machine-op-inspct", "
↪Craft-repair", " Transport-moving/cr",
                    " Farming-fishing", " Other-service", " Priv-house-serv",
↪" Protective-serv", " Armed-Forces") ~ "Manual/Service",
  TRUE ~ "Other"  # for any unexpected levels
 )
 )
```

[29]:

```r
head(df_gm)
```

A data.frame: 6 × 14

	age	workclass	fnlwgt	education	education_num	marital_status	oc
	<int>	<chr>	<int>	<chr>	<int>	<chr>	<c
1	39	Government	77516	Bachelors	13	Not Married	Of
2	50	Self-employed	83311	Bachelors	13	Married	Ma
3	38	Private	215646	High School	9	Not Married	Ma
4	53	Private	234721	High School	7	Married	Ma
5	28	Private	338409	Bachelors	13	Married	Ma
6	37	Private	284582	Masters	14	Married	Ma

[30]:

```r
summary(df_gm)
```

```
      age          workclass              fnlwgt          education
 Min.   :17.00   Length:29936       Min.   :  13769   Length:29936
 1st Qu.:28.00   Class :character   1st Qu.: 117679   Class :character
 Median :37.00   Mode  :character   Median : 178514   Mode  :character
 Mean   :38.39                      Mean   : 189846
 3rd Qu.:47.00                      3rd Qu.: 237647
 Max.   :90.00                      Max.   :1484705
 education_num   marital_status      occupation         relationship
 Min.   : 1.00   Length:29936       Length:29936        Length:29936
 1st Qu.: 9.00   Class :character   Class :character    Class :character
 Median :10.00   Mode  :character   Mode  :character    Mode  :character
 Mean   :10.11
 3rd Qu.:12.00
 Max.   :16.00
     race                sex             capital_gain      capital_loss
 Length:29936       Length:29936       Min.   :    0.0   Min.   :   0.00
 Class :character   Class :character   1st Qu.:    0.0   1st Qu.:   0.00
 Mode  :character   Mode  :character   Median :    0.0   Median :   0.00
```

```
                                          Mean   :  603.6   Mean   :  88.68
                                          3rd Qu.:    0.0   3rd Qu.:   0.00
                                          Max.   :41310.0   Max.   :4356.00
 hours_per_week  income_group
 Min.   : 1.00   Length:29936
 1st Qu.:40.00   Class :character
 Median :40.00   Mode  :character
 Mean   :40.73
 3rd Qu.:45.00
 Max.   :98.00
```

[24]:

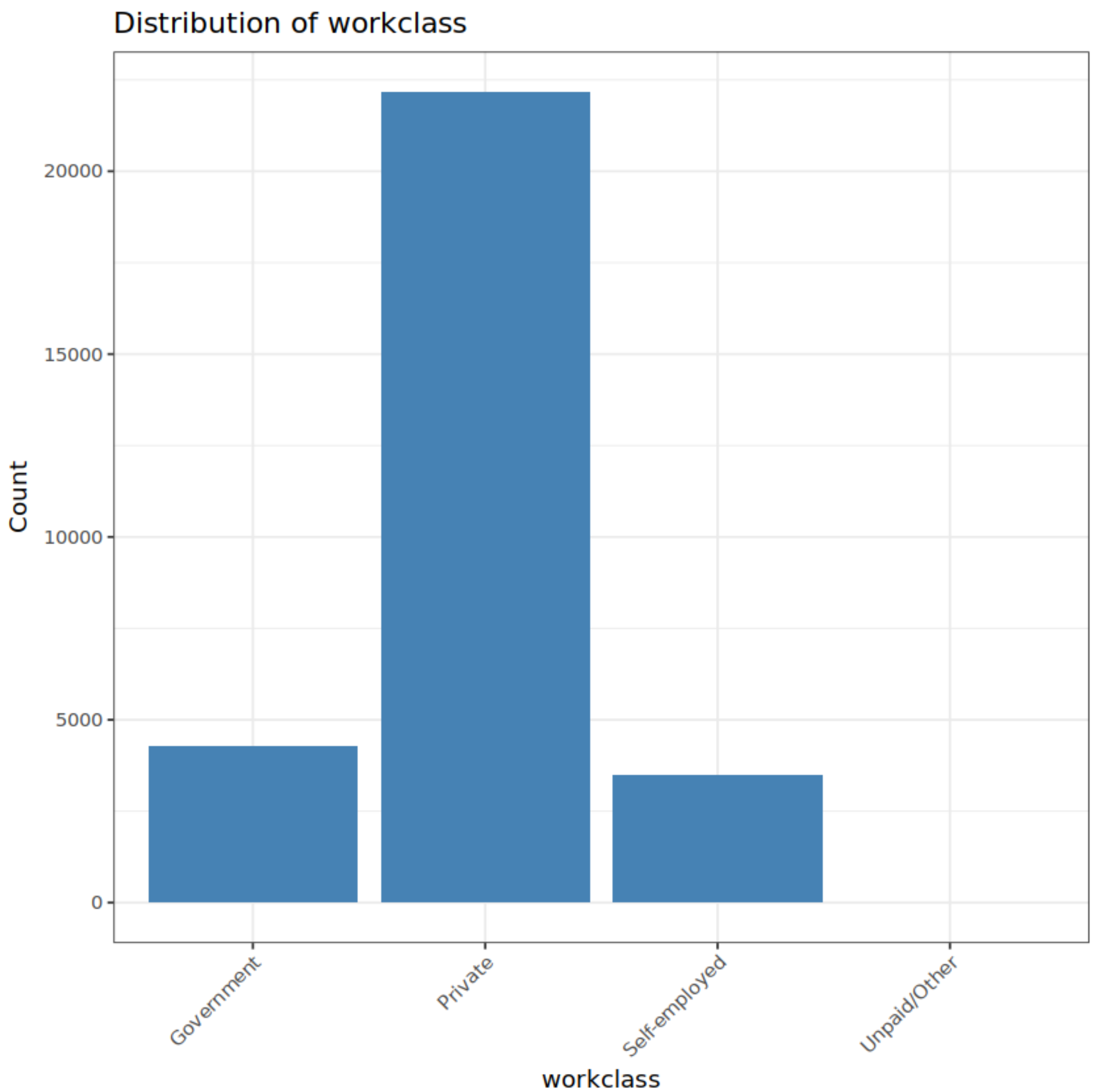
```
#reinitiating cols data
cat_cols <- sapply(df_gm, function(x) !is.numeric(x))
cat_data <- df_gm[, cat_cols]

for (col in names(cat_data)) {
  print(paste("Table for", col))
  print(prop.table(table(df_gm[[col]])))
}

#reduced levels are indeed much more sensible and informative than before while
 ↪also having reduced complexities
```

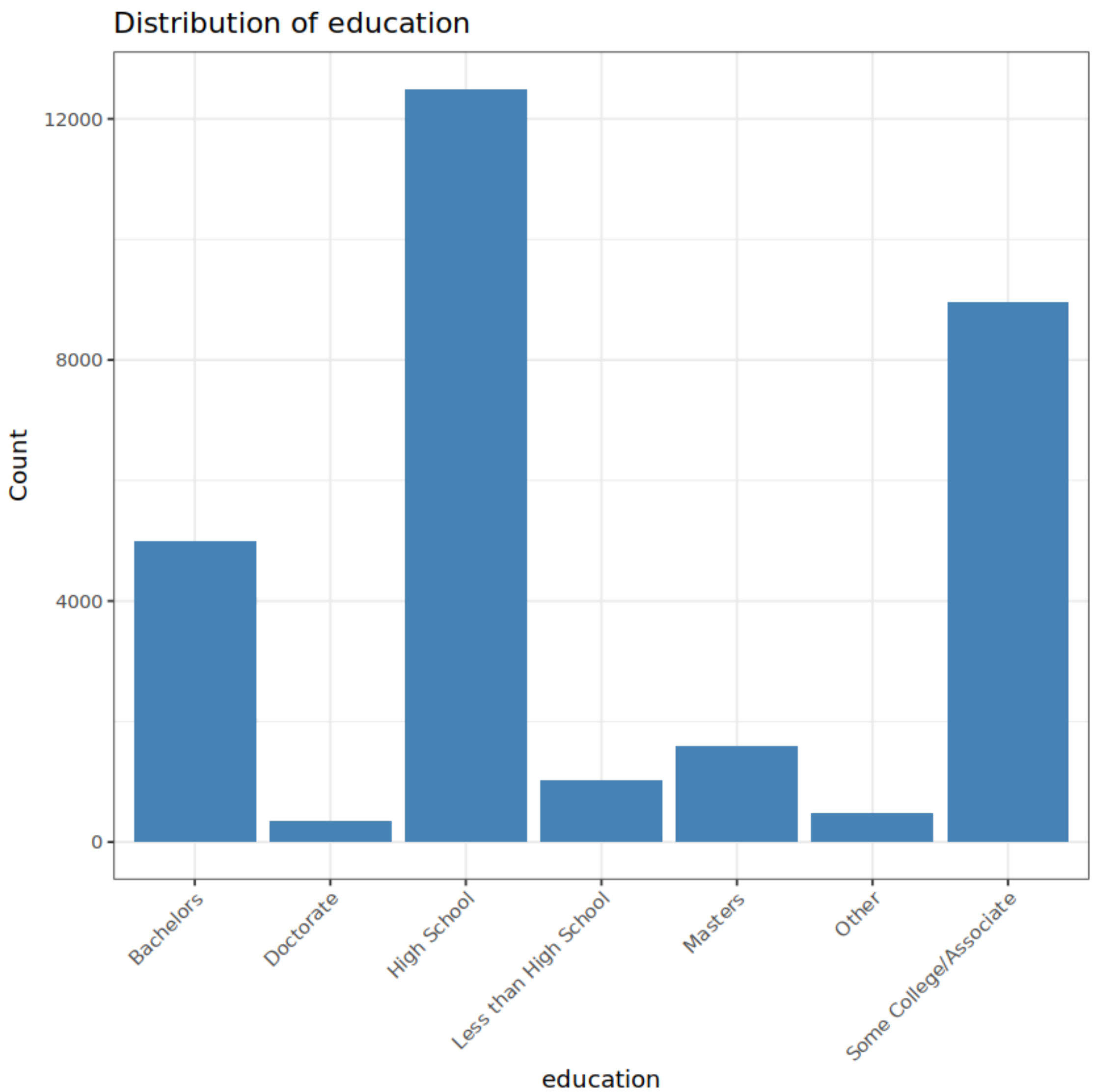
```
[1] "Table for workclass"

   Government       Private Self-employed  Unpaid/Other
 0.1428380545  0.7404128808  0.1162814003  0.0004676644
[1] "Table for education"

            Bachelors             Doctorate           High School
           0.16695617            0.01192544            0.41729022
 Less than High School               Masters                 Other
           0.03460716            0.05354757            0.01640166
Some College/Associate
           0.29927178
[1] "Table for marital_status"

    Married  Not Married
   0.477051     0.522949
[1] "Table for occupation"

Management/Professional          Manual/Service Office/Clerical & Sales
             0.29419428              0.41074292              0.24291823
                  Other
             0.05214458
[1] "Table for relationship"

          Child Other/Nonfamily  Spouse/Partner
```

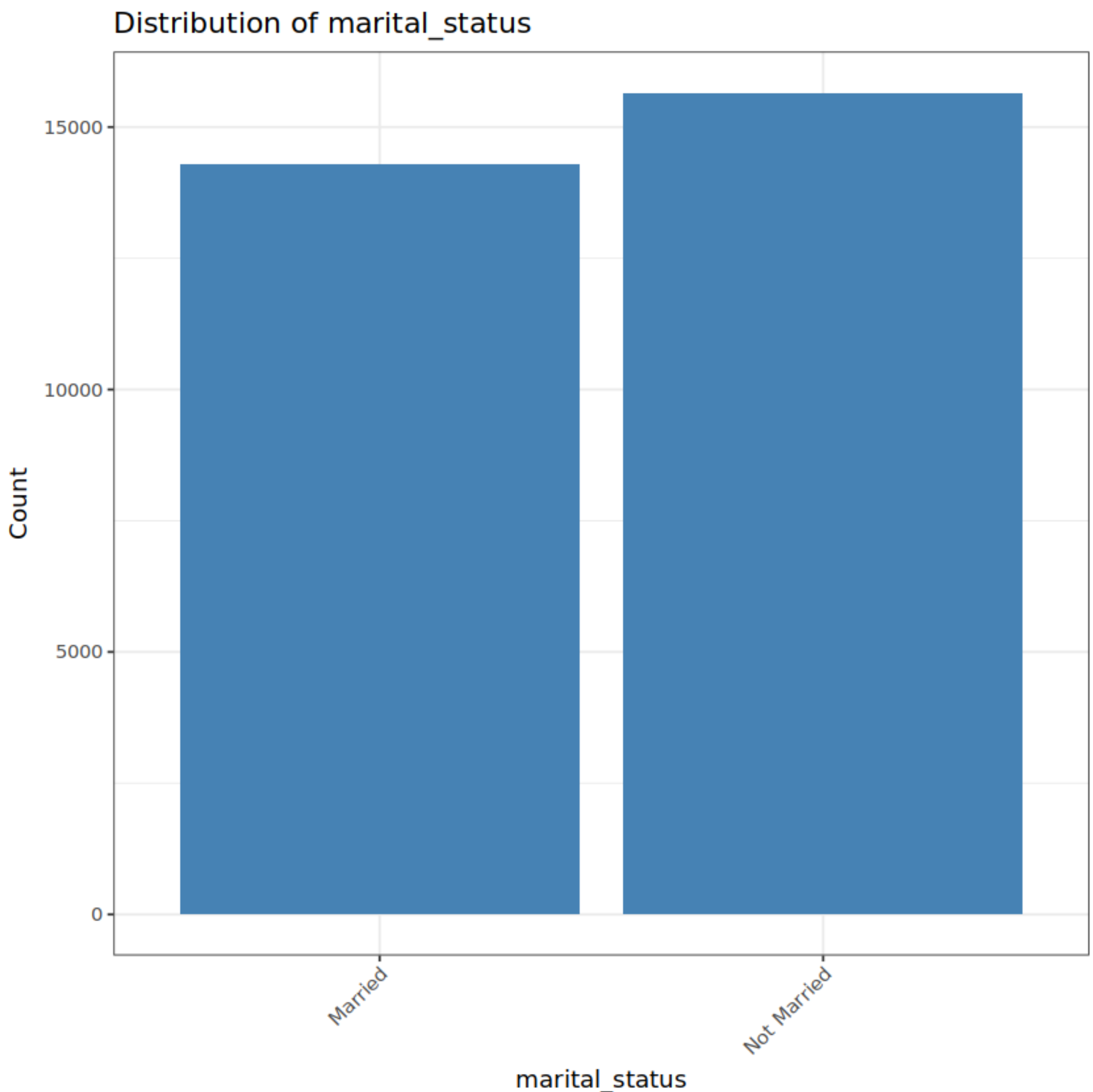
```
      0.1490179       0.3934059       0.4575762
[1] "Table for race"

 Amer-Indian-Eskimo  Asian-Pac-Islander               Black               Other
        0.009553715         0.029529663         0.093666489         0.007649653
              White
        0.859600481
[1] "Table for sex"

   Female      Male
0.3255946 0.6744054
[1] "Table for income_group"

    <=50K      >50K
0.7549773 0.2450227
```

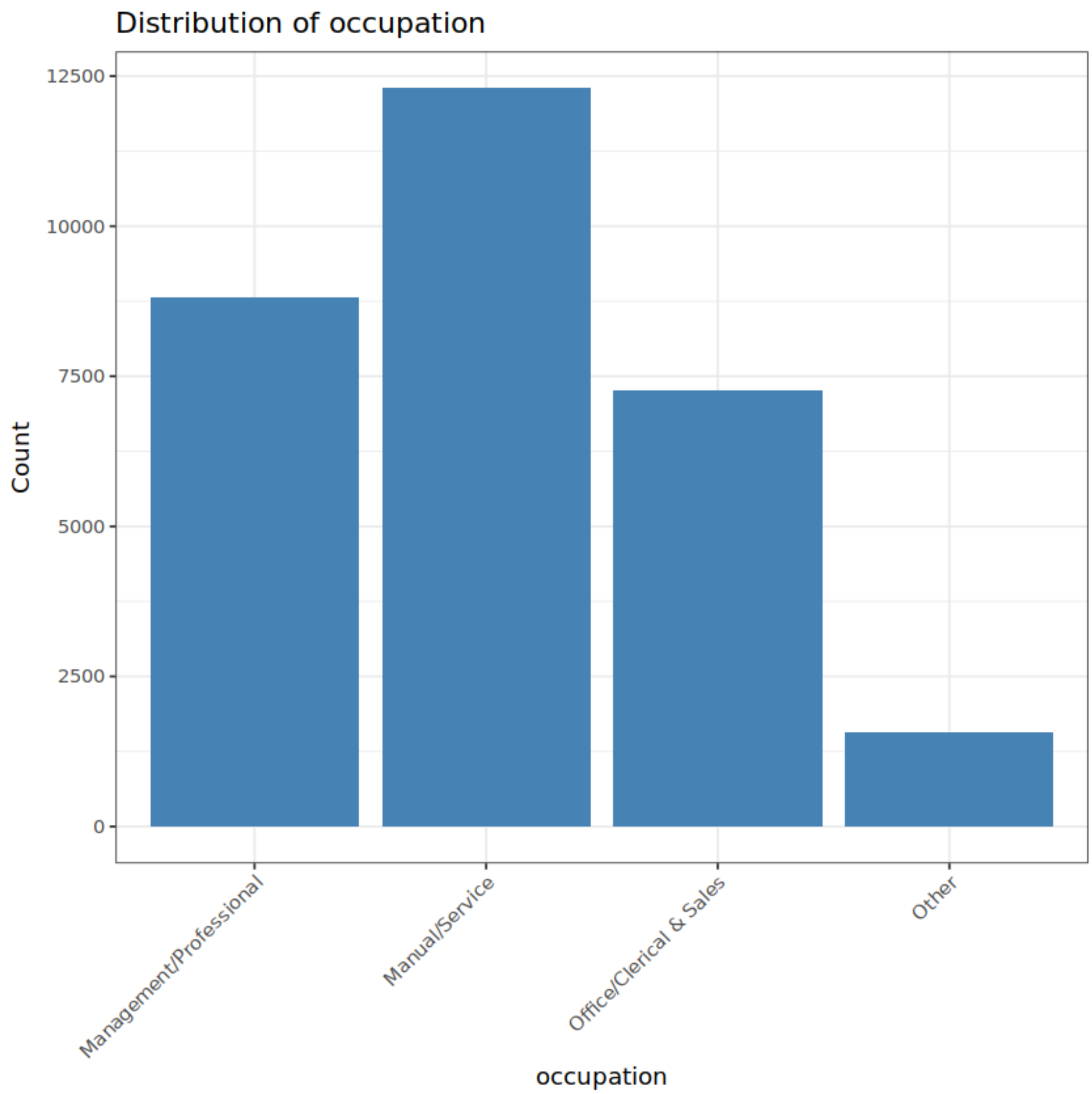
Now we will plot the categorial and numerical variables and analyse the data in hand.

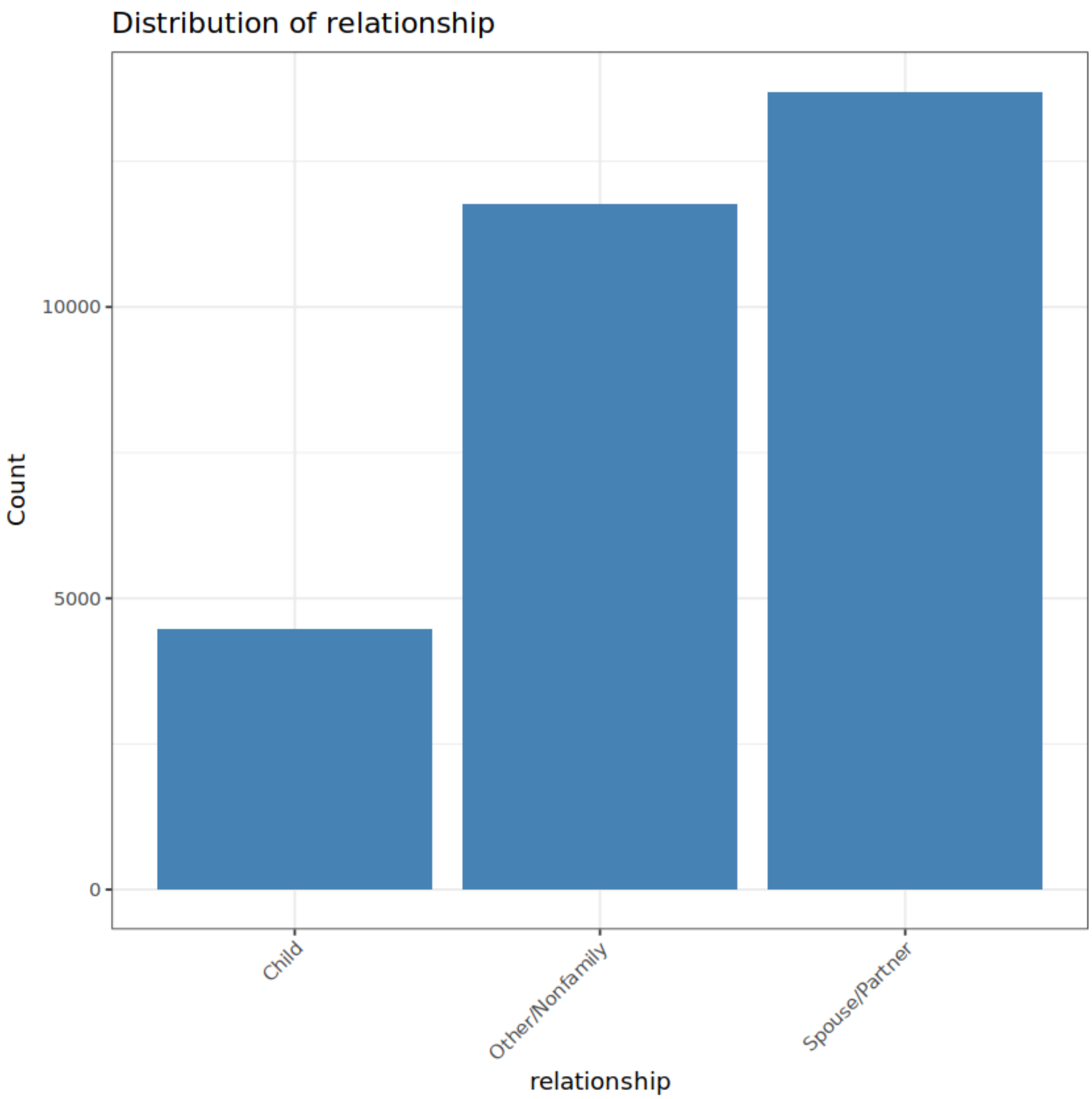
[27]:
```r
for (var in names(cat_data)) {
  p <- ggplot(df_gm, aes_string(x = var)) +
    geom_bar(fill = "steelblue") +
    theme_bw() +
    labs(title = paste("Distribution of", var),
         x = var, y = "Count") +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))

  print(p)
}
```

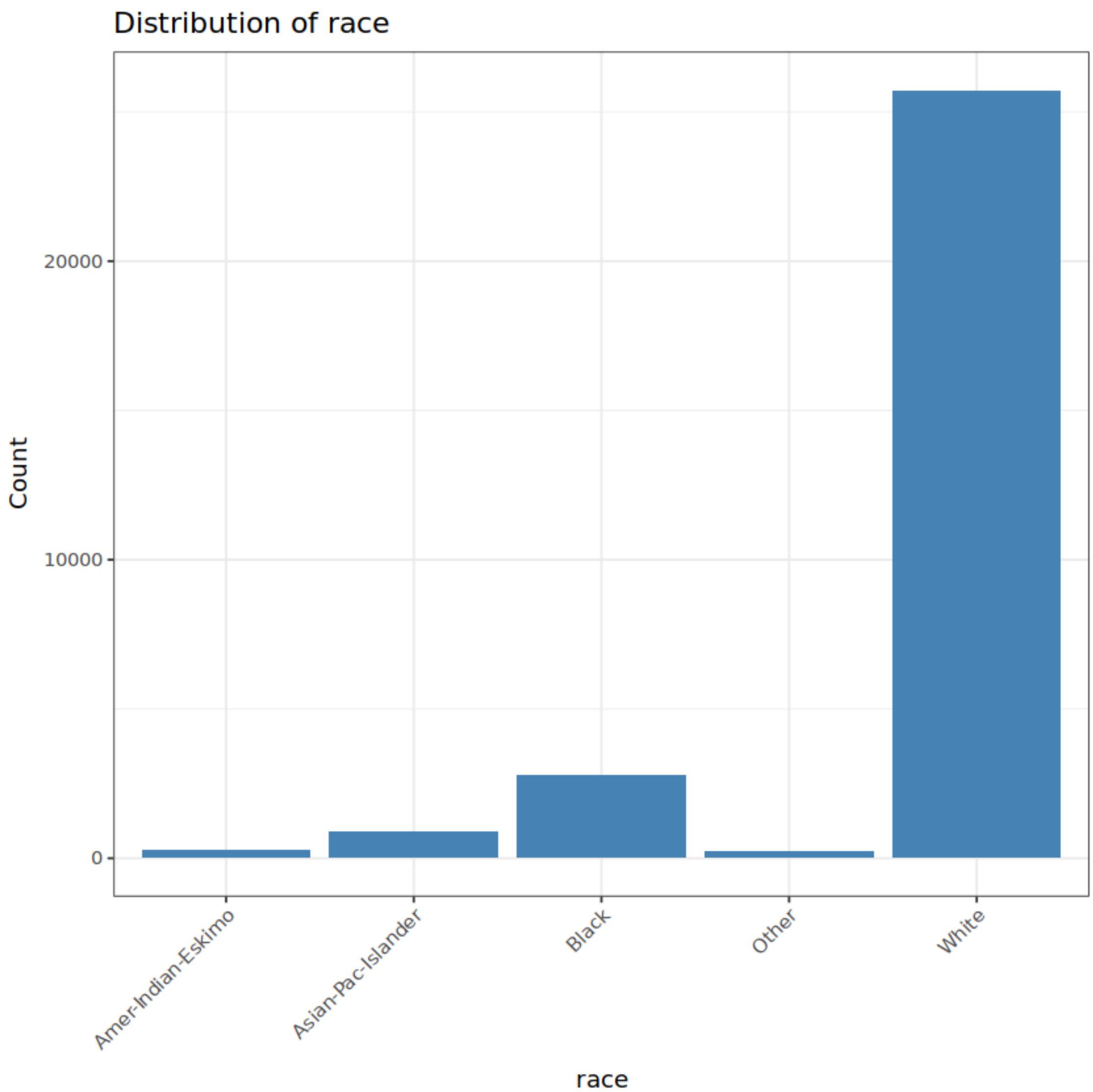

Distribution of workclass
Count
20000
15000
10000
5000
0
Government
Private
Self-employed
Unpaid/Other
workclass

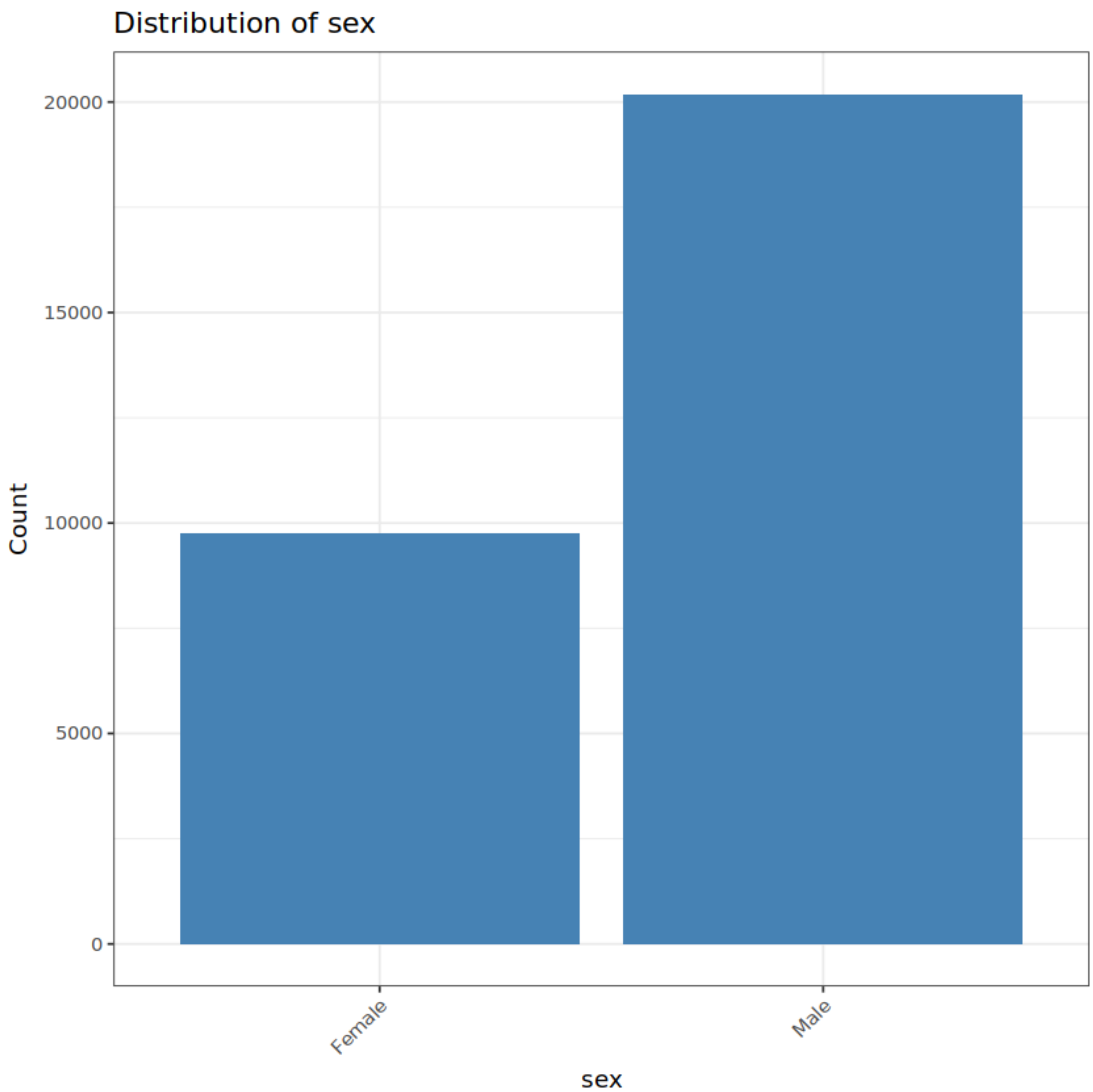

Distribution of education
Count
12000
8000
4000
0
Bachelors
Doctorate
High School
Less than High School
Masters
Other
Some College/Associate
education


Distribution of marital_status
15000
10000
5000
0
Count
Married
Not Married
marital_status

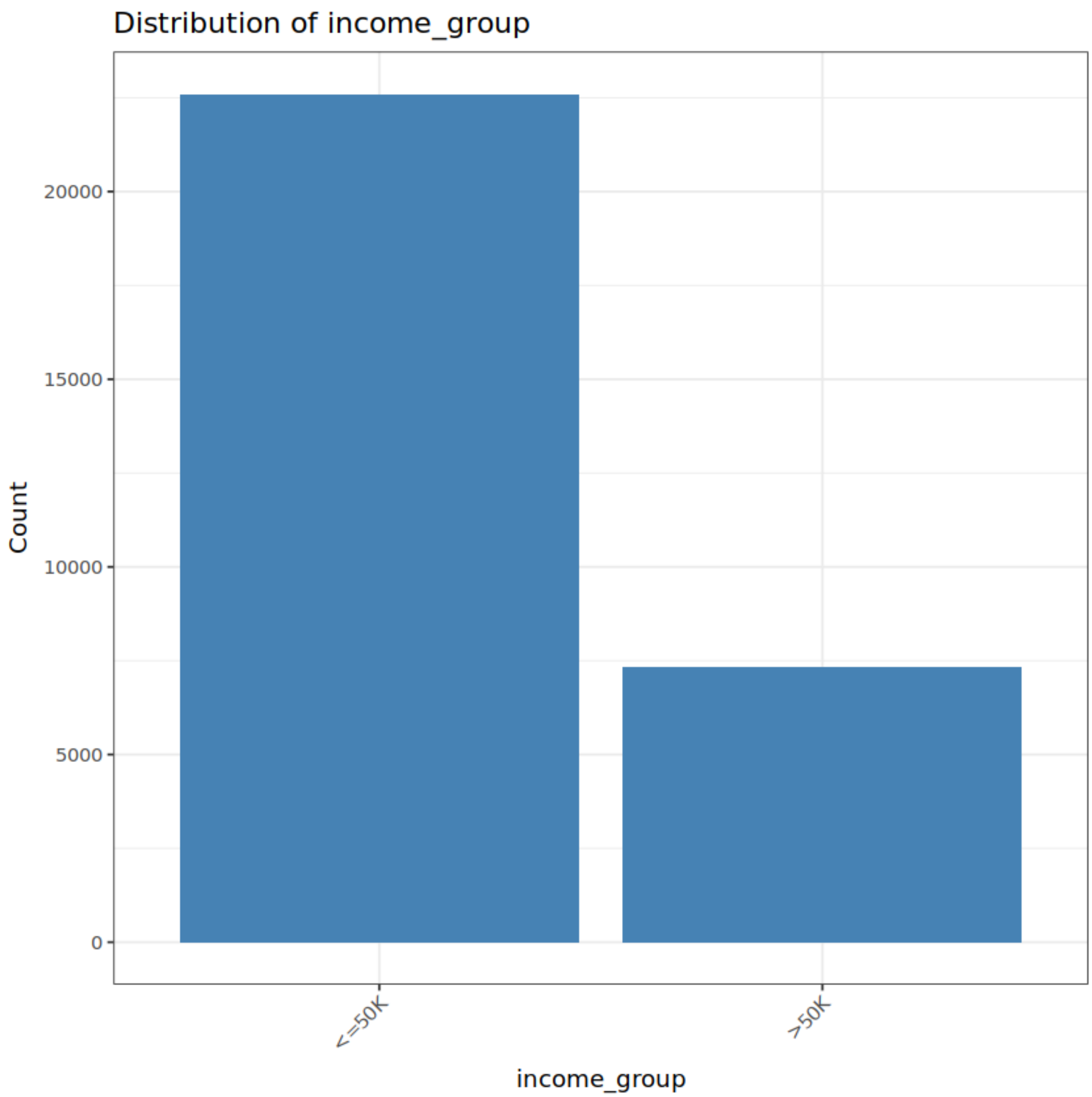

Distribution of occupation
12500
10000
7500
5000
2500
0
Count
Management/Professional
Manual/Service
Office/Clerical & Sales
Other
occupation


Distribution of relationship
Count
10000
5000
0
Child
Other/Nonfamily
Spouse/Partner
relationship


Distribution of race
Count
20000
10000
0
Amer-Indian-Eskimo
Asian-Pac-Islander
Black
Other
White
race



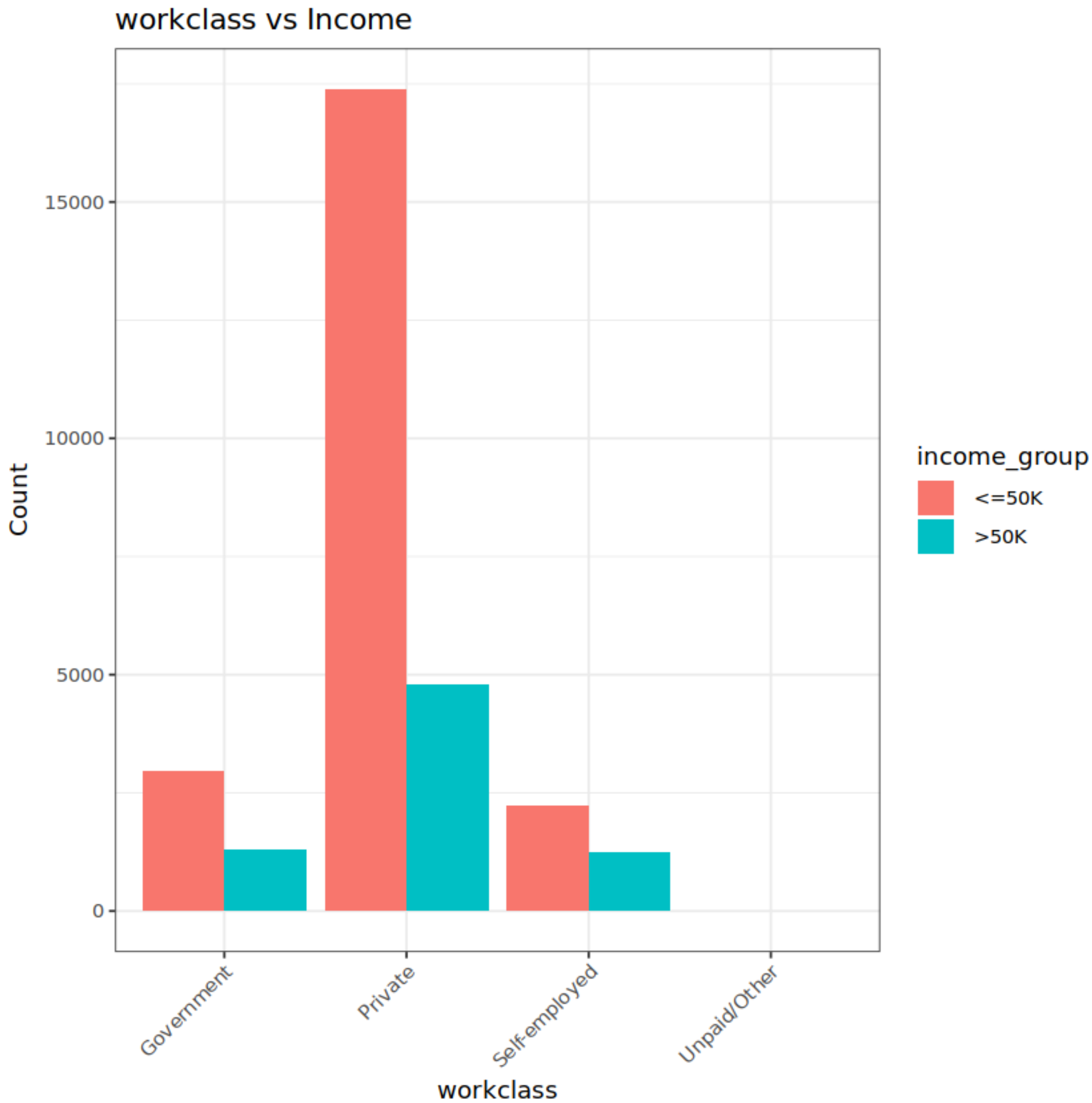
Distribution of sex
20000
15000
10000
5000
0
Count
Female
Male
sex

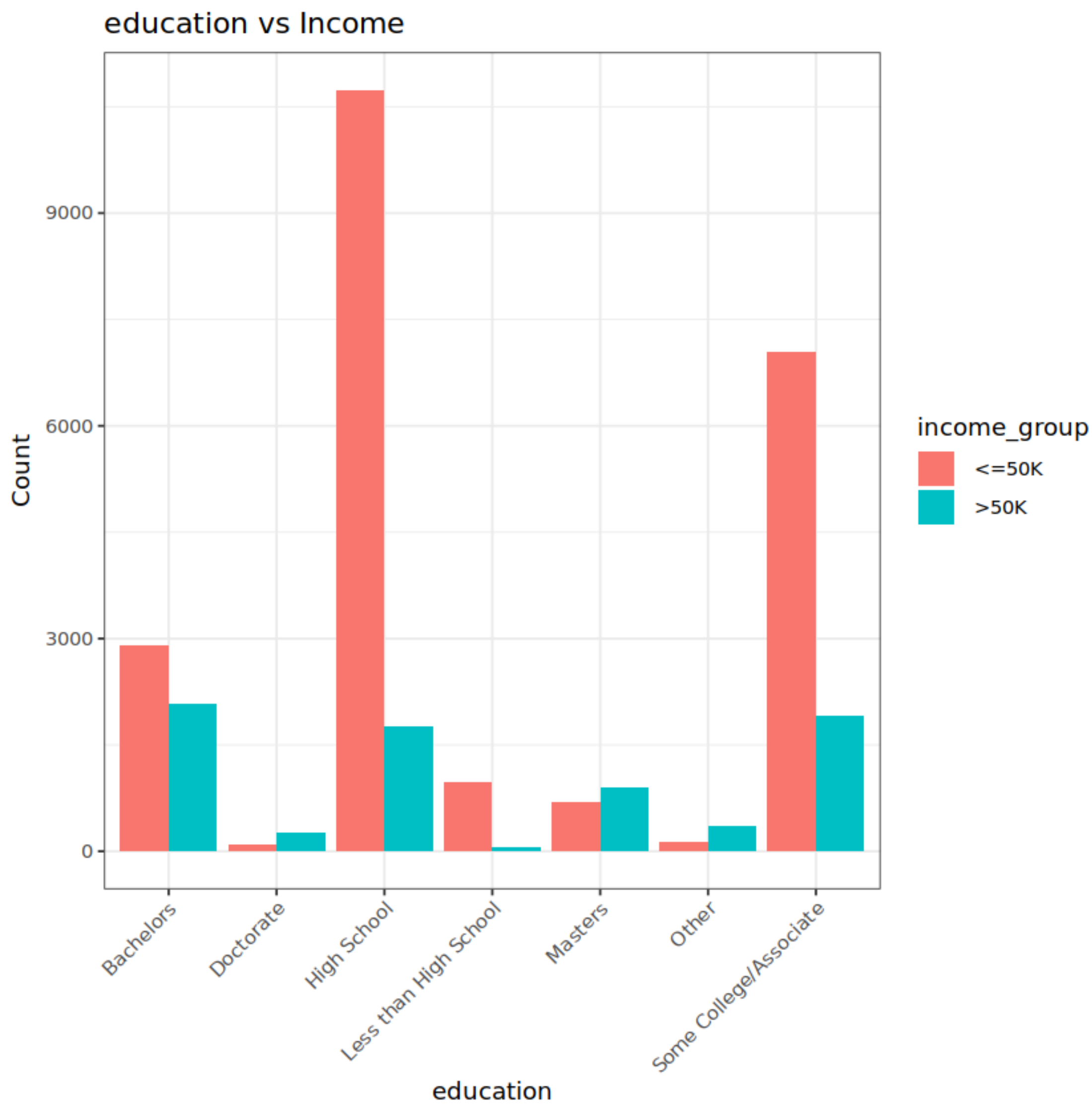


From above graphs we have following observations: 1. Most of the workforce have been employed in private sector with 74% of the sample falling in this group. 2. Most of the workforce have highschool as highest education followed by at least some college education, 3. Proportion of married vs not married is quite similar with 52% people not married, not that this category include divorced, widowed, and other similar categories besides not marrying at all. 4. Most of the workforce were employed manual/service sector. 5. Around 85% of our sample is represented by white race which is not a unique observation overall. However it would be interesting to observe unemployment across the races too with this data, sadly we don't have similar dataset across similar timeline on our hand. 6. 67% of workforce in the dataset are men which is quite unexpected as there should have been some difference but not this much of a difference. 7. Around 24% of the observations have more than 50k income a year. It is to be noted that the observations are weighted so this value is not a representative of the population.

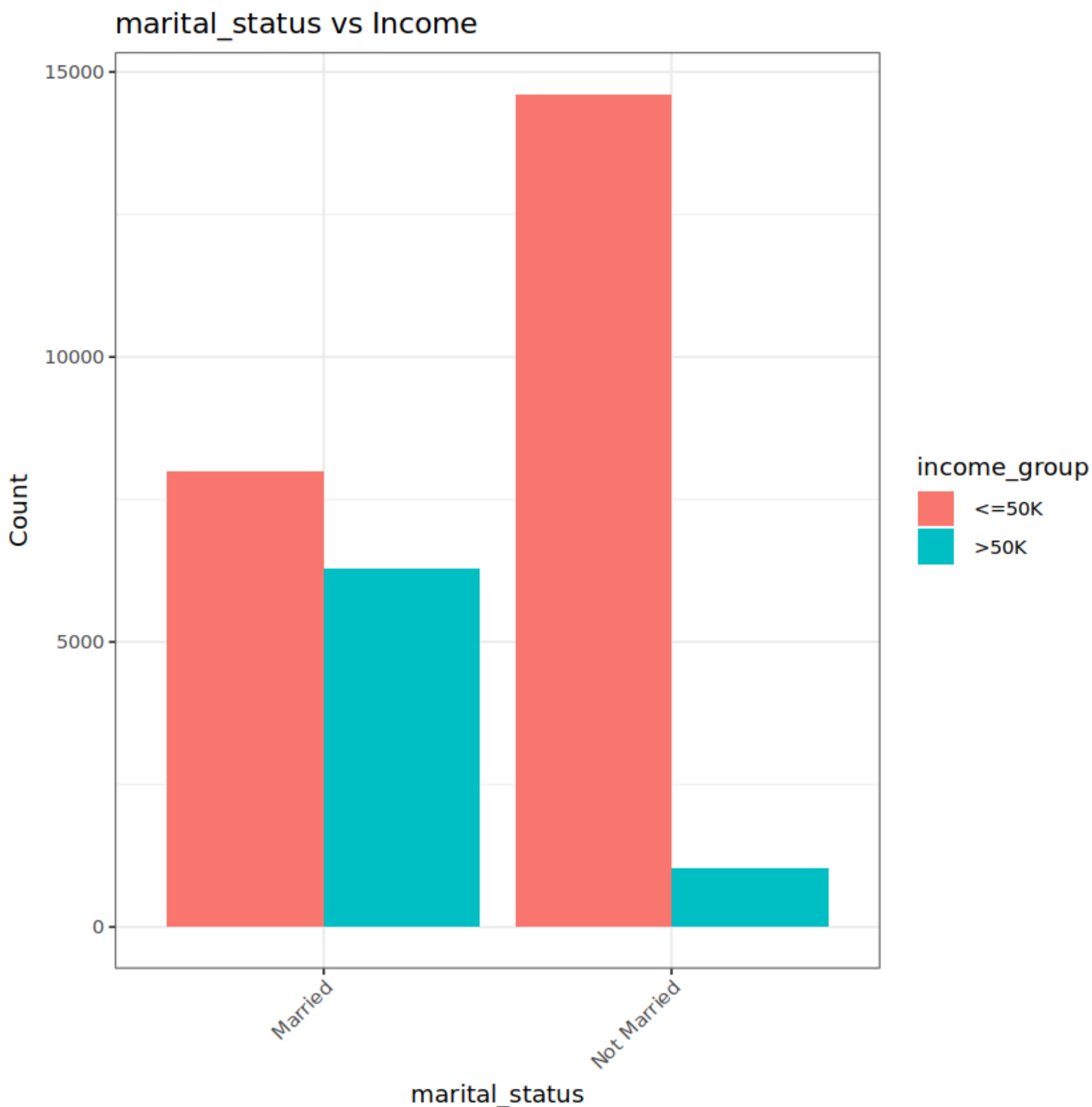
[26]:

```
# Check the association between workclass and income
for (var in names(cat_data)) {
    if (var != 'income_group'){
  p <- ggplot(df_gm, aes_string(x = var, fill = 'income_group')) +
    geom_bar(position = 'dodge') +
    theme_bw() +
    labs(title = paste(var,'vs Income'),
         x = var, y = "Count") +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))

  print(p)
}}
```

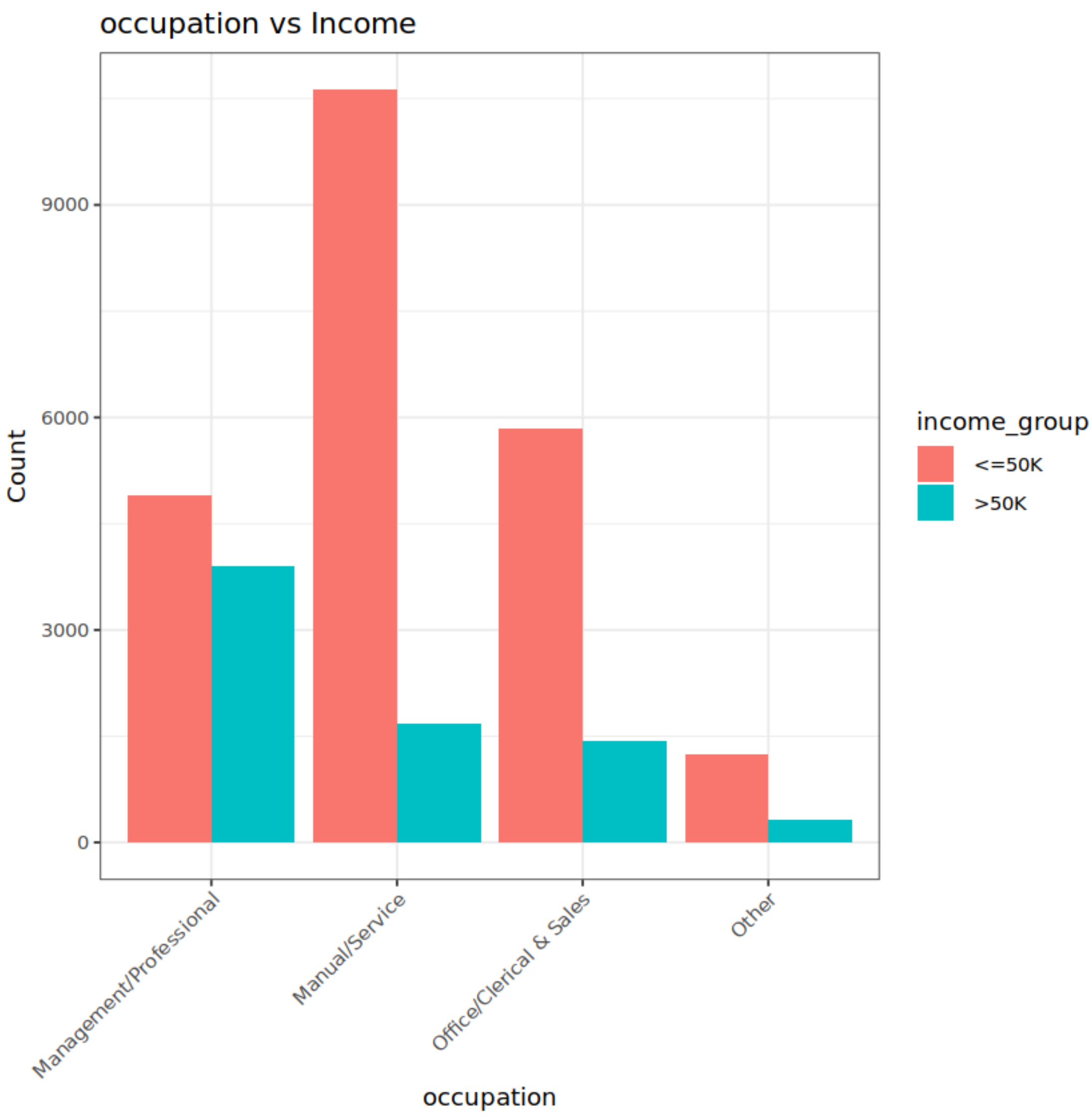


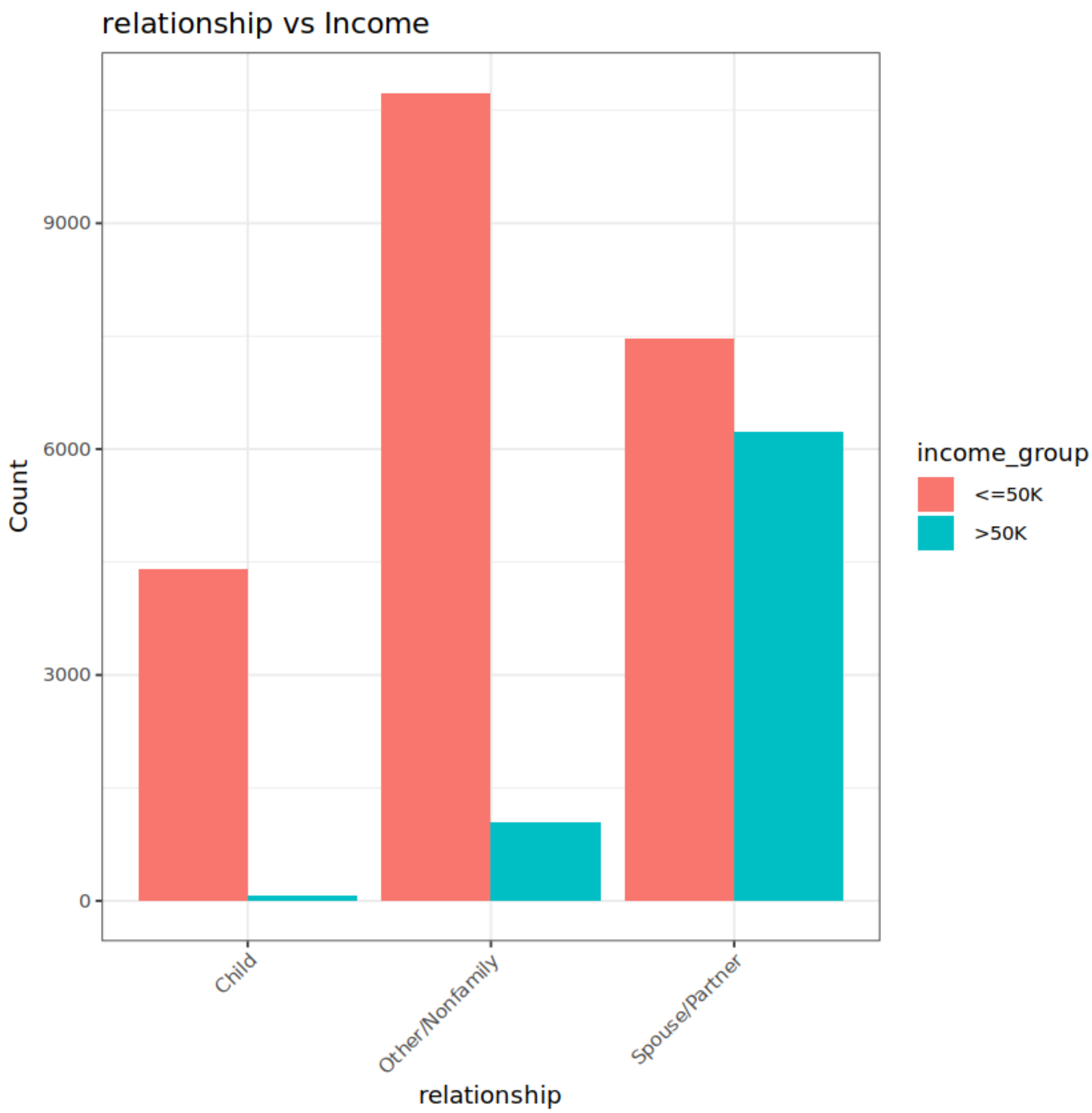


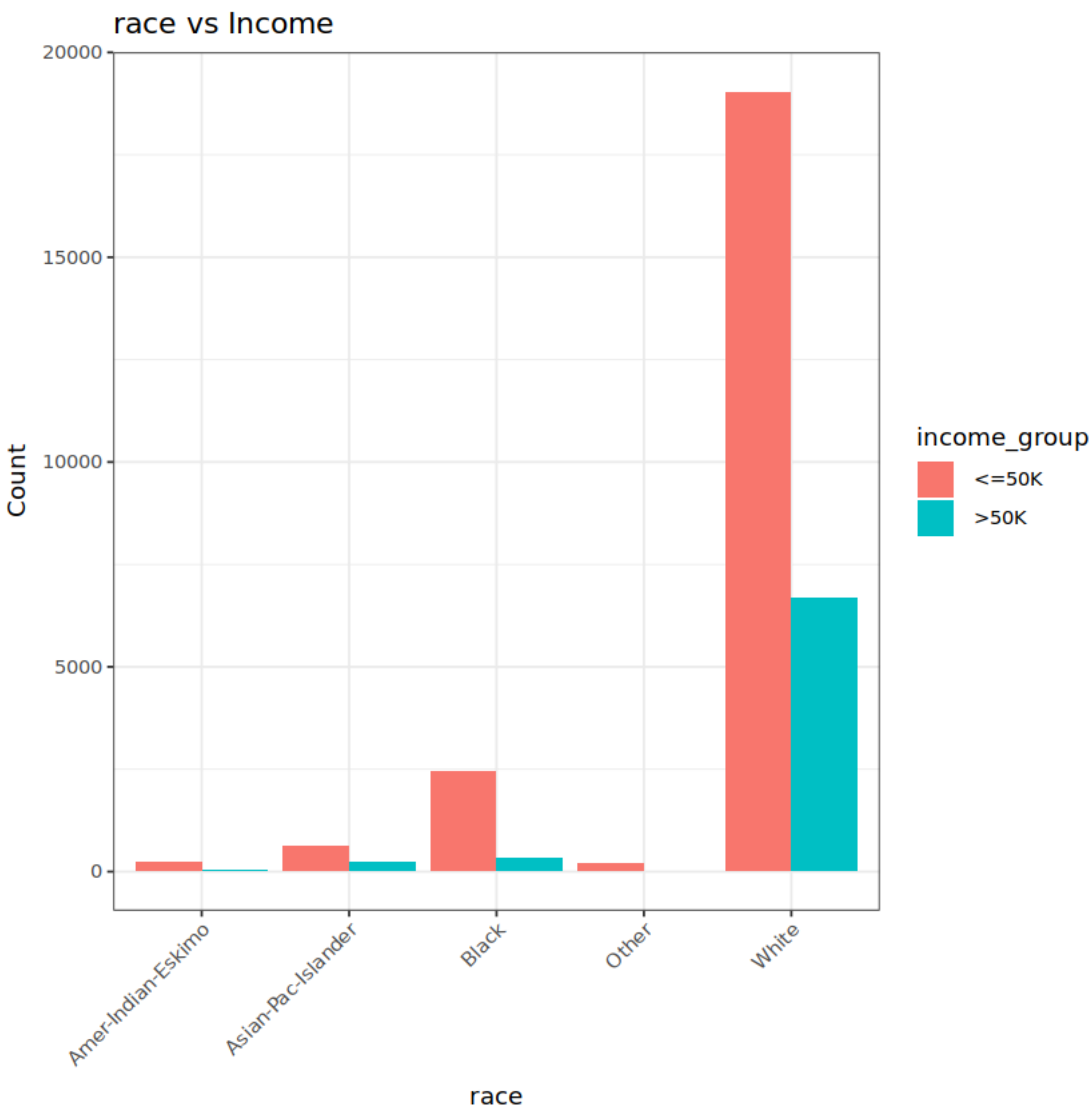
education vs Income
Count
9000
6000
3000
0
Bachelors
Doctorate
High School
Less than High School
Masters
Other
Some College/Associate
education
income_group
<=50K
>50K



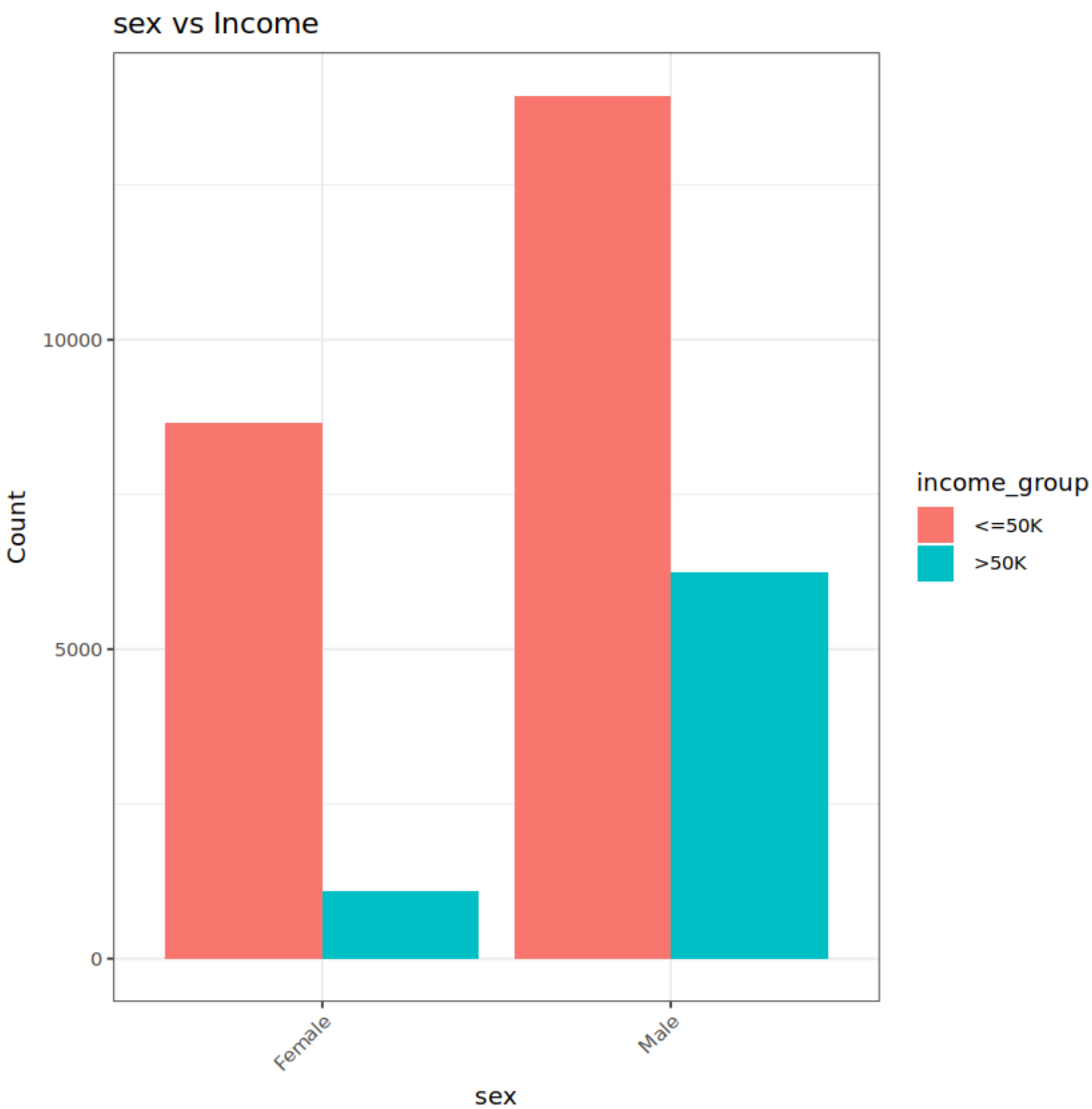
marital_status vs Income
15000
10000
5000
0
Count
Married
Not Married
marital_status
income_group
<=50K
>50K


occupation vs Income
Count
9000
6000
3000
0
Management/Professional
Manual/Service
Office/Clerical & Sales
Other
occupation
income_group
<=50K
>50K


relationship vs Income
Count
9000
6000
3000
0
Child
Other/Nonfamily
Spouse/Partner
relationship
income_group
<=50K
>50K



race vs Income
20000
15000
10000
5000
0
Count
Amer-Indian-Eskimo
Asian-Pac-Islander
Black
Other
White
race
income_group
<=50K
>50K



Above barplots provide us wiht more important observations than previous plots: 1. Masters, Doctrates, and Professional degrees incldued in other are the only categories where we have more number of workforce having more than 50k income. This also supports a general consensus that higher education leads to better salary overall. 2. Proportion of people having income greater than 50k is way less in not married class than the married class. This is an interesting observation as proportions should have been equal overall. It shows that married couples have more income overall. 3. Management and Professional vocations have larger proportion of the workforce in income above 50k. While present day data may differ it shows that in 1990s management and professional vocations indeed were more lucrative than manual, sales, and other jobs.

Readers can observe some of the proportions contained in the contingency tables below for higher precisions.

[309]:
```
CrossTable(df_gm$marital_status, df_gm$income_group, prop.chisq = FALSE)
```

```
   Cell Contents
|-------------------------|
|                       N |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  29936


                     | df_gm$income_group
df_gm$marital_status |     <=50K |      >50K | Row Total |
---------------------|-----------|-----------|-----------|
             Married |      7987 |      6294 |     14281 |
                     |     0.559 |     0.441 |     0.477 |
                     |     0.353 |     0.858 |           |
                     |     0.267 |     0.210 |           |
---------------------|-----------|-----------|-----------|
         Not Married |     14614 |      1041 |     15655 |
                     |     0.934 |     0.066 |     0.523 |
                     |     0.647 |     0.142 |           |
                     |     0.488 |     0.035 |           |
---------------------|-----------|-----------|-----------|
        Column Total |     22601 |      7335 |     29936 |
                     |     0.755 |     0.245 |           |
---------------------|-----------|-----------|-----------|
```

[313]:

```
CrossTable(df_gm$occupation, df_gm$income_group, prop.chisq = FALSE)
```

```
   Cell Contents
|-------------------------|
|                       N |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  29936
```

```
                        | df_gm$income_group
       df_gm$occupation |     <=50K |      >50K | Row Total |
------------------------|-----------|-----------|-----------|
Management/Professional |      4897 |      3910 |      8807 |
                        |     0.556 |     0.444 |     0.294 |
                        |     0.217 |     0.533 |           |
                        |     0.164 |     0.131 |           |
------------------------|-----------|-----------|-----------|
         Manual/Service |     10622 |      1674 |     12296 |
                        |     0.864 |     0.136 |     0.411 |
                        |     0.470 |     0.228 |           |
                        |     0.355 |     0.056 |           |
------------------------|-----------|-----------|-----------|
Office/Clerical & Sales |      5837 |      1435 |      7272 |
                        |     0.803 |     0.197 |     0.243 |
                        |     0.258 |     0.196 |           |
                        |     0.195 |     0.048 |           |
------------------------|-----------|-----------|-----------|
                  Other |      1245 |       316 |      1561 |
                        |     0.798 |     0.202 |     0.052 |
                        |     0.055 |     0.043 |           |
                        |     0.042 |     0.011 |           |
------------------------|-----------|-----------|-----------|
           Column Total |     22601 |      7335 |     29936 |
                        |     0.755 |     0.245 |           |
------------------------|-----------|-----------|-----------|
```

[312]:
```
CrossTable(df_gm$workclass, df_gm$income_group, prop.chisq = FALSE)
```

```
   Cell Contents
|-------------------------|
|                       N |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  29936


                | df_gm$income_group
df_gm$workclass |     <=50K |      >50K | Row Total |
```

```
----------------|-----------|-----------|-----------|
     Government |      2966 |      1310 |      4276 |
                |     0.694 |     0.306 |     0.143 |
                |     0.131 |     0.179 |           |
                |     0.099 |     0.044 |           |
----------------|-----------|-----------|-----------|
        Private |     17383 |      4782 |     22165 |
                |     0.784 |     0.216 |     0.740 |
                |     0.769 |     0.652 |           |
                |     0.581 |     0.160 |           |
----------------|-----------|-----------|-----------|
  Self-employed |      2238 |      1243 |      3481 |
                |     0.643 |     0.357 |     0.116 |
                |     0.099 |     0.169 |           |
                |     0.075 |     0.042 |           |
----------------|-----------|-----------|-----------|
   Unpaid/Other |        14 |         0 |        14 |
                |     1.000 |     0.000 |     0.000 |
                |     0.001 |     0.000 |           |
                |     0.000 |     0.000 |           |
----------------|-----------|-----------|-----------|
   Column Total |     22601 |      7335 |     29936 |
                |     0.755 |     0.245 |           |
----------------|-----------|-----------|-----------|
```

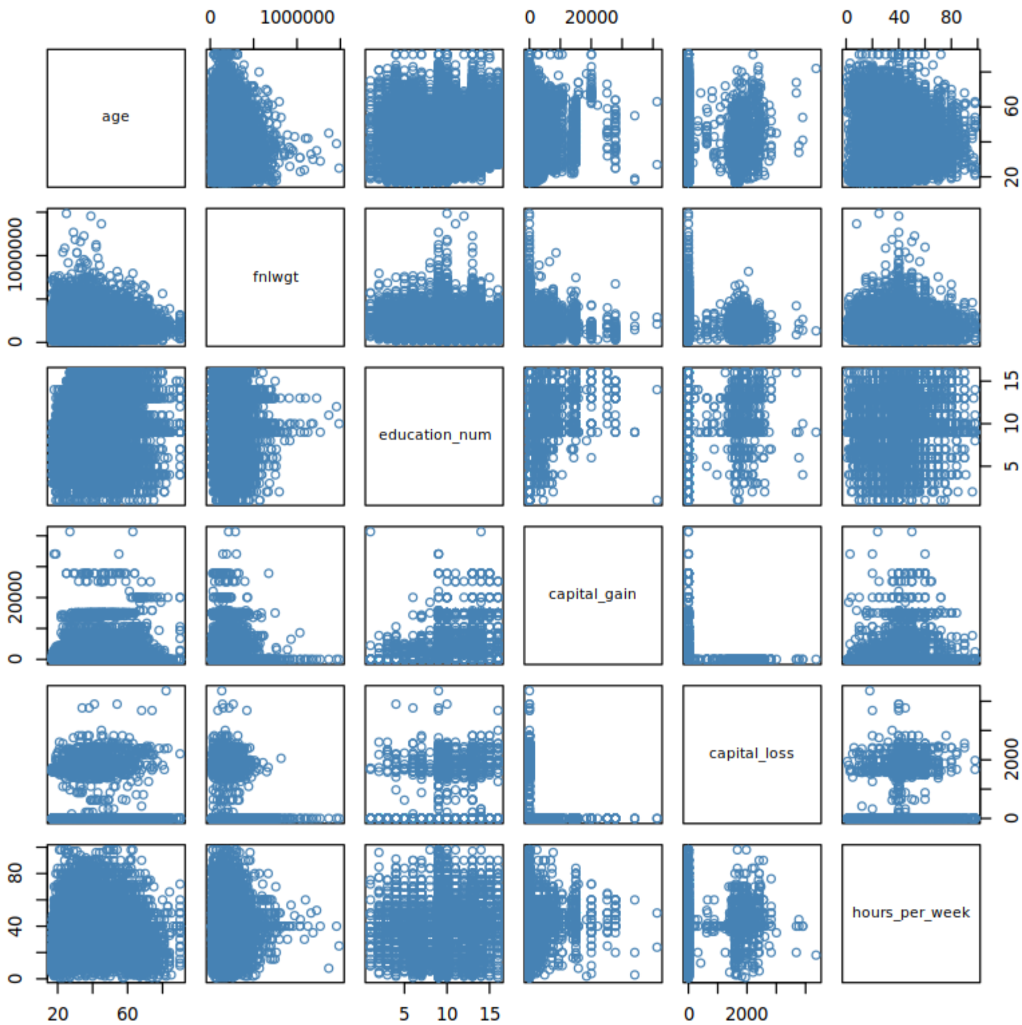
Above crosstables are only some of the further analysis we can do for the categorial variables. It is showing similar observations to the graphs before it generally, for example Self employed has highest proportion of the higher income group with 35.7% of them falling in this category, more than 80% of people employed in Manual and Sales based vocations have less than 50k income per year, and so on.

Now we will analyze numerical data.

[58]:
```
#subsetting dataset for numerical variables

df_gm_n <- df_gm[,sapply(df_gm, is.numeric)]
summary(df_gm_n)
pairs(df_gm_n, col = 'steelblue')
```

```
      age            fnlwgt         education_num    capital_gain
 Min.   :17.00   Min.   :  13769   Min.   : 1.00   Min.   :    0.0
 1st Qu.:28.00   1st Qu.: 117679   1st Qu.: 9.00   1st Qu.:    0.0
 Median :37.00   Median : 178514   Median :10.00   Median :    0.0
 Mean   :38.39   Mean   : 189846   Mean   :10.11   Mean   :  603.6
 3rd Qu.:47.00   3rd Qu.: 237647   3rd Qu.:12.00   3rd Qu.:    0.0
 Max.   :90.00   Max.   :1484705   Max.   :16.00   Max.   :41310.0
  capital_loss     hours_per_week
```

```
Min.   :   0.00   Min.   : 1.00
1st Qu.:   0.00   1st Qu.:40.00
Median :   0.00   Median :40.00
Mean   :  88.68   Mean   :40.73
3rd Qu.:   0.00   3rd Qu.:45.00
Max.   :4356.00   Max.   :98.00
```



Pair plots give only a basic overview here with few relationship observable, this may imply to the fact that across the numerical variables we don't have high degrees of correlation, we will analyze this fact further with correlation plots and table.

[448]:
```
#numerical variables summary across the workclass groups
```

```
df_gm %>%
  group_by(workclass) %>%
  summarize(mean_age = mean(age),
            median_age = median(age),
            mean_edu_num = mean(education_num),
            mean_hpw = mean(hours_per_week),
            mean_cap_gain = mean(capital_gain),
            count = n()) %>%
            print()
```

```
# A tibble: 4 × 7
  workclass     mean_age median_age mean_edu_num mean_hpw mean_cap_gain count
  <chr>            <dbl>
<dbl>        <dbl>
<dbl>         <dbl>
<int>
1 Government        41.2         41         11.1      40.4
624.  4276
2 Private           36.8         35          9.86     40.1
530. 22165
3 Self-employed     45.3         44         10.5      45.2
1046.  3481
4 Unpaid/Other      47.8         57          9.07     32.7
488.    14
```

Mean age and capital gain seems to be different between most of the groups, we can perform post hoc hypothesis tests for it but its not the object of interest in our project and effect seems big enough to be true. Education mean is higher for government but for other categories its similar, and mean hours per week is quite different for self employed and unpaid employees

[52]:
```
#numerical variables summary across the workclass groups

df_gm %>%
  group_by(sex) %>%
  summarize(mean_age = mean(age),
            median_age = median(age),
            mean_edu_num = mean(education_num),
            mean_hpw = mean(hours_per_week),
            mean_cap_gain = mean(capital_gain),
            count = n()) %>%
            print()
```

```
# A tibble: 2 × 7
  sex       mean_age median_age mean_edu_num mean_hpw mean_cap_gain count
  <chr>        <dbl>
<int>        <dbl>
<dbl>         <dbl>
<int>
```

```
1 " Female"     36.9          35          10.1
36.8          367.  9747
2 " Male"       39.1          38          10.1
42.6          718. 20189
```

[49]:

```
#weight summary across the sex or genders

df_gm %>%
  group_by(sex) %>%
  summarize(mean_weight = mean(fnlwgt),
            median_weight = median(age),
            count = n()) %>%
            print()
```

```
# A tibble: 2 × 4
  sex       mean_weight median_weight count
  <chr>           <dbl>
<int> <int>
1 " Female"     186024.
35  9747
2 " Male"       191692.
38 20189
```

[54]:

```
#weight summary across the sex or genders

df_gm %>%
  group_by(sex, income_group) %>%
  summarize(mean_weight = mean(fnlwgt),
            median_weight = median(fnlwgt),
            max_weight = max(fnlwgt),
            count = n()) %>%
            print()
```

```
`summarise()` has grouped output by 'sex'. You can override using the
`.groups` argument.
```

```
# A tibble: 4 × 6
# Groups:   sex [2]
  sex       income_group mean_weight median_weight max_weight count
  <chr>     <chr>
<dbl>         <dbl>
<int> <int>
1 " Female" " <=50K"
186242.        176448
1484705  8656
2 " Female" " >50K"
184296.        174640
953588  1091
```

```
3 " Male"    " <=50K"
193031.         181810
1455435 13945
4 " Male"    " >50K"
188701.         176683
1226583  6244
```

Above three grouped summaries are important for this project. Proportion of women in workforce was easily more than 40% in 1990s in US however for our sample we got only around 25% of women in workforce so we also checked the weights for the groups above and found that weights for female and male less than 50k is almost equal in all respect with a big difference in max weights for only high income group across the gender. This implies that overall weights are distributed equally across the sample observations and female grouped observations were given similar weights like men, of course maybe authors of the dataset made some mistake in this respect and underrepresented female population in US for the 1994 census. This aspect needs more analysis preferable from different census data and other US Employment based data for the same period.

Regardless, mean age, capital gain, and hours per week of women employees are quite lower than their male counterpart with similar education years.

[56]:
```r
#numerical summary across the income_group

df_gm %>%
  group_by(income_group) %>%
  summarize(mean_age = mean(age),
            mean_edu_num = mean(education_num),
            mean_hpw = mean(hours_per_week),
            mean_cap_gain = mean(capital_gain),
            mean_cap_loss = mean(capital_loss),
            count = n()) %>%
            print()
```

```
# A tibble: 2 × 7
  income_group mean_age mean_edu_num mean_hpw mean_cap_gain mean_cap_loss count
  <chr>           <dbl>
<dbl>    <dbl>
<dbl>         <dbl>
<int>
1 " <=50K"         36.6         9.63     39.2
149.          53.6 22601
2 " >50K"          43.9        11.6      45.4
2005.         197.   7335
```

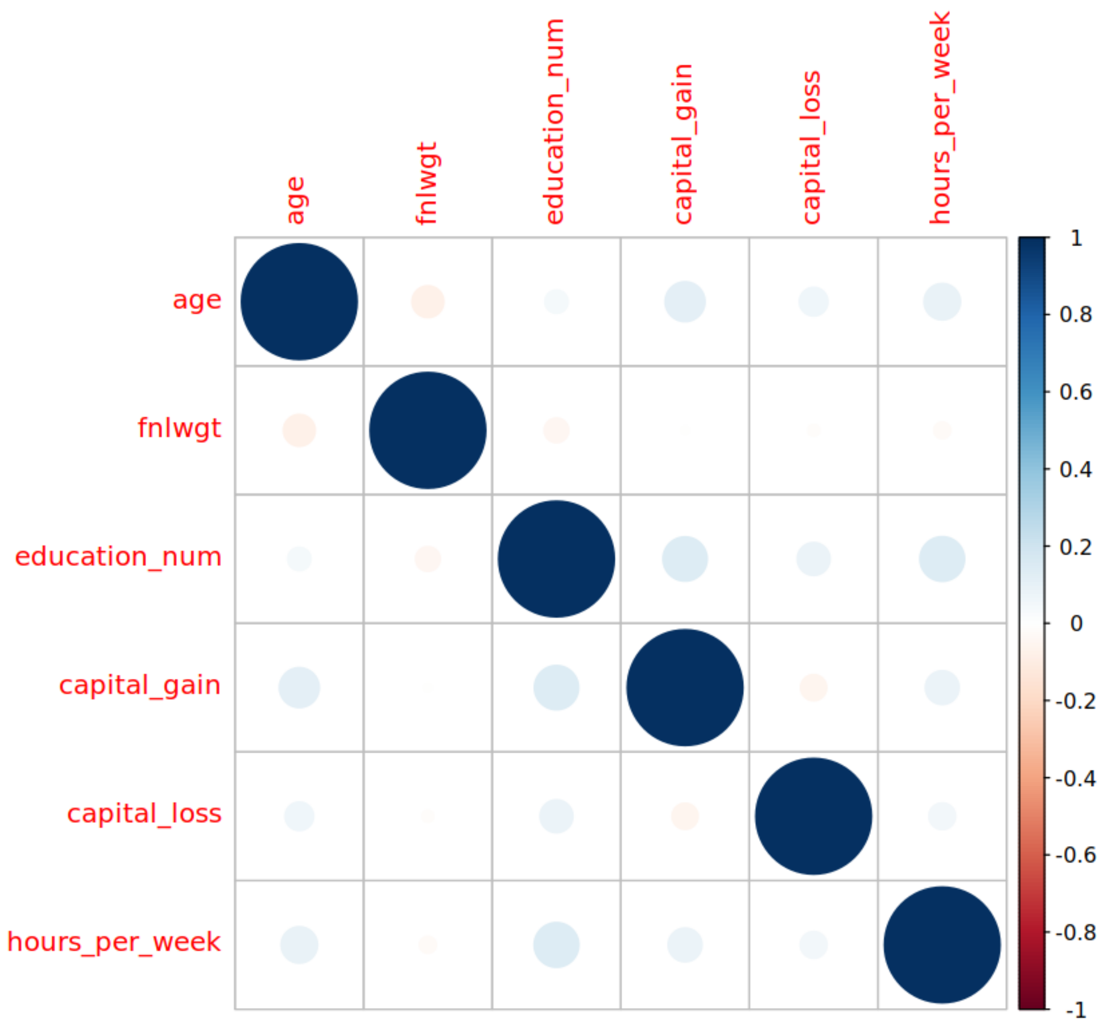
There seems to be a difference in age, education num and hours per week for both groups. Generally higher income group seems to be having higher education years, higher hours per week worked, higher mean age, and obviously higher capital gain and capital loss. These variables seems to have a visible impact on our classification for income group, we will analyze this further in modeling.

[59]:
```r
cor(df_gm_n)
```

A matrix: 6 × 6 of type dbl

	age	fnlwgt	education_num	capital_gain	ca
age	1.00000000	-0.076149227	0.04032180	0.119557860	0.0
fnlwgt	-0.07614923	1.000000000	-0.04559915	-0.005091701	-0.
education_num	0.04032180	-0.045599154	1.00000000	0.146564125	0.0
capital_gain	0.11955786	-0.005091701	0.14656412	1.000000000	-0.
capital_loss	0.06139910	-0.010234307	0.08121796	-0.051309920	1.0
hours_per_week	0.09967975	-0.021213210	0.14947912	0.085879548	0.0

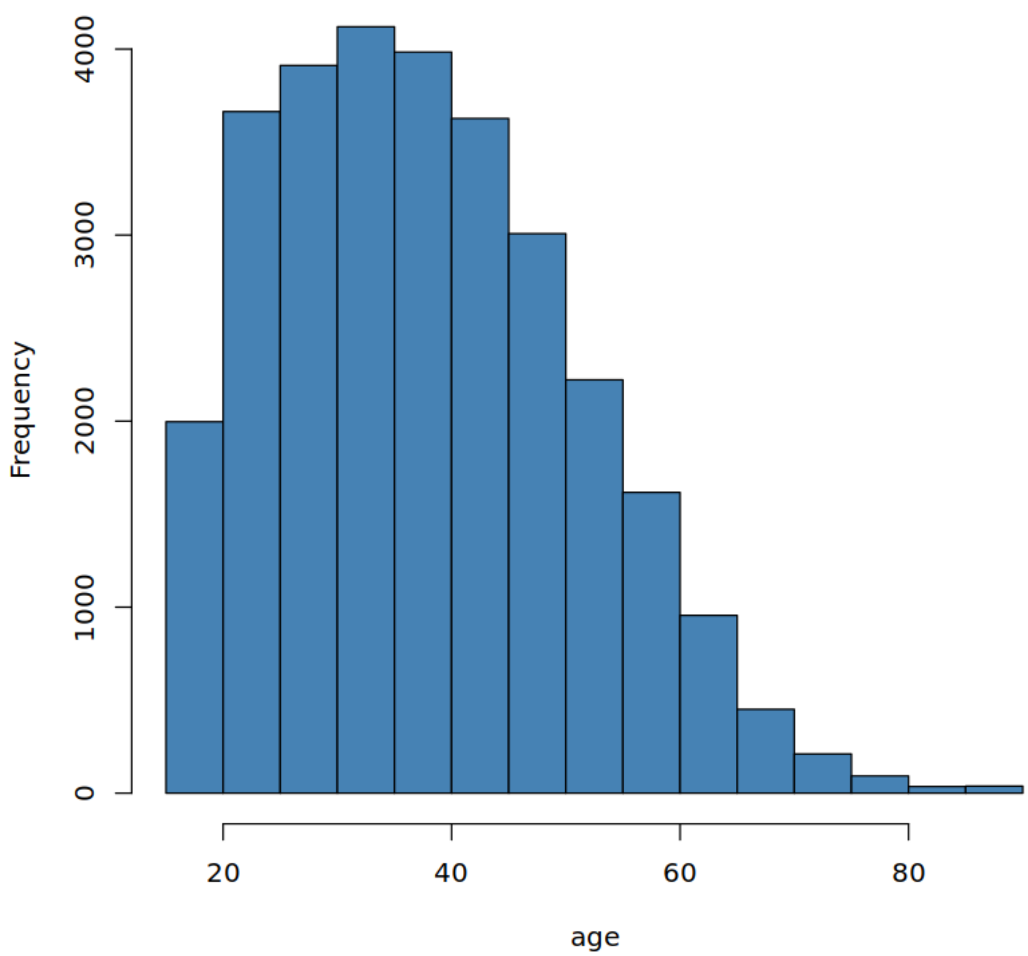
```
[60]: corrplot(cor(df_gm_n))
```



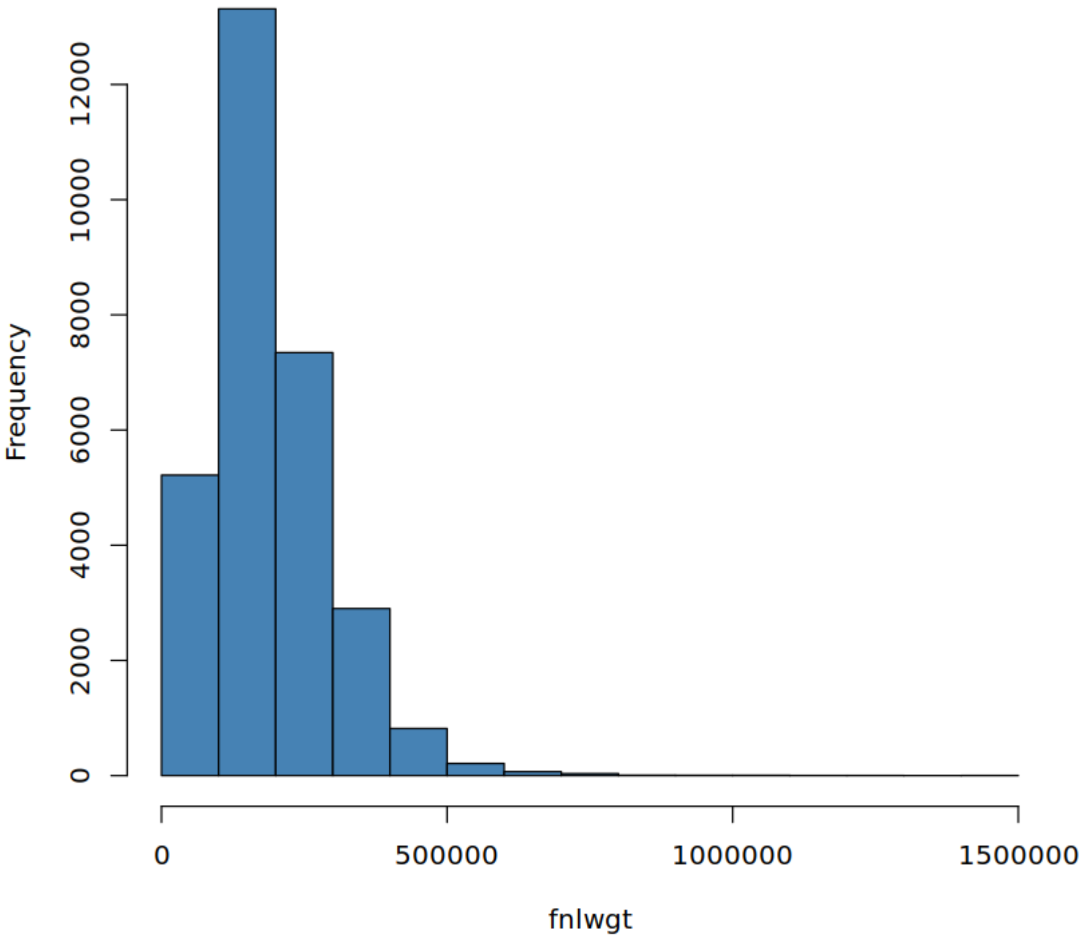
There is generally no correlation across the numerical values which is quite surprising to observe. Hours per week and Education Numbers should have had negative correlation as per general belief that more educated workers have lower hours per week with more income. Similarly Age and education num, capital gain, hours_per week, and capital loss should have some correlation between them but there does not exist any such correlation.

[62]:
```
for (i in 1:dim(df_gm_n)[2]){
    hist(df_gm_n[,i], main = cat('Plot for', colnames(df_gm_n)[i]), col =␣
 ↪'steelblue', xlab = colnames(df_gm_n)[i])
}
```

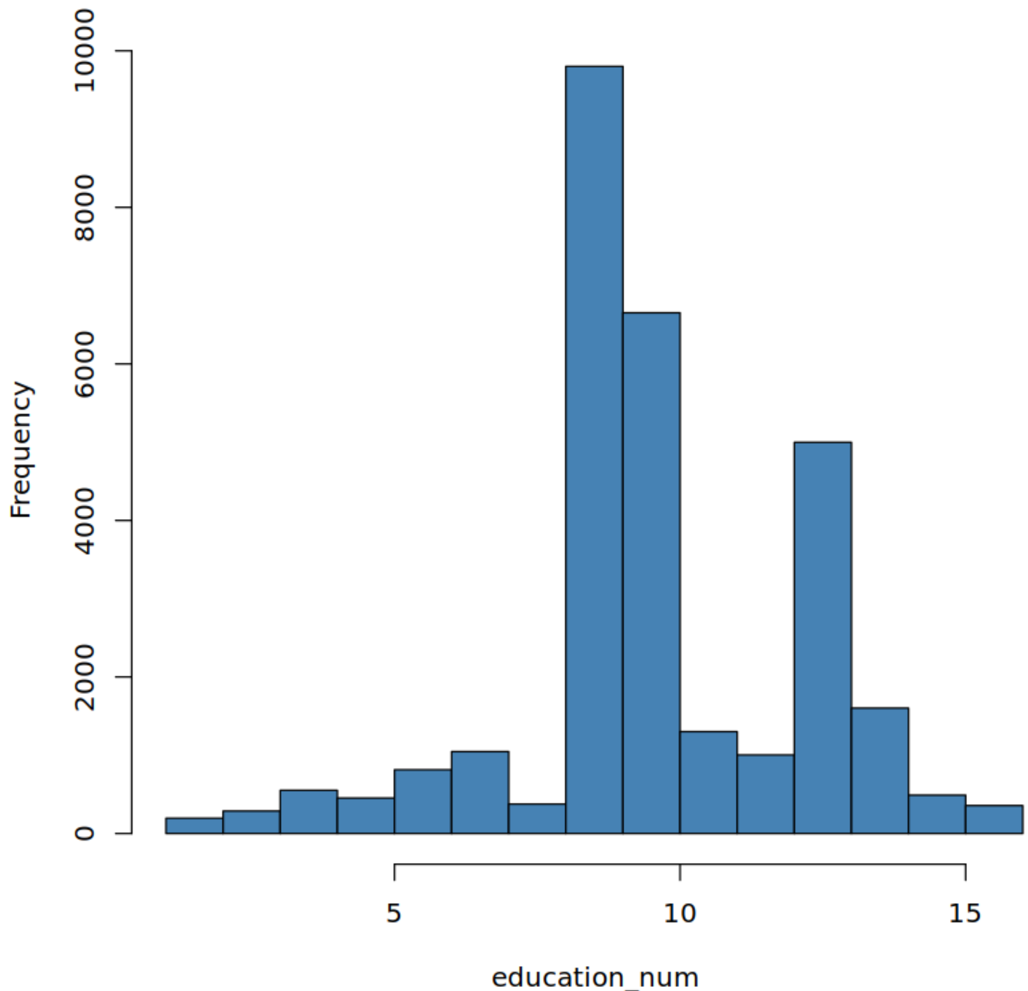
```
Plot for age
```



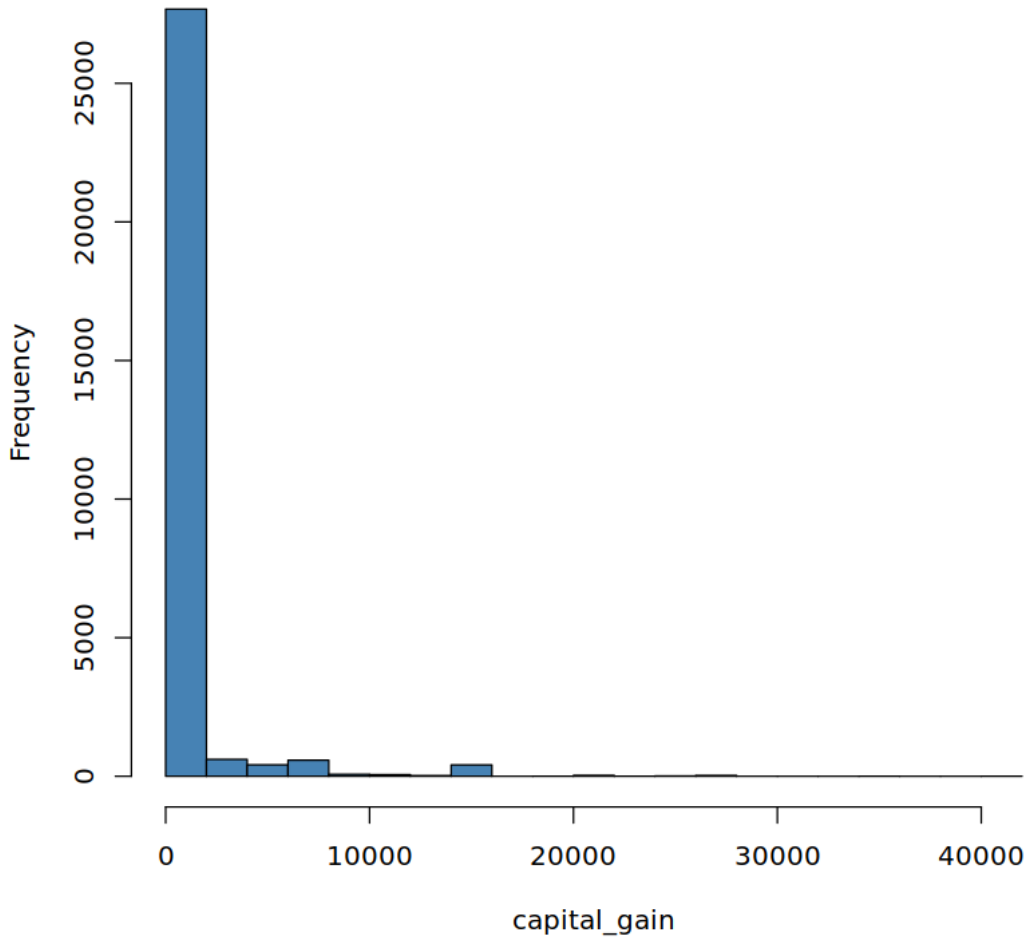

```
Plot for fnlwgt
```



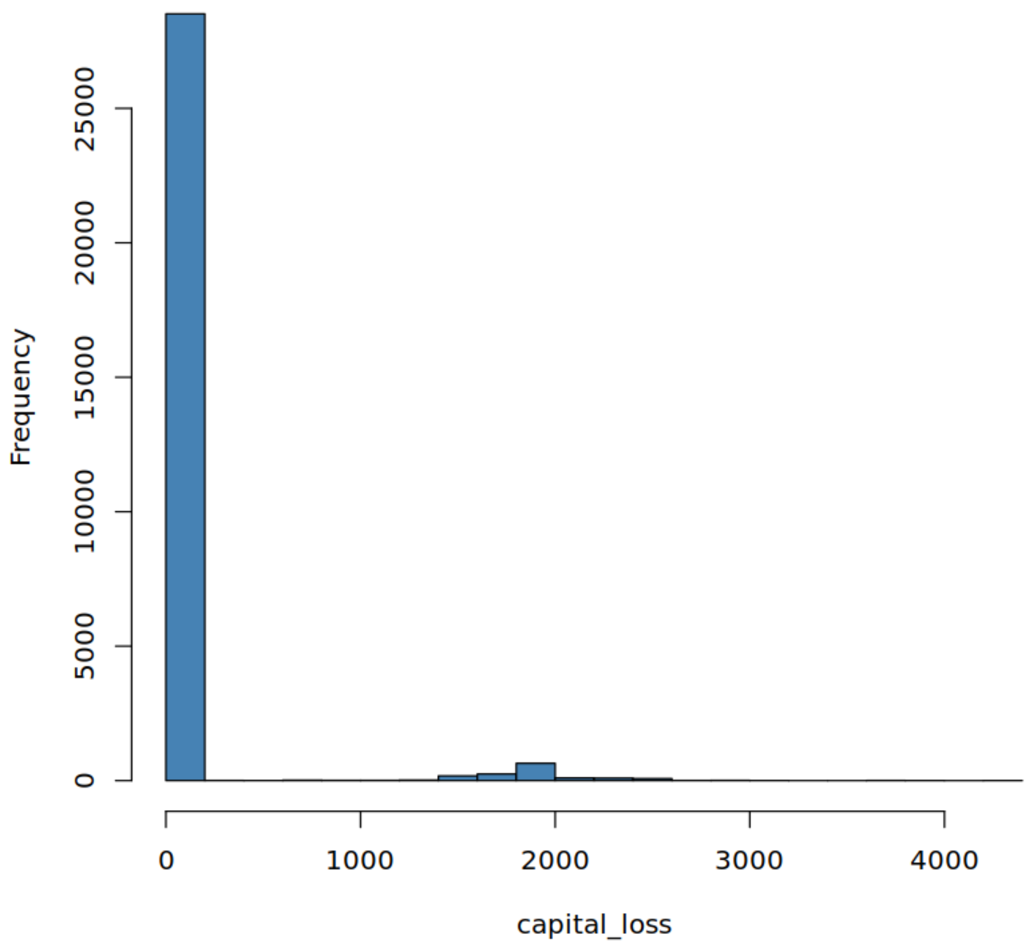

```
Plot for education_num
```



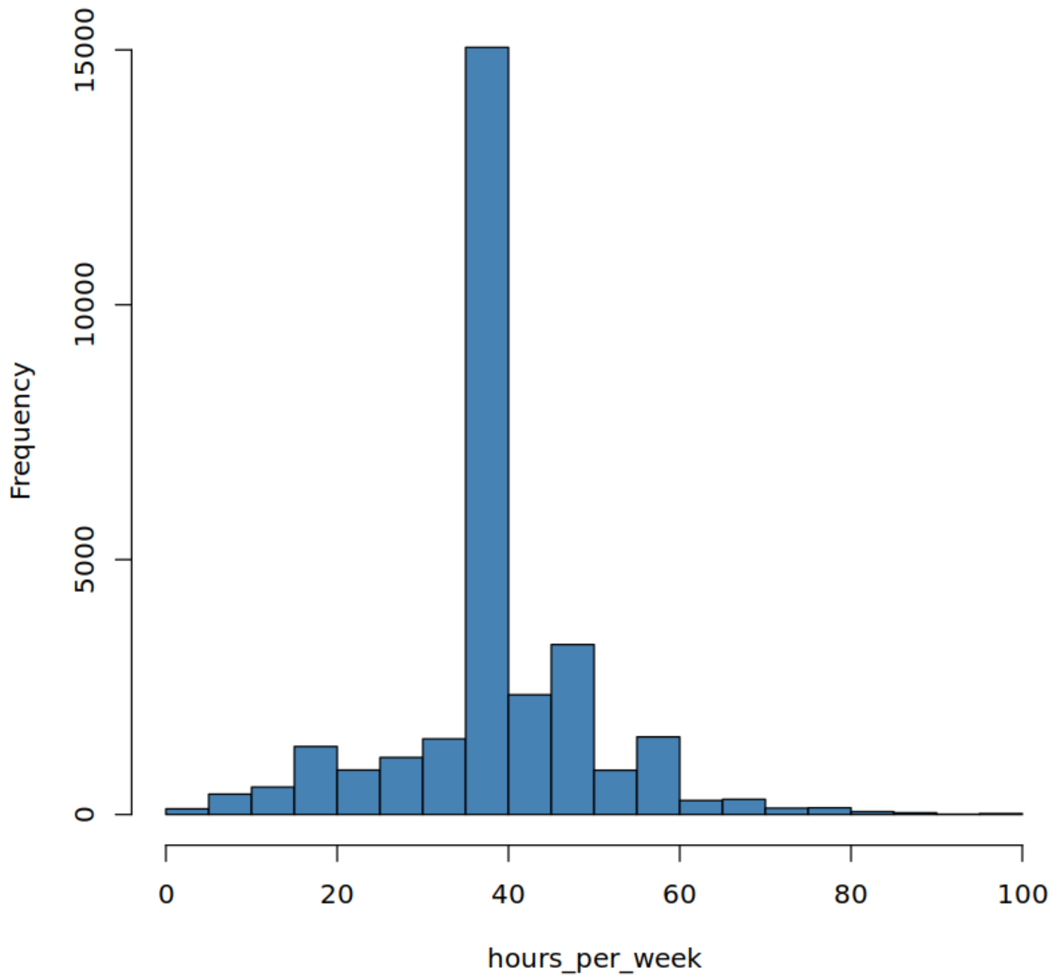
Plot for capital_gain



Plot for capital_loss



Plot for hours_per_week



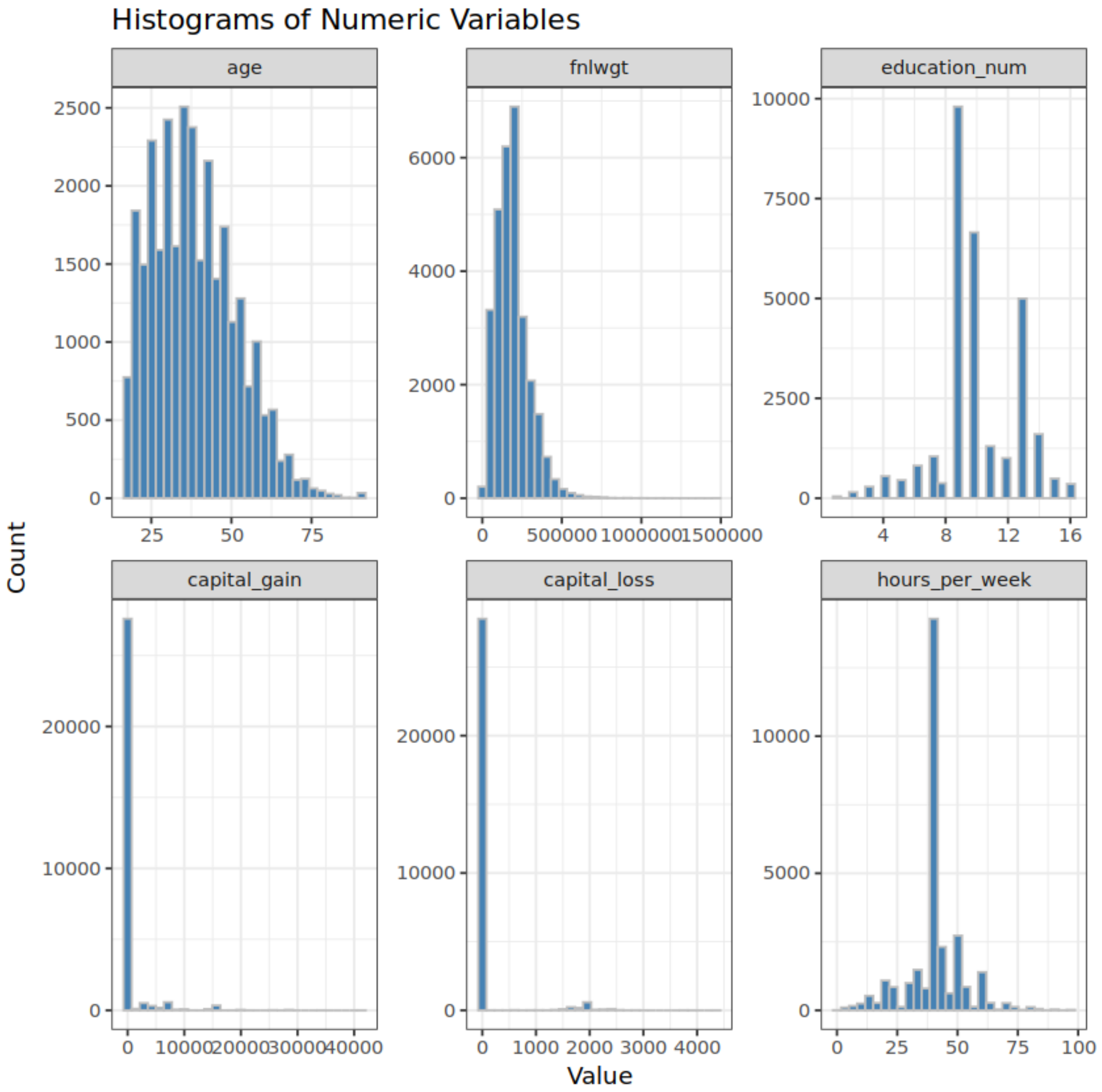
From the above plots we can observe following observations: 1. Hours Per week, education_num, and age appears to be normally distributed with age having a cut off at 17 in this dataset. 2. Capital gain and capital loss are highly rightskewed and appears to be zero inflated distribution as 0s have highest count. 3. Fnlwgt is right skewed with locational parameters cenetered near 0 but it also have extreme outliers too.

[63]:
```
melted_numeric <- melt(df_gm_n)
```

```
No id variables; using all as measure variables
```

[64]:
```
ggplot(melted_numeric, aes(x = value)) +
  geom_histogram(fill = "steelblue", bins = 30, color = "gray") +
```

```
  facet_wrap(~ variable, scales = "free") +
  theme_bw() +
  labs(title = "Histograms of Numeric Variables",
       x = "Value", y = "Count")
```

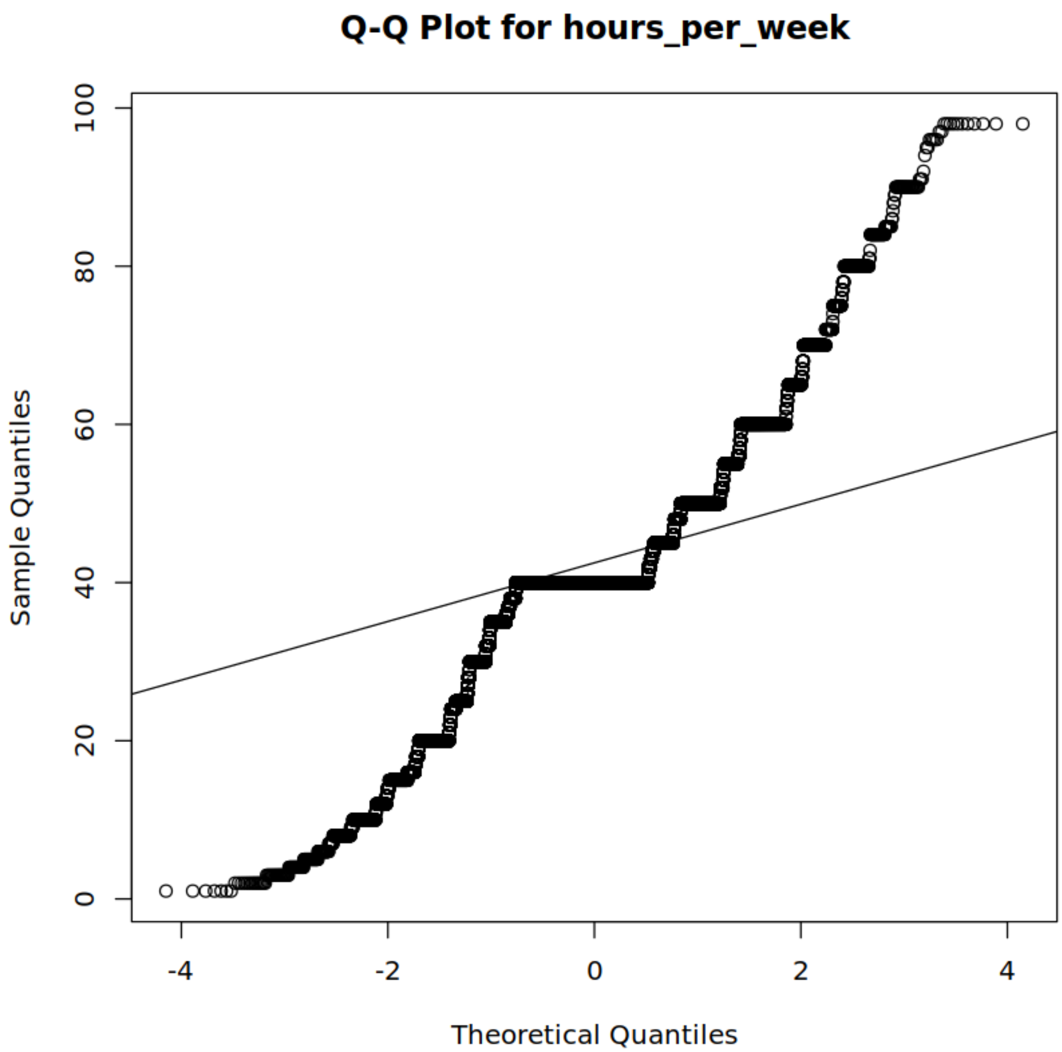


[65]:
```
qqnorm(df_gm_n$age, main = "Q-Q Plot for Age")
qqline(df_gm_n$age)
```

[73]:
```
shapiro.test(df_gm_n$age[1:5000])
```

```
        Shapiro-Wilk normality test

data:  df_gm_n$age[1:5000]
W = 0.96897, p-value < 2.2e-16
```

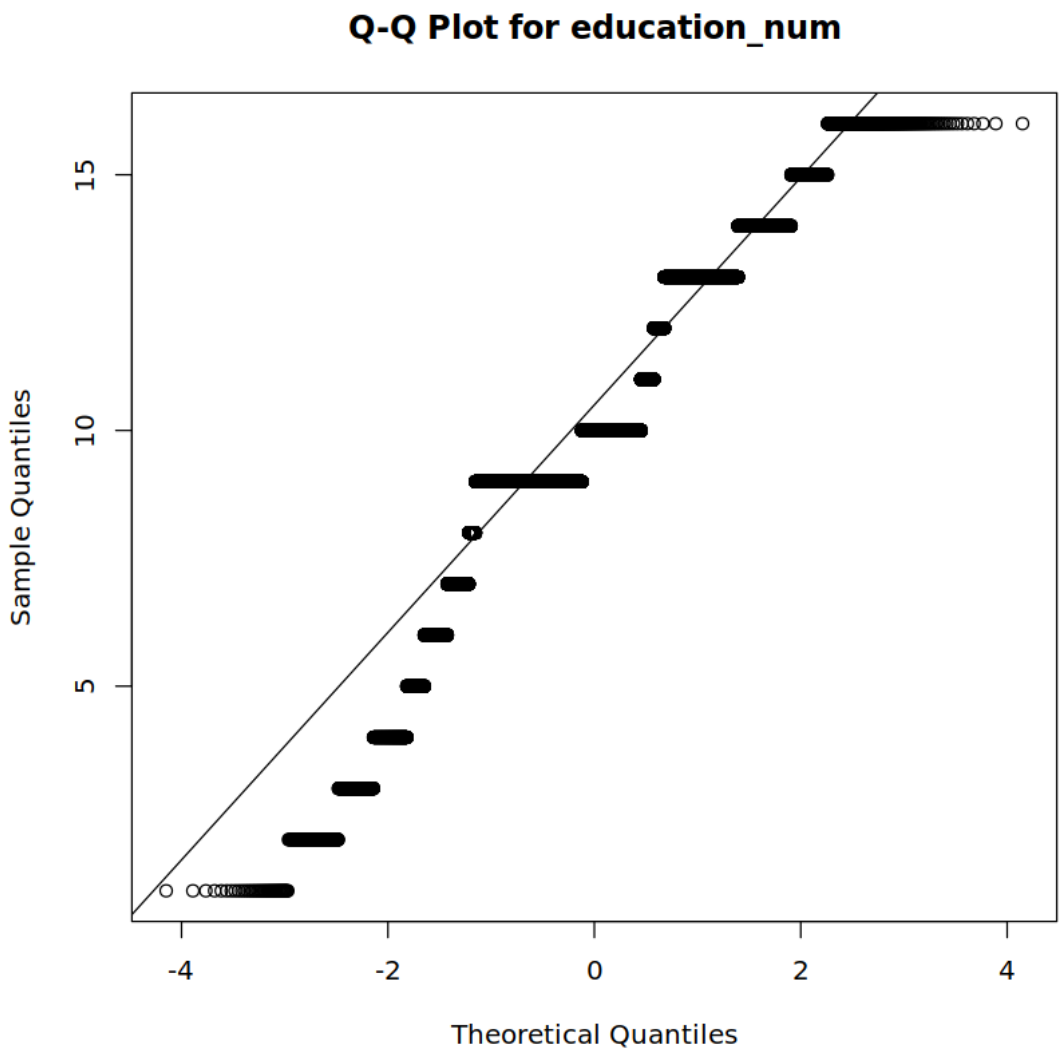
[66]:
```
qqnorm(df_gm_n$hours_per_week, main = "Q-Q Plot for hours_per_week")
qqline(df_gm_n$hours_per_week)
```



[72]:
```
shapiro.test(df_gm_n$hours_per_week[1:5000])
```

```
	Shapiro-Wilk normality test

data:  df_gm_n$hours_per_week[1:5000]
W = 0.88825, p-value < 2.2e-16
```

[67]:
```
qqnorm(df_gm_n$education_num, main = "Q-Q Plot for education_num")
qqline(df_gm_n$education_num)
```

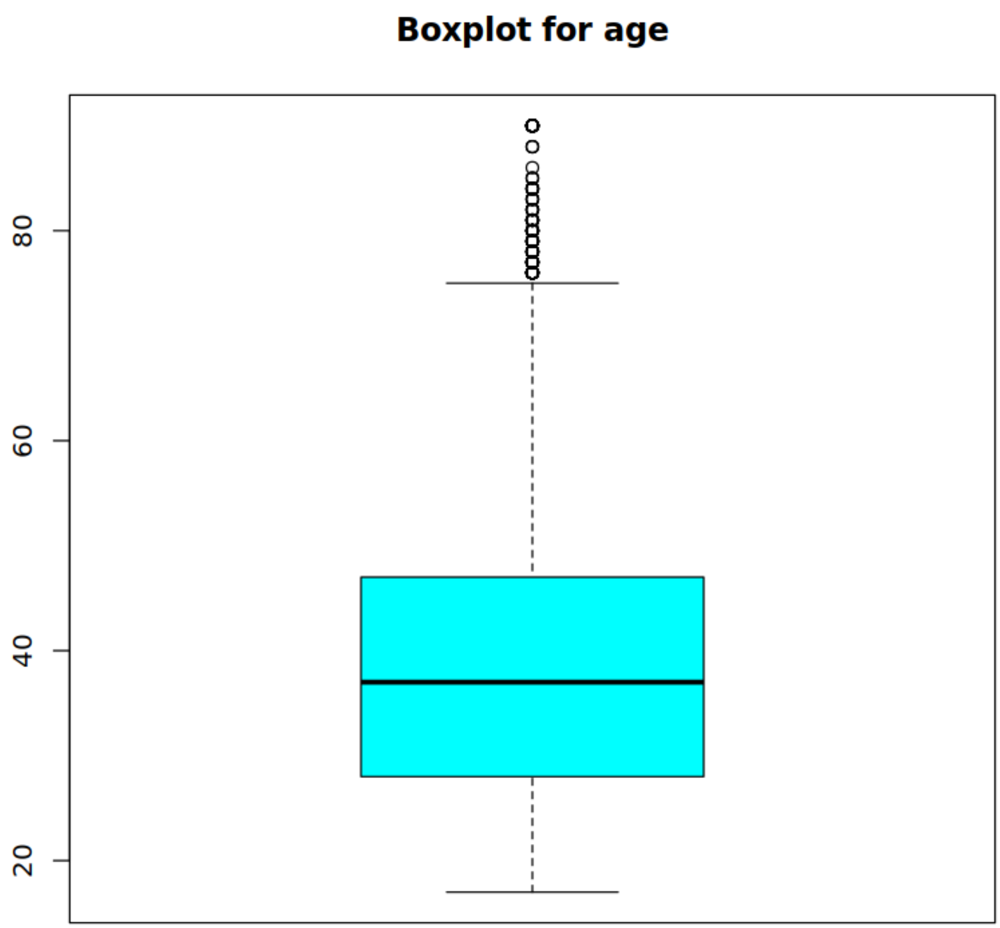


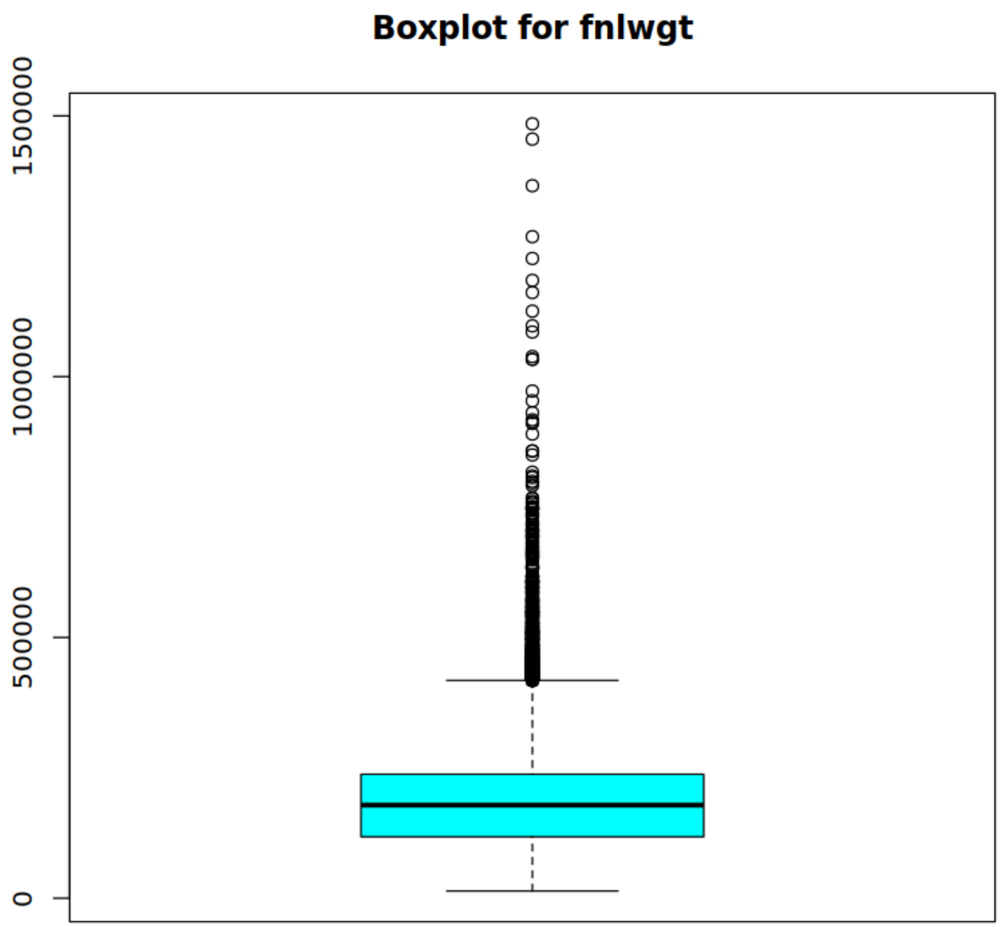
[71]:
```
shapiro.test(df_gm_n$education_num[1:5000])
```

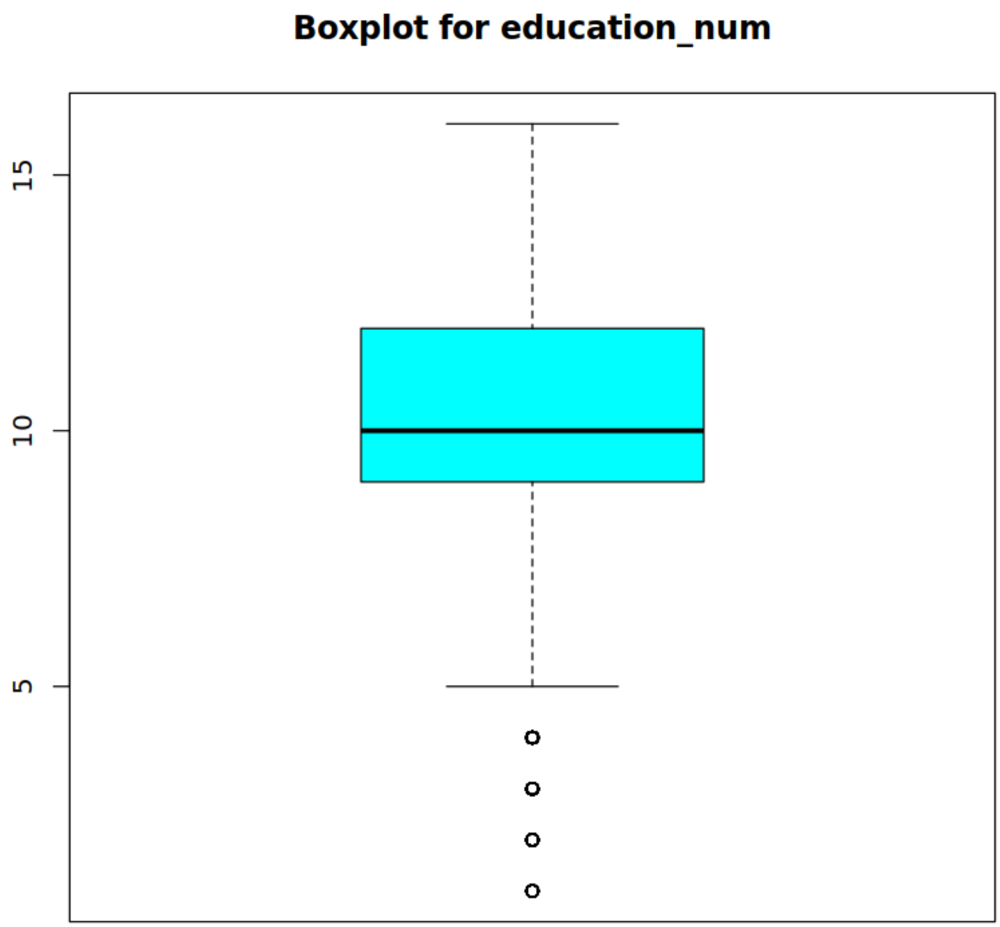
```
        Shapiro-Wilk normality test

data:  df_gm_n$education_num[1:5000]
W = 0.92432, p-value < 2.2e-16
```

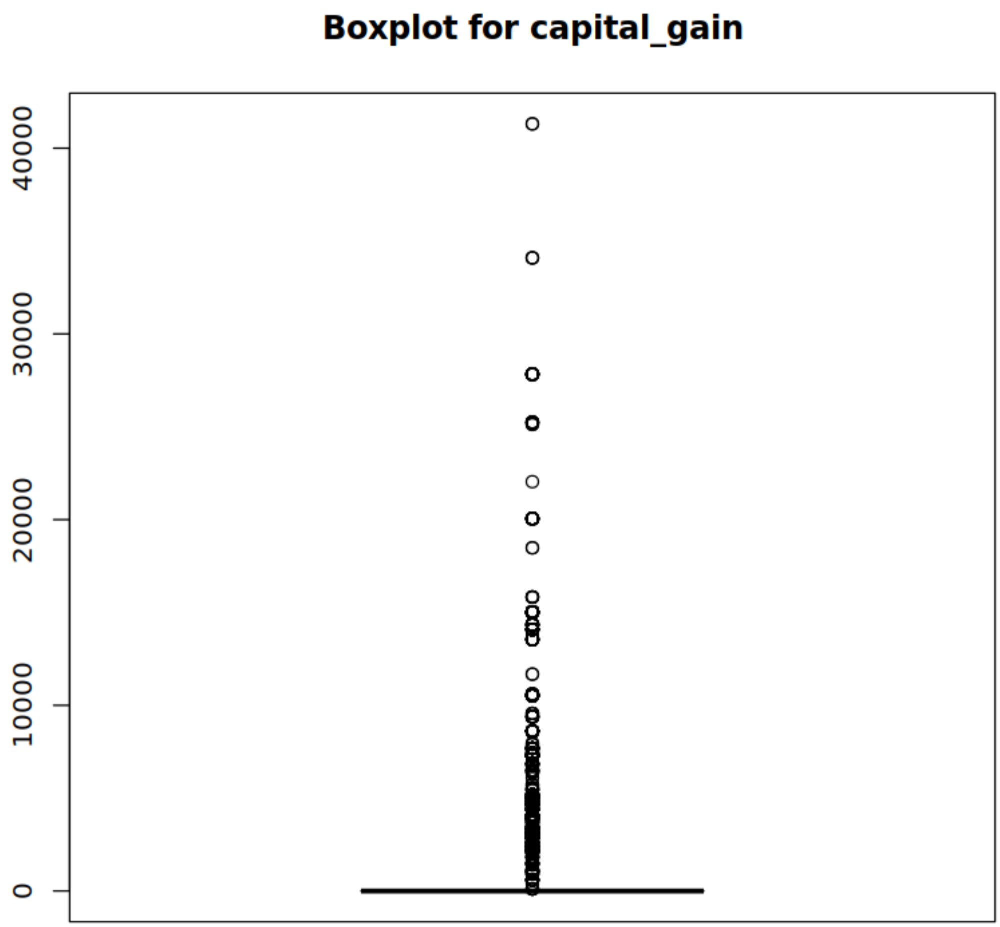
After checking the qq plots for the three observations we observed as normal initially we can see that only educaation_num has a proper qqplot. Age has issues with normality at lower end because of the cut off and hours per week is not a good fit for normality at all. We also got low p value for all of them so its very likely that none of them are normally distributed overall.

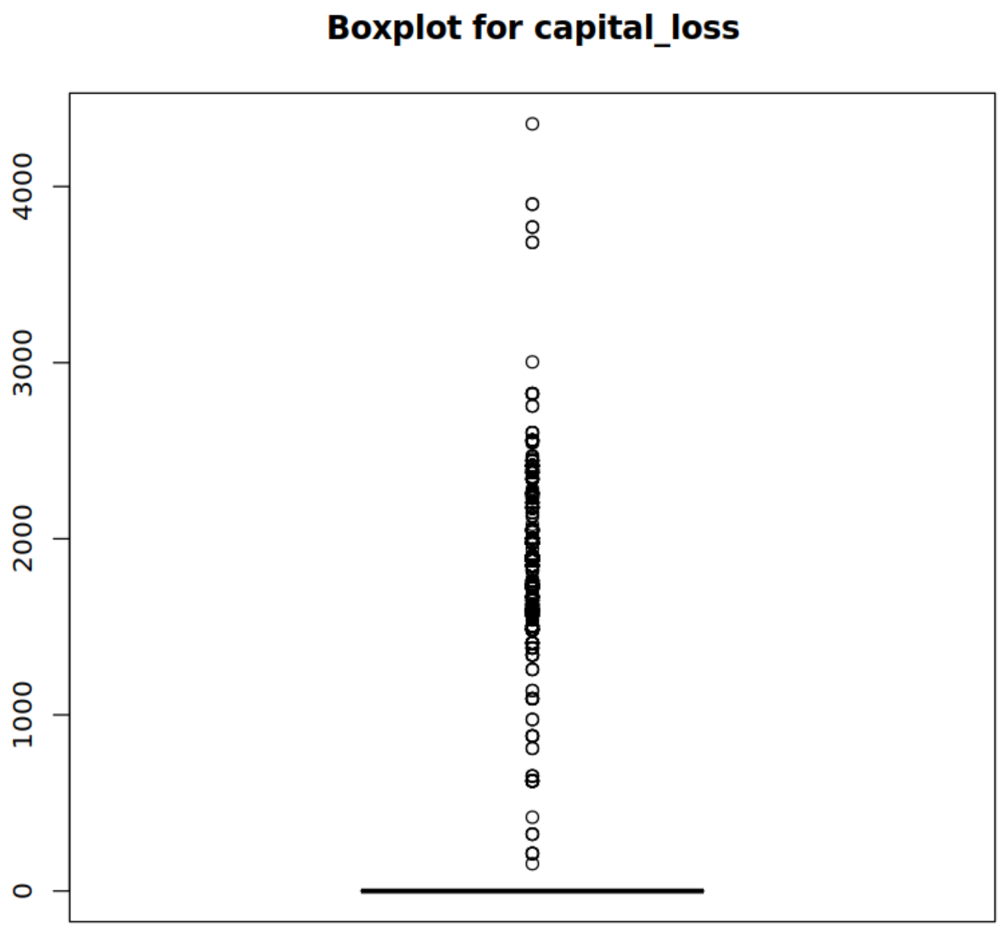
[326]:
```
for (i in colnames(df_gm_n)){
    boxplot(df_gm_n[,i], col = 'cyan', main = paste('Boxplot for',i))
}
```

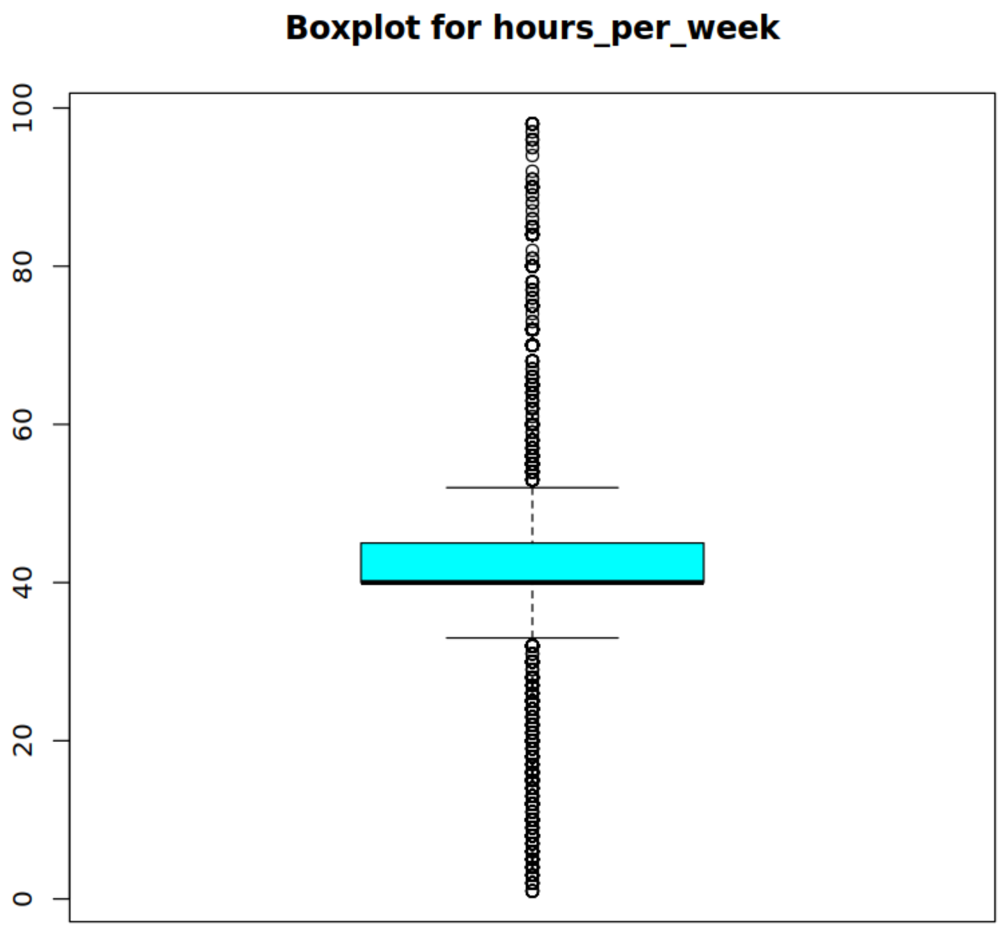



Boxplot for fnlwgt
0
500000
1000000
1500000


Boxplot for education_num
15
10
5

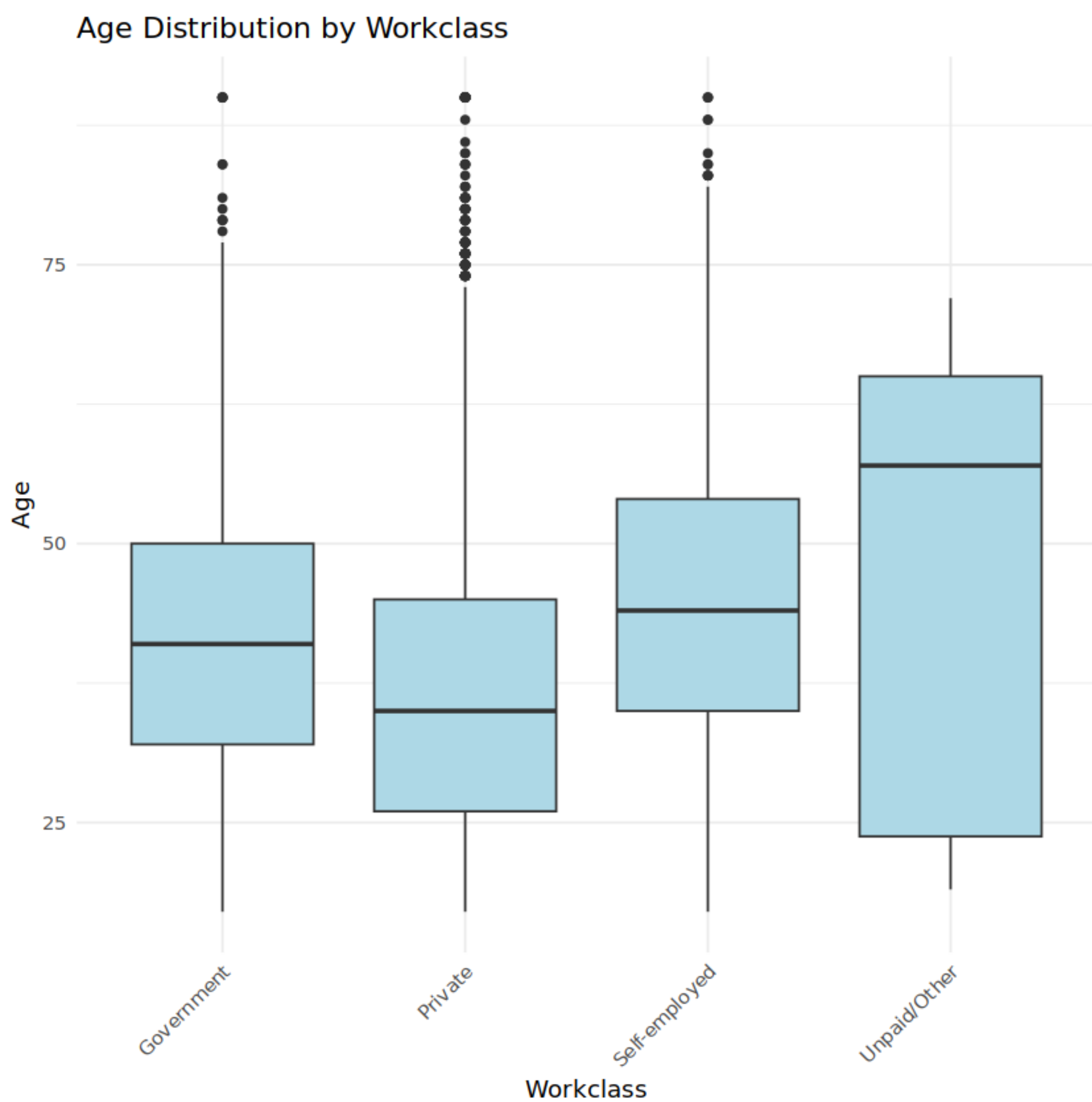

Boxplot for capital_gain
0
10000
20000
30000
40000


Boxplot for capital_loss
0
1000
2000
3000
4000

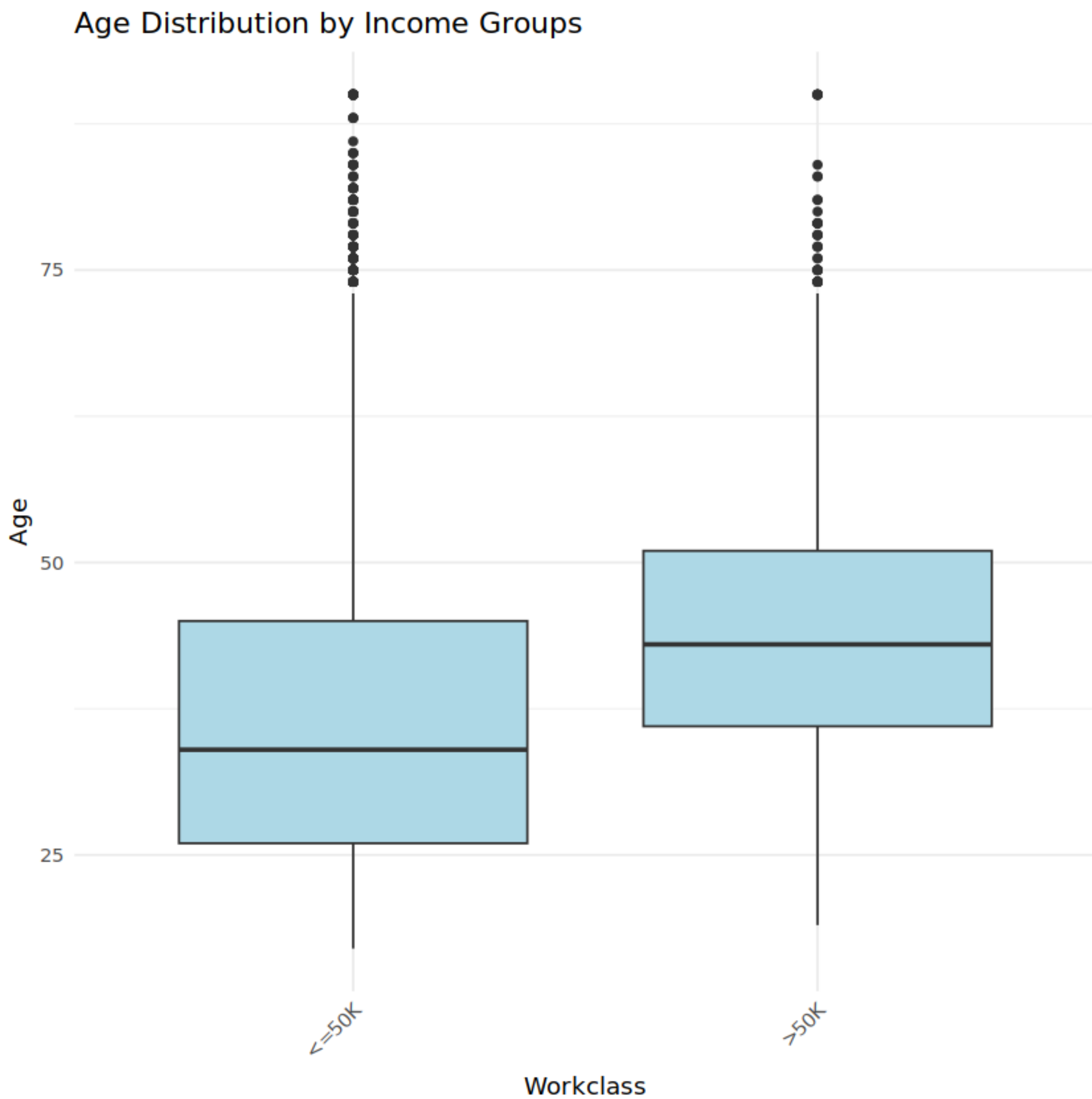


We have a lot of outliers in the dataset which impact the project overall. We can't remove them but we will scale them for model fitting for an efficient fit overall.

[431]:
```
ggplot(df_gm, aes(x = workclass, y = age)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Age Distribution by Workclass",
       x = "Workclass",
       y = "Age") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

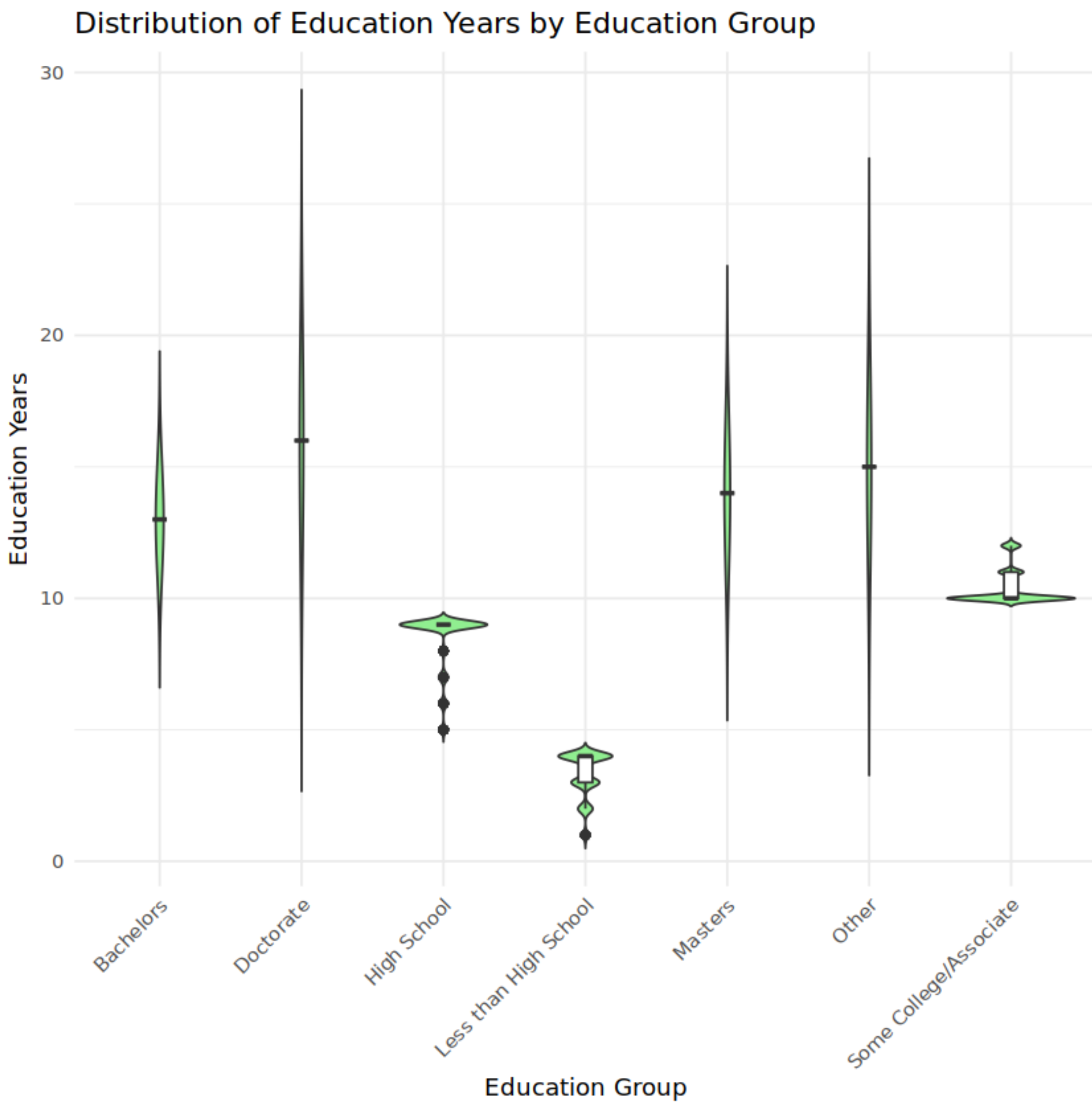


[433]:
```
ggplot(df_gm, aes(x = income_group, y = age)) +
  geom_boxplot(fill = "lightblue") +
  labs(title = "Age Distribution by Income Groups",
       x = "Workclass",
       y = "Age") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



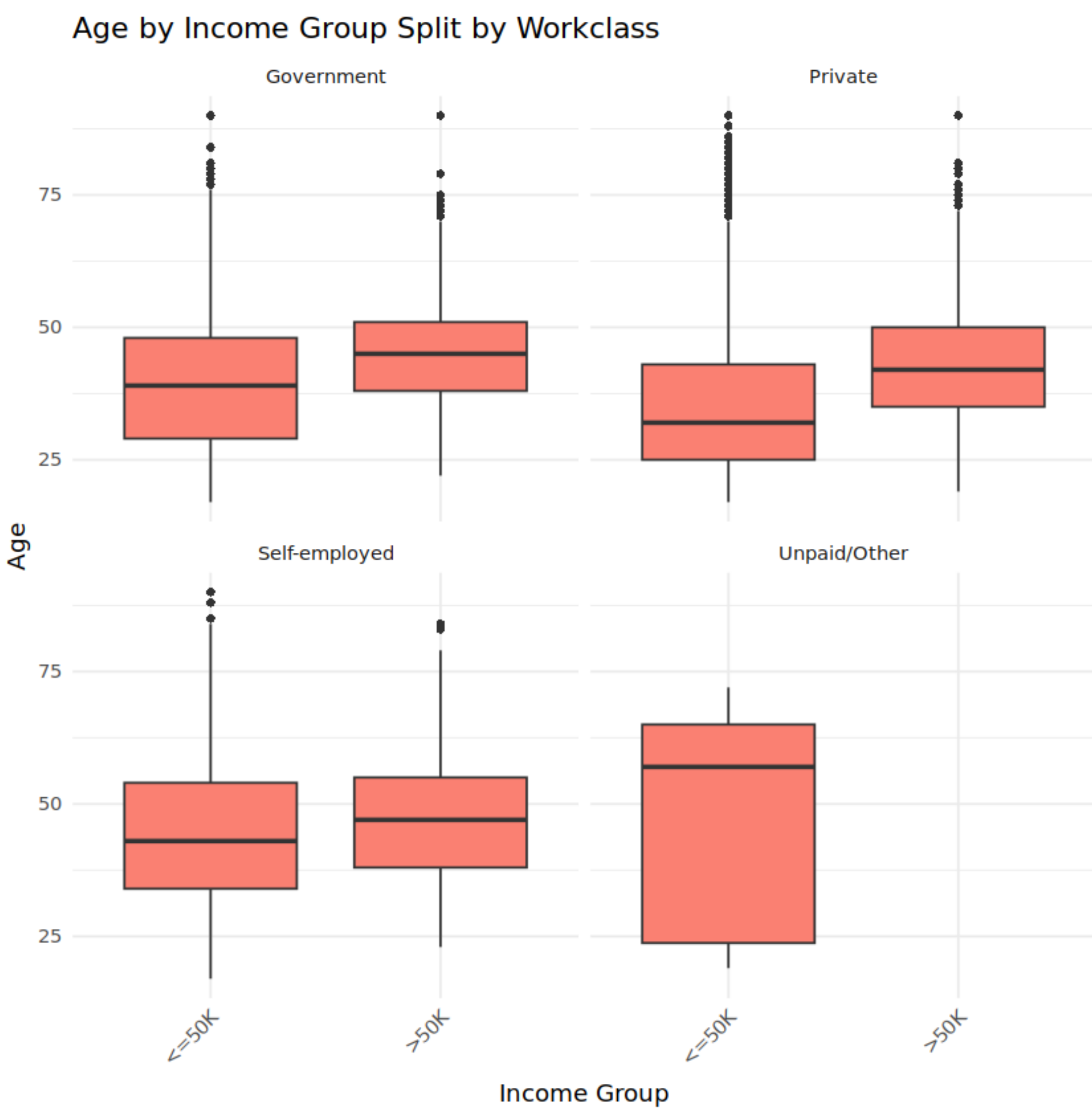
Distribution of income groups across the wages seems to be different with high income group having smaller range and less outliers too.

[434]:
```
ggplot(df_gm, aes(x = education, y = education_num)) +
  geom_violin(fill = "lightgreen", trim = FALSE) +  # shows the density
  geom_boxplot(width = 0.1, fill = "white") +       # overlays a boxplot
  labs(title = "Distribution of Education Years by Education Group",
       x = "Education Group",
       y = "Education Years") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

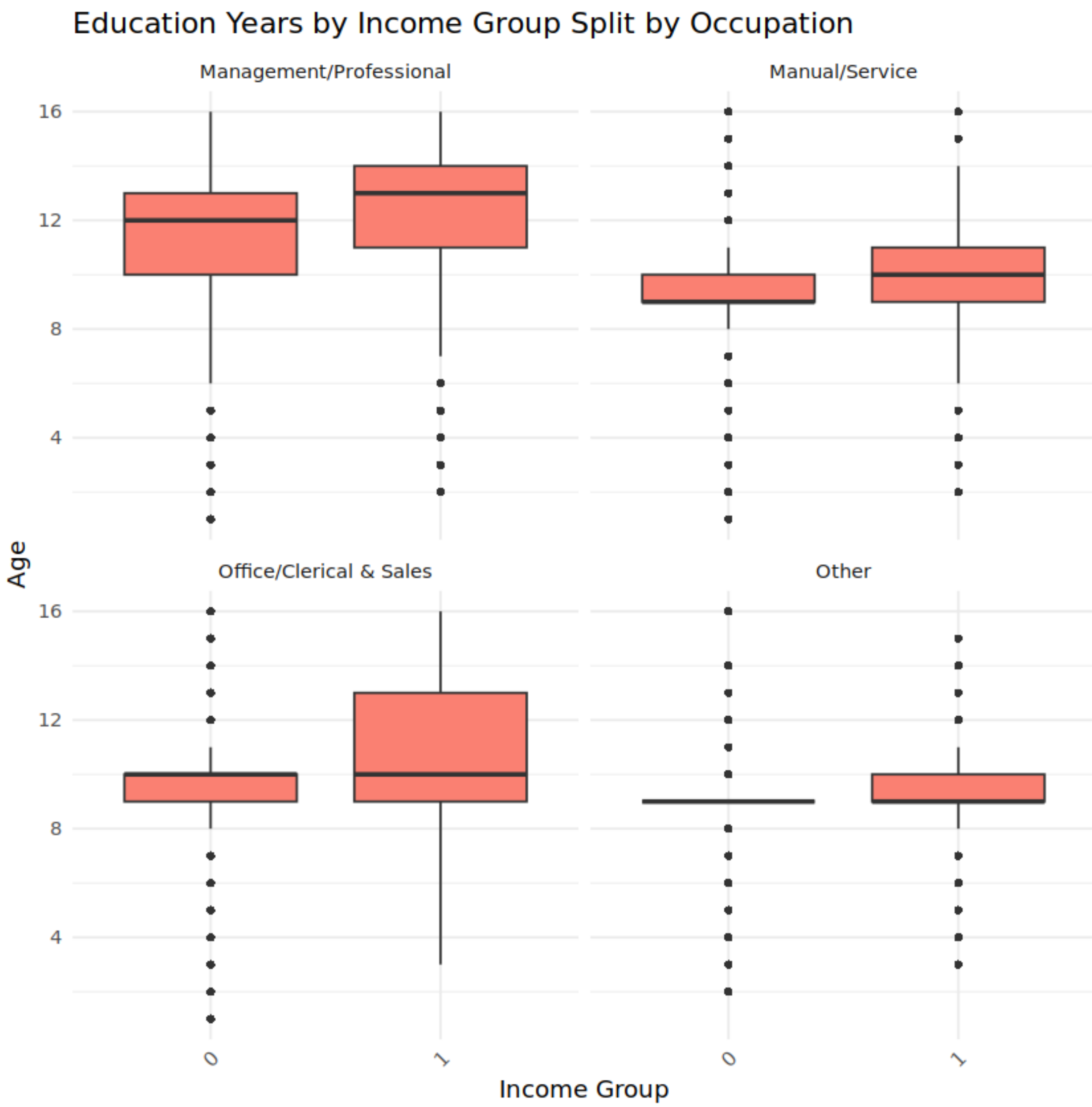


Education groups and years seems to be proprly distributed with less than high school having much smaller distribution range compared to doctrate which has highest spread reaching to near 30 years of education as its maximum.

[454]:
```
ggplot(df_gm, aes(x = income_group, y = age)) +
  geom_boxplot(fill = "salmon", outlier.shape = 16) +
  facet_wrap(~ workclass) +
  labs(title = "Age by Income Group Split by Workclass",
       x = "Income Group",
       y = "Age") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

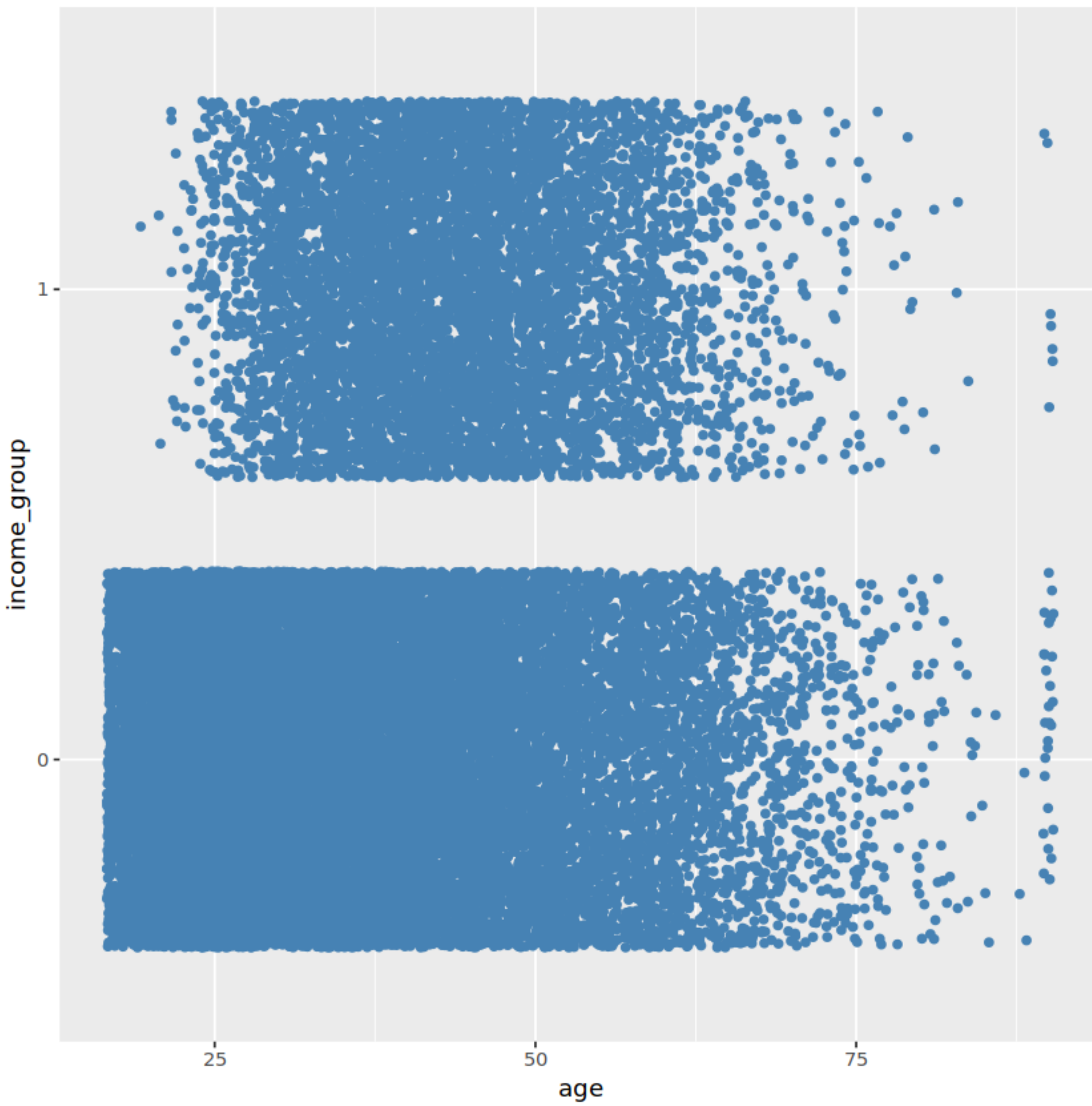



```
[559]: ggplot(df_gm, aes(x = income_group, y = education_num)) +
         geom_boxplot(fill = "salmon", outlier.shape = 16) +
         facet_wrap(~ occupation) +
         labs(title = "Education Years by Income Group Split by Occupation",
              x = "Income Group",
              y = "Age") +
         theme_minimal() +
         theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Education Years across different occupations and income groups seems to be captured properly here with distribution of age for management/professional having wider or larger distributions compared to other groups.

[558]:
```
ggplot(data = df_gm, aes(y = income_group, x = age)) +
geom_jitter(col = 'steelblue')
```



1.1.1 Hypothesis Tests

For hypothesis tests for the the planned hypothesis we will first do it based on the sample dataset and then use the weights presented in fnlwgt which represents the weight of each observation in true census data collected during the period. By using the survey with such weights we could analyze these hypothesis for the population estimates properly and observe if there indeed exist some difference between using these weights or not.

We will cover three planned hypothesis here with following null hypothesis: 1. Average Education Numbers for US Workforce in 1994 = 11.5 2. Proportion of Women in High income group is same as proportion of women in workforce as per 1990 data. That is proportion is 44.1%. 3. Proportion of workforce in high income group is 15%

[80]:
```r
# Create a survey design object using the sampling weight "fnlwgt"
des <- svydesign(ids = ~1, data = df_gm, weights = ~fnlwgt)
```

[79]:
```r
#First Hypothesis sample t test

print(t.test(df_gm$education_num, mu = 11.5))
```

```
        One Sample t-test

data:  df_gm$education_num
t = -95.003, df = 29935, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 11.5
95 percent confidence interval:
 10.07655 10.13410
sample estimates:
mean of x
 10.10532
```

[412]:
```r
# Compute weighted mean for education_num
res_mean <- svymean(~ education_num, des)
print(res_mean)

# Extract estimate and its standard error
est1 <- coef(res_mean)[1]
se1  <- SE(res_mean)[1]

# Degrees of freedom (approximate effective degrees of freedom for the design)
df1 <- degf(des)

# Compute the t-statistic and p-value
t_stat1 <- (est1 - 11.5) / se1
p_value1 <- 2 * pt(-abs(t_stat1), df = df1)

cat("Hypothesis 1 (Mean Education Years):\n")
cat("  Weighted mean =", round(est1, 3), "\n")
cat("  Standard Error =", round(se1, 3), "\n")
cat("  t-statistic =", round(t_stat1, 3), "\n")
cat("  p-value =", round(p_value1, 4), "\n\n")
cat('Confidence Interval:', est1 - 1.96*se1, est1 + 1.96*se1)
```

```
                mean     SE
education_num 10.041 0.0173
Hypothesis 1 (Mean Education Years):
  Weighted mean = 10.041
  Standard Error = 0.017
  t-statistic = -84.136
```

```
  p-value = 0
```

```
Confidence Interval: 10.00687 10.07486
```

Mean estimate is similar for both cases and in both cases, with and without weights, we got a p value less than 0.05, allowing us to reject the Null Hypothesis that average number of Education year is 11.5. And observing the condidence interval we can say that average number of Education year in US for year 1996 is around 10 years.

[408]:
```
table(df_gm[df_gm$income_group == ' >50K',]$sex)
prop_h2 <- dim(df_gm[(df_gm['income_group'] == ' >50K' & df_gm['sex'] == '␣
 ↪Female'),])[1]/dim(df_gm)[1]

print(paste('Proportion of women having income greater thaan 50K:', prop_h2))
```

```
 Female    Male
   1091    6244
```

```
[1] "Proportion of women having income greater thaan 50K: 0.0364444147514698"
```

[81]:
```
#Second Hypothesis Test

n_female <- dim(df_gm[(df_gm['income_group'] == ' >50K' & df_gm['sex'] == '␣
 ↪Female'),])[1]

# Perform a binomial test for the null proportion
prop_test_result <- prop.test(n_female, dim(df_gm)[1], p = 0.441)
print(prop_test_result)
```

```
	1-sample proportions test with continuity correction

data:  n_female out of dim(df_gm)[1], null probability 0.441
X-squared = 19873, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.441
95 percent confidence interval:
 0.03436418 0.03864454
sample estimates:
         p
0.03644441
```

[82]:
```
# Subset the design to individuals with >50K income
des_hi <- subset(des, income_group == " >50K")

# Compute the weighted proportion of females in the >50K group
res_prop_female <- svymean(~I(sex == " Female"), des_hi)
print(res_prop_female)
```

```r
# Extract estimate and standard error (the indicator returns 1 if female, 0␣
↪otherwise)
est2 <- coef(res_prop_female)[2]
se2  <- SE(res_prop_female)[1]

# Degrees of freedom for the high-income subset
df2 <- degf(des_hi)

# t-statistic and p-value to test H0: p = 0.441
t_stat2 <- (est2 - 0.441) / se2
p_value2 <- 2*pt(abs(t_stat2), df = df2, lower.tail = FALSE)

cat("Hypothesis 2 (Proportion of Women in >50K Income Group):\n")
cat("  Weighted proportion =", round(est2, 3), "\n")
cat("  Standard Error =", round(se2, 3), "\n")
cat("  t-statistic =", round(t_stat2, 3), "\n")
cat("  p-value =", round(p_value2, 4), "\n\n")
cat('Confidence Interval:', est2 - 1.96*se2, est2 + 1.96*se2)
```

```
                            mean     SE
I(sex == " Female")FALSE 0.85423 0.0047
I(sex == " Female")TRUE  0.14577 0.0047
Hypothesis 2 (Proportion of Women in >50K Income Group):
  Weighted proportion = 0.146
  Standard Error = 0.005
  t-statistic = -63.485
  p-value = 0

Confidence Interval: 0.1366586 0.1548879
```

Weight adjusted proportion of Women having Income greater than 50k income is 0.15 or 15% which is quite different from the 3% proportion we obtained from our dataset. This means that weighted adjustment indeed worked well in this hypothesis test, however even in that case we got a pvalue of less than 0.05 due to which we reject the null hypothesis that the proportion of women having yearly income greater than 50k is 0.441 or 44% which itself was the proportion of women in workforce in US in 1990.

This implies that while weighted adjustment increased the proportion quite a lot, as per census data proprotion of women in high income group is way less than the overall proportion of women in workforce in US during that period, this can be one of the evidence for some real income disparity that existed among both genders during that period.

[387]:
```r
#Third Hypothesis Test

# Count the total number of observations and those with >50K income
total_obs <- nrow(df_gm)
high_income_obs <- sum(df_gm$income_group == " >50K")
```

```r
# Perform a binomial test for the overall proportion
overall_prop_test <- binom.test(high_income_obs, total_obs, p = 0.15)
print(overall_prop_test)
```

```
        Exact binomial test

data:  high_income_obs and total_obs
number of successes = 7335, number of trials = 29936, p-value < 2.2e-16
alternative hypothesis: true probability of success is not equal to 0.15
95 percent confidence interval:
 0.2401588 0.2499362
sample estimates:
probability of success
             0.2450227
```

[414]:
```r
# Compute the weighted proportion for individuals with income >50K (indicator␣
↪variable)
res_prop_hi <- svymean(~I(income_group == " >50K"), des)
print(res_prop_hi)

# Extract the estimate and standard error
est3 <- coef(res_prop_hi)[2]
se3  <- SE(res_prop_hi)[1]

df3 <- degf(des)

# t-statistic and p-value for H0: p = 0.15
t_stat3 <- (est3 - 0.15) / se3
p_value3 <- 2 * pt(-abs(t_stat3), df = df3)

cat("Hypothesis 3 (Overall Proportion Earning >50K):\n")
cat("  Weighted proportion =", round(est3, 3), "\n")
cat("  Standard Error =", round(se3, 3), "\n")
cat("  t-statistic =", round(t_stat3, 3), "\n")
cat("  p-value =", round(p_value3, 4), "\n")
cat('Confidence Interval:', est3 - 1.96*se3, est3 + 1.96*se3)
```

```
                                  mean     SE
I(income_group == " >50K")FALSE 0.7573 0.0028
I(income_group == " >50K")TRUE  0.2427 0.0028
Hypothesis 3 (Overall Proportion Earning >50K):
  Weighted proportion = 0.243
  Standard Error = 0.003
  t-statistic = 32.889
```

```
  p-value = 0
Confidence Interval: 0.2371742 0.2482229
```

For our third hypothesis test weighted proportions are not changing the test much and both tests are quite similar. In both tests we got a high test statistic/critical value and got p value less than 0.05, allowing us to reject the Null Hypothesis that the Proportion of workforce in high income bracket is 15%. Our null hypothesis was based on the multiple factors and data at 1990, and with this rejection we can observe that this proportion of people in high income bracket in US was around 24% as per Census data in year 1994.

Overall, we rejected all our hypothesis. Did we establish weak hypothesis? We can't say so. Average number of education years for US citizens was around 11.5 for 1990s however as per the census-income dataset we have we can see that for the workforce that number is around 10 years. Proportion of women in workforce was 44% according to multiple studies during that time, we assumed that this proportion will be similar in the high income bracket but from the hypothesis test we failed to see our intended effect, instead we got adjusted proportion of only 15%.

Similarly proportion of people in workforce having high income is greater than our expected 15%, it is to be reminded to the reader that inflation adjusted amount is around 100k USD and even in 2024-25 this amount is recorded at quite conservative value at around 10%. Still, there should not have been such a big difference of around 10% in such case and this should be analyzed further and corroborated with other studies and data during that period.

Note: Weight adjusted hypothesis helped in one hypothesis only here, it helped in second hypothesis because number of women in high income group is highly unbalanced. From the weights analysis before we saw that while both such subgroups have similar weights distribution, due to data imbalance in across the groups sample alone is not good in analyzing the proportions when we are using much comparisions across multiple groups. Still our sample proportions are quite similar to weighted sample proportions if we don't use proportions spread across multiple levels for different groups.

1.2 Linear Models

In this section we will fit Linear Models, specifically Logistic Regression models to our dataset. We will fit Generalized Linear Models and Generalized Additive models and test them across different metrics.

We will first optimize the dataset further

[84]:
```
copy(df_gm)
```

```
Error in copy(df_gm): could not find function "copy"
Traceback:
```

[86]:
```
df_gm2 <- data.frame(df_gm)
```

[88]:
```
df_gm2$income_group <- ifelse(df_gm2$income_group == ' >50K',1,0)
```

[91]:
```
df_gm2$income_group <- factor(as.character(df_gm2$income_group))
df_gm2$workclass <- factor(df_gm2$workclass)
df_gm2$education  <- factor(df_gm2$education)
df_gm2$marital_status <- factor(df_gm2$marital_status)
df_gm2$relationship  <- factor(df_gm2$relationship)
df_gm2$occupation <- factor(df_gm2$occupation)
df_gm2$race <- factor(df_gm2$race)
df_gm2$sex <- factor(df_gm2$sex)
```

[154]:
```
df_gm2$age <- scale(df_gm2$age)
df_gm2$education_num <- scale(df_gm2$education_num)
df_gm2$hours_per_week <- scale(df_gm2$hours_per_week)
df_gm2$capital_gain <- scale(df_gm2$capital_gain)
df_gm2$capital_loss <- scale(df_gm2$capital_loss)
```

[155]:
```
str(df_gm2)
```

```
'data.frame':   29936 obs. of  14 variables:
 $ age           : num [1:29936, 1] 0.0466 0.8842 -0.0296 1.1126 -0.791 …
  ..- attr(*, "scaled:center")= num 1.43e-16
  ..- attr(*, "scaled:scale")= num 1
 $ workclass     : Factor w/ 4 levels "Government","Private",..: 1 3 2 2 2 2 2 3
2 2 …
 $ fnlwgt        : int  77516 83311 215646 234721 338409 284582 160187 209642
45781 159449 …
 $ education     : Factor w/ 7 levels "Bachelors","Doctorate",..: 1 1 3 3 1 5 3
3 5 1 …
 $ education_num : num [1:29936, 1] 1.14 1.14 -0.435 -1.223 1.14 …
  ..- attr(*, "scaled:center")= num 10.1
  ..- attr(*, "scaled:scale")= num 2.54
 $ marital_status: Factor w/ 2 levels " Married"," Not Married": 2 1 2 1 1 1 1 1
2 1 …
 $ occupation    : Factor w/ 4 levels "Management/Professional",..: 3 1 2 2 1 1
2 1 1 1 …
 $ relationship  : Factor w/ 3 levels "Child","Other/Nonfamily",..: 2 3 2 3 3 3
2 3 2 3 …
 $ race          : Factor w/ 5 levels " Amer-Indian-Eskimo",..: 5 5 5 3 3 5 3 5
5 5 …
 $ sex           : Factor w/ 2 levels " Female"," Male": 2 2 2 2 1 1 1 2 1 2 …
 $ capital_gain  : num [1:29936, 1] 0.609 -0.234 -0.234 -0.234 -0.234 …
  ..- attr(*, "scaled:center")= num 3e-17
  ..- attr(*, "scaled:scale")= num 1
 $ capital_loss  : num [1:29936, 1] -0.219 -0.219 -0.219 -0.219 -0.219 …
  ..- attr(*, "scaled:center")= num 1.8e-19
  ..- attr(*, "scaled:scale")= num 1
 $ hours_per_week: num [1:29936, 1] -0.0633 -2.3907 -0.0633 -0.0633 -0.0633 …
  ..- attr(*, "scaled:center")= num 1.82e-16
```

```
  ..- attr(*, "scaled:scale")= num 1
 $ income_group  : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 2 2 …
```

[156]:
```
set.seed(123)
train_index <- sample(1:(nrow(df_gm2)), size = 0.8*nrow(df_gm2))
train_gam <- df_gm2[train_index,]
test_gam <- df_gm2[-train_index,]
print(dim(train_gam))
print(dim(test_gam))
```

```
[1] 23948    14
[1] 5988   14
```

[157]:
```
str(train_gam)
```

```
'data.frame':   23948 obs. of  14 variables:
 $ age           : num [1:23948, 1] 1.493 0.351 1.341 -0.182 -0.258 …
 $ workclass     : Factor w/ 4 levels "Government","Private",..: 3 1 2 2 2 1 2 2
2 1 …
 $ fnlwgt        : int  450544 598995 109015 28572 189922 317733 161708 220860
80933 255835 …
 $ education     : Factor w/ 7 levels "Bachelors","Doctorate",..: 1 1 7 7 3 1 7
3 3 7 …
 $ education_num : num [1:23948, 1] 1.1396 1.1396 -0.0415 -0.0415 -0.4352 …
 $ marital_status: Factor w/ 2 levels " Married"," Not Married": 1 2 1 1 1 1 2 2
1 2 …
 $ occupation    : Factor w/ 4 levels "Management/Professional",..: 1 1 1 2 2 1
2 4 2 3 …
 $ relationship  : Factor w/ 3 levels "Child","Other/Nonfamily",..: 3 2 3 3 3 3
1 2 3 2 …
 $ race          : Factor w/ 5 levels " Amer-Indian-Eskimo",..: 5 3 5 5 5 5 5 5
5 5 …
 $ sex           : Factor w/ 2 levels " Female"," Male": 2 1 1 2 2 2 1 2 2 1 …
 $ capital_gain  : num [1:23948, 1] -0.234 -0.234 2.749 -0.234 -0.234 …
 $ capital_loss  : num [1:23948, 1] -0.219 -0.219 -0.219 -0.219 -0.219 …
 $ hours_per_week: num [1:23948, 1] -0.0633 0.1091 0.7987 0.7987 -0.0633 …
 $ income_group  : Factor w/ 2 levels "0","1": 2 1 2 1 1 2 1 1 1 1 …
```

[657]:
```
summary(glm(data = train_gam, formula = income_group ~.,
                              family = binomial))
```

```
Call:
glm(formula = income_group ~ ., family = binomial, data = train_gam)

Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -3.313e+00  3.500e-01  -9.466  < 2e-16 ***
```

```
age                                3.676e-01  2.346e-02  15.670  < 2e-16 ***
workclassPrivate                   1.500e-02  5.606e-02   0.268 0.789054
workclassSelf-employed            -3.305e-01  7.296e-02  -4.530 5.90e-06 ***
workclassUnpaid/Other             -1.228e+01  1.378e+02  -0.089 0.928973
fnlwgt                             7.419e-07  1.908e-07   3.888 0.000101 ***
educationDoctorate                 1.056e-01  1.899e-01   0.556 0.578117
educationHigh School              -5.807e-02  1.385e-01  -0.419 0.674980
educationLess than High School    -2.456e-01  3.280e-01  -0.749 0.453930
educationMasters                   5.855e-02  9.017e-02   0.649 0.516101
educationOther                     4.220e-01  1.618e-01   2.608 0.009110 **
educationSome College/Associate   -3.172e-02  9.335e-02  -0.340 0.734036
education_num                      6.137e-01  7.483e-02   8.202 2.37e-16 ***
marital_status Not Married        -8.483e-01  1.773e-01  -4.784 1.72e-06 ***
occupationManual/Service          -9.739e-01  5.449e-02 -17.872  < 2e-16 ***
occupationOffice/Clerical & Sales -5.168e-01  5.537e-02  -9.334  < 2e-16 ***
occupationOther                   -8.774e-01  9.311e-02  -9.423  < 2e-16 ***
relationshipOther/Nonfamily        1.072e+00  1.625e-01   6.594 4.27e-11 ***
relationshipSpouse/Partner         2.525e+00  2.283e-01  11.060  < 2e-16 ***
race Asian-Pac-Islander            4.314e-01  2.714e-01   1.589 0.111977
race Black                         3.840e-01  2.590e-01   1.483 0.138063
race Other                        -8.449e-02  3.941e-01  -0.214 0.830243
race White                         5.627e-01  2.471e-01   2.278 0.022750 *
sex Male                           2.598e-01  5.774e-02   4.499 6.81e-06 ***
capital_gain                       8.124e-01  2.935e-02  27.677  < 2e-16 ***
capital_loss                       2.534e-01  1.722e-02  14.716  < 2e-16 ***
hours_per_week                     3.587e-01  2.198e-02  16.319  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 26781  on 23947  degrees of freedom
Residual deviance: 16193  on 23921  degrees of freedom
AIC: 16247

Number of Fisher Scoring iterations: 12
```

Based on the full model logistic regression we can see that while many of t tests for parameters and blocks appear to be valid there are few categories where we fail to reject the null hypothesis that parameters for them are 0, this is especially true for the race and education classes. Moreover, even after reducing the classes across the categories to a great level we have more than 25 t tests here, meaning that we have more than 73% of at least one of the above tests being false.

On top of that as we observed before the numeric variables themselves are not not normally distributed specifically and there does not appear to be direct linear relationship between them and the response variable, for this we will fit a Generalized Additive Models where we will fit cubic splines for each of the numerical variables.

[658]:
```r
gam_full <- gam(data = train_gam, formula = income_group ~ s(age, bs = "cr") +
                              s(education_num, bs = "cr") +
                              s(hours_per_week, bs = "cr") +
                              s(capital_gain, bs = "cr") +
                              s(capital_loss, bs = "cr") +
                              workclass + education +
                              marital_status + relationship +
                              occupation + sex + race,
                              family = binomial)
summary(gam_full)
```

```
Family: binomial
Link function: logit

Formula:
income_group ~ s(age, bs = "cr") + s(education_num, bs = "cr") +
    s(hours_per_week, bs = "cr") + s(capital_gain, bs = "cr") +
    s(capital_loss, bs = "cr") + workclass + education + marital_status +
    relationship + occupation + sex + race

Parametric coefficients:
                                   Estimate Std. Error z value Pr(>|z|)
(Intercept)                      -5.562e+00  1.424e+00  -3.907 9.35e-05 ***
workclassPrivate                  1.268e-02  5.830e-02   0.217 0.827862
workclassSelf-employed           -2.721e-01  7.686e-02  -3.540 0.000400 ***
workclassUnpaid/Other            -4.382e+01  2.023e+07   0.000 0.999998
educationDoctorate                7.383e-01  4.391e-01   1.682 0.092640 .
educationHigh School             -4.909e-01  2.052e-01  -2.392 0.016757 *
educationLess than High School   -4.032e-01  3.855e-01  -1.046 0.295492
educationMasters                  1.809e-01  1.487e-01   1.217 0.223745
educationOther                    8.679e-01  2.977e-01   2.915 0.003556 **
educationSome College/Associate  -3.840e-01  1.653e-01  -2.323 0.020179 *
marital_status Not Married       -8.915e-01  1.805e-01  -4.939 7.86e-07 ***
relationshipOther/Nonfamily       6.714e-01  1.761e-01   3.814 0.000137 ***
relationshipSpouse/Partner        2.041e+00  2.359e-01   8.650  < 2e-16 ***
occupationManual/Service         -9.463e-01  5.666e-02 -16.700  < 2e-16 ***
occupationOffice/Clerical & Sales -4.416e-01  5.794e-02  -7.622 2.50e-14 ***
occupationOther                  -8.677e-01  9.668e-02  -8.975  < 2e-16 ***
sex Male                          2.346e-01  6.121e-02   3.832 0.000127 ***
race Asian-Pac-Islander           5.835e-01  2.887e-01   2.021 0.043296 *
race Black                        5.264e-01  2.753e-01   1.912 0.055860 .
race Other                        2.041e-01  4.182e-01   0.488 0.625575
race White                        7.478e-01  2.638e-01   2.835 0.004585 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Approximate significance of smooth terms:
                  edf Ref.df Chi.sq  p-value    
s(age)           7.963  8.618 456.35  < 2e-16 ***
s(education_num) 2.222  2.751  46.87 1.56e-06 ***
s(hours_per_week) 5.925  6.966 203.09  < 2e-16 ***
s(capital_gain)  7.971  7.999 428.53  < 2e-16 ***
s(capital_loss)  8.844  8.989 320.04  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

R-sq.(adj) =  0.465   Deviance explained = 44.5%
UBRE = -0.37436  Scale est. = 1         n = 23948
```

We fitted a full additive model above while ignoring fnlwt for now. From the Efficitive Degrees of Freedom we can see that all the numeric variables in the model have edf > 1.5 or so, implying that non parametric relationship with the log odds for income greater than 50K indeed exist for all 5 variables and their p value is low enough for us to reject the null hypothesis that f(variables) are 0 or that these variables has no additional effect on the log odds for income greater than 50K. All this implies that Generalized Additive Model is indeed a better fit for our census income dataset.

We have not included the fnlwt for now, we will test the effect of having weights after hyperparameter tuning and model selection for this additive model. Generally, weights will help us to form population esitmates with better credibility but still the models without it would be enough to explain the linear additive relationship across the variables and response.

Some categories still have high p values for the t test, we can't remove them specifically because some parts of them are indeed having proper p value for our analysis but they could be appearing due to faulty t tests as there are atleast 73% chances that one of the test will be wrong. To test this we will form reduced model where we will remove education, workclass, and race categories as all of them have classes where we are seeing this issue and compare them via anova function to analyze if the bigger model improve the model explainability or not.

[660]:
```
#testing reduced mode, removing race, workclass, and education

reduced_gam <- gam(data = train_gam, formula = income_group ~ s(age, bs = "cr")
 +
                              s(education_num, bs = "cr") +
                              s(hours_per_week, bs = "cr") +
                              s(capital_gain, bs = "cr") +
                              s(capital_loss, bs = "cr") +
                              marital_status + relationship +
                              occupation + sex,
                              family = binomial)

anova(reduced_gam, gam_full, test = 'Chisq')
```

A anova: 2 × 5

	Resid. Df <dbl>	Resid. Dev <dbl>	Df <dbl>	Deviance <dbl>	Pr(>Chi) <dbl>
1	23902.71	14930.22	NA	NA	NA
2	23891.68	14875.03	11.03671	55.18721	7.394996e-08

[661]:
```
#testing other categories too
reduced_gam <- gam(data = train_gam, formula = income_group ~ s(age, bs = "cr")␣
↪+
                              s(education_num, bs = "cr") +
                              s(hours_per_week, bs = "cr") +
                              s(capital_gain, bs = "cr") +
                              s(capital_loss, bs = "cr") +
                              race+ workclass +education,
                              family = binomial)

anova(reduced_gam, gam_full, test = 'Chisq')
```

A anova: 2 × 5

	Resid. Df <dbl>	Resid. Dev <dbl>	Df <dbl>	Deviance <dbl>	Pr(>Chi) <dbl>
1	23897.78	17857.72	NA	NA	NA
2	23891.68	14875.03	6.107635	2982.689	0

From above anova test we can observe that Full Model has reduced residual deviance, less residual deviance here implies that full model with adding the categories where not all levels pass the t tests indeed improve the capability of the model. Along with very low p value we can indeed reject the null hypothesis that the full model does not improve the model information while adding the parameters to be estimated.

And with other anova test we can also clearly see that marital_status, relationship, occupation, and sex categories have more explainibilty than the categories with dubious t tests, still by adding all the categories we are indeed getting better model so we will use them, and since every numeric variable is also valid for our test we will use a full model.

Still, our model fitting is not complete as all the cubic spline functions have a smoothing parameter we can tune for better modelling fitting.

[662]:
```
#hyperparameter tuning for the numerical factors

k_candidates <- c(5, 7, 9, 11)
aic_values <- numeric(length(k_candidates))

for(i in seq_along(k_candidates)){
  mod_temp <- gam(income_group ~
                    s(age, k = k_candidates[i], bs = "cr") +
                    s(education_num, bs = "cr") +
                    s(hours_per_week, bs = "cr") +
                    s(capital_gain, bs = "cr") +
                    s(capital_loss, bs = "cr") +
                    workclass + education + marital_status + relationship +␣
↪occupation + sex,
                    data = train_gam,
                    family = binomial)
  aic_values[i] <- AIC(mod_temp)
}
```

```r
best_k_age <- k_candidates[which.min(aic_values)]
cat("Best k for age smooth:", best_k_age, "\n")
```

```
Best k for age smooth: 9
```

[663]:
```r
k_candidates <- c(5, 7, 9, 11)
aic_values <- numeric(length(k_candidates))

for(i in seq_along(k_candidates)){
  mod_temp <- gam(income_group ~
                    s(age, bs = "cr") +
                    s(education_num, k = k_candidates[i], bs = "cr") +
                    s(hours_per_week, bs = "cr") +
                    s(capital_gain, bs = "cr") +
                    s(capital_loss, bs = "cr") +
                    workclass + education + marital_status + relationship +␣
 ↪occupation + sex,
                    data = train_gam,
                    family = binomial)
  aic_values[i] <- AIC(mod_temp)
}

best_k_edu <- k_candidates[which.min(aic_values)]
cat("Best k for education_num smooth:", best_k_edu, "\n")
```

```
Best k for education_num smooth: 9
```

[664]:
```r
k_candidates <- c(5, 7, 9, 11)
aic_values <- numeric(length(k_candidates))

for(i in seq_along(k_candidates)){
  mod_temp <- gam(income_group ~
                    s(age, bs = "cr") +
                    s(education_num, bs = "cr") +
                    s(hours_per_week, k = k_candidates[i], bs = "cr") +
                    s(capital_gain, bs = "cr") +
                    s(capital_loss, bs = "cr") +
                    workclass + education + marital_status + relationship +␣
 ↪occupation + sex,
                    data = train_gam,
                    family = binomial)
  aic_values[i] <- AIC(mod_temp)
}

best_k_hpw <- k_candidates[which.min(aic_values)]
cat("Best k for hours_per_week smooth:", best_k_hpw, "\n")
```

```
Best k for hours_per_week smooth: 7
```

[665]:
```r
k_candidates <- c(5, 7, 9, 11, 13, 15)
aic_values <- numeric(length(k_candidates))

for(i in seq_along(k_candidates)){
  mod_temp <- gam(income_group ~
                    s(age, bs = "cr") +
                    s(education_num, bs = "cr") +
                    s(hours_per_week, bs = "cr") +
                    s(capital_gain,  k = k_candidates[i], bs = "cr") +
                    s(capital_loss, bs = "cr") +
                    workclass + education + marital_status + relationship +␣
 ↪occupation + sex,
                    data = train_gam,
                    family = binomial)
  aic_values[i] <- AIC(mod_temp)
}

best_k_cg <- k_candidates[which.min(aic_values)]
cat("Best k for capital gain smooth", best_k_cg, "\n")
```

```
Best k for capital gain smooth 11
```

[666]:
```r
k_candidates <- c(5, 7, 9, 11, 13, 15)
aic_values <- numeric(length(k_candidates))

for(i in seq_along(k_candidates)){
  mod_temp <- gam(income_group ~
                    s(age, bs = "cr") +
                    s(education_num, bs = "cr") +
                    s(hours_per_week, bs = "cr") +
                    s(capital_gain, bs = "cr") +
                    s(capital_loss, k = k_candidates[i], bs = "cr") +
                    workclass + education + marital_status + relationship +␣
 ↪occupation + sex,
                    data = train_gam,
                    family = binomial)
  aic_values[i] <- AIC(mod_temp)
}

best_k_cl <- k_candidates[which.min(aic_values)]
cat("Best k for capital loss smooth:", best_k_cl, "\n")
```

```
Best k for capital loss smooth: 19
```

[667]:
```
#using tuned k values for full additive model

gam_full_adj <- gam(data = train_gam, formula = income_group ~ s(age, k = 9, bs␣
↪= "cr") +
                              s(education_num, k = 9, bs = "cr") +
                              s(hours_per_week, k = 7, bs = "cr") +
                              s(capital_gain, k = 11, bs = "cr") +
                              s(capital_loss, k = 19, bs = "cr") +
                              workclass + education +
                              marital_status + relationship +
                              occupation + sex + race,
                              family = binomial)
```

[668]:
```
anova(gam_full, gam_full_adj, test = 'Chisq')
```

A anova: 2 × 5

	Resid. Df <dbl>	Resid. Dev <dbl>	Df <dbl>	Deviance <dbl>	Pr(>Chi) <dbl>
1	23891.68	14875.03	NA	NA	NA
2	23884.30	14685.73	7.376726	189.3052	4.01352e-37

We didn't expect the tuning to have this much effect, with same full model with adjusted smoothing bandwidth values we got a deviance difference of 189 with minimal reduction in the degrees of freedom. With p value so low can be clear that new model is a better fit overall. Now we will compare it with the weights adjusted model.

[669]:
```
gam_full_adj_w <- gam(data = train_gam, formula = income_group ~ s(age, k = 9,␣
↪bs = "cr") +
                              s(education_num, k = 9, bs = "cr") +
                              s(hours_per_week, k = 7, bs = "cr") +
                              s(capital_gain, k = 11, bs = "cr") +
                              s(capital_loss, k = 19, bs = "cr") +
                              workclass + education +
                              marital_status + relationship +
                              occupation + sex + race,
                              family = binomial,
                              weights = fnlwgt)
```

```
Warning message in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L
= G$L, :
"Iteration limit reached without full convergence - check carefully"
```

[670]:
```
aic_weighted <- AIC(gam_full_adj_w)
aic_unweighted <- AIC(gam_full_adj)
cat("AIC for weighted model:", aic_weighted, "\n")
cat("AIC for unweighted model:", aic_unweighted, "\n")
```

```
AIC for weighted model: 2708412179
AIC for unweighted model: 14810.01
```

[671]:
```r
pred_probs_unw <- predict(gam_full_adj, newdata = test_gam, type = "response")
pred_probs_w <- predict(gam_full_adj_w, newdata = test_gam, type = "response")

pred_class_w   <- ifelse(pred_probs_w >= 0.5, 1, 0)
pred_class_unw <- ifelse(pred_probs_unw >= 0.5, 1, 0)

actual <- as.numeric(as.character(test_gam$income_group))
```

[672]:
```r
conf_w <- table(Observed = actual, Predicted = pred_class_w)
conf_unw <- table(Observed = actual, Predicted = pred_class_unw)

# Print confusion matrices:
print(conf_w)
print(conf_unw)
```

```
        Predicted
Observed    0    1
       0 4311  260
       1  530  887
        Predicted
Observed    0    1
       0 4323  248
       1  560  857
```

[673]:
```r
accuracy_w <- sum(diag(conf_w)) / sum(conf_w)
accuracy_unw <- sum(diag(conf_unw)) / sum(conf_unw)

cat("Accuracy for weighted model:", round(accuracy_w, 4), "\n")
cat("Accuracy for unweighted model:", round(accuracy_unw, 4), "\n")
```

```
Accuracy for weighted model: 0.8681
Accuracy for unweighted model: 0.8651
```

[674]:
```r
roc_w <- roc(actual, pred_probs_w)
roc_unw <- roc(actual, pred_probs_unw)
cat("AUC for weighted model:", auc(roc_w), "\n")
cat("AUC for unweighted model:", auc(roc_unw), "\n")
```

```
Setting levels: control = 0, case = 1

Setting direction: controls < cases

Setting levels: control = 0, case = 1

Setting direction: controls < cases


AUC for weighted model: 0.9174194
```

```
AUC for unweighted model: 0.9168934
```

From the above workings we have following observations: 1. AIC can not be used to compare the models as weighted model impacts the likelihood each observation contributes depending upon the weights. AIC could have been used to compare the models if we had a full on population dataset but its outside the scope for this project. 2. Accuracy and AUC are similar for both models so for prediction purposes both models are quite similar.

Depending on the above observations we conclude that while both models have simialr prediction power with good accuracy overall weighted model could be better fit if we want to use the model for the population level inference while sample confined model is still a good model for explaining the overall relationship across the variables.

[724]:
```r
#we observed that weighted and non weighted models are quite similar for accurcy and other criterias so we will not use weights further on for
#model fitting as it indeed adds complexity to the model and impact the model plots and further analysis.

gam_fit <- gam(data = train_gam, formula = income_group ~ s(age, k = 9, bs = "cr") +
                                    s(education_num, k = 9, bs = "cr") +
                                    s(hours_per_week, k = 7, bs = "cr") +
                                    s(capital_gain, k = 11, bs = "cr") +
                                    s(capital_loss, k = 19, bs = "cr") +
                                    workclass + education +
                                    marital_status + relationship +
                                    occupation + sex + race,
                                    family = binomial, method = "ML")
```

[725]:
```r
#comparing gam with glm with the help of AIC
glm_final <- glm(data = train_gam, formula = income_group ~., family = binomial)
print(AIC(glm_final))
print(AIC(gam_fit))
#AIC of gam is much lower
```

```
[1] 16247
[1] 14859.81
```

[730]:
```r
concurvity(gam_fit, full = TRUE)
```

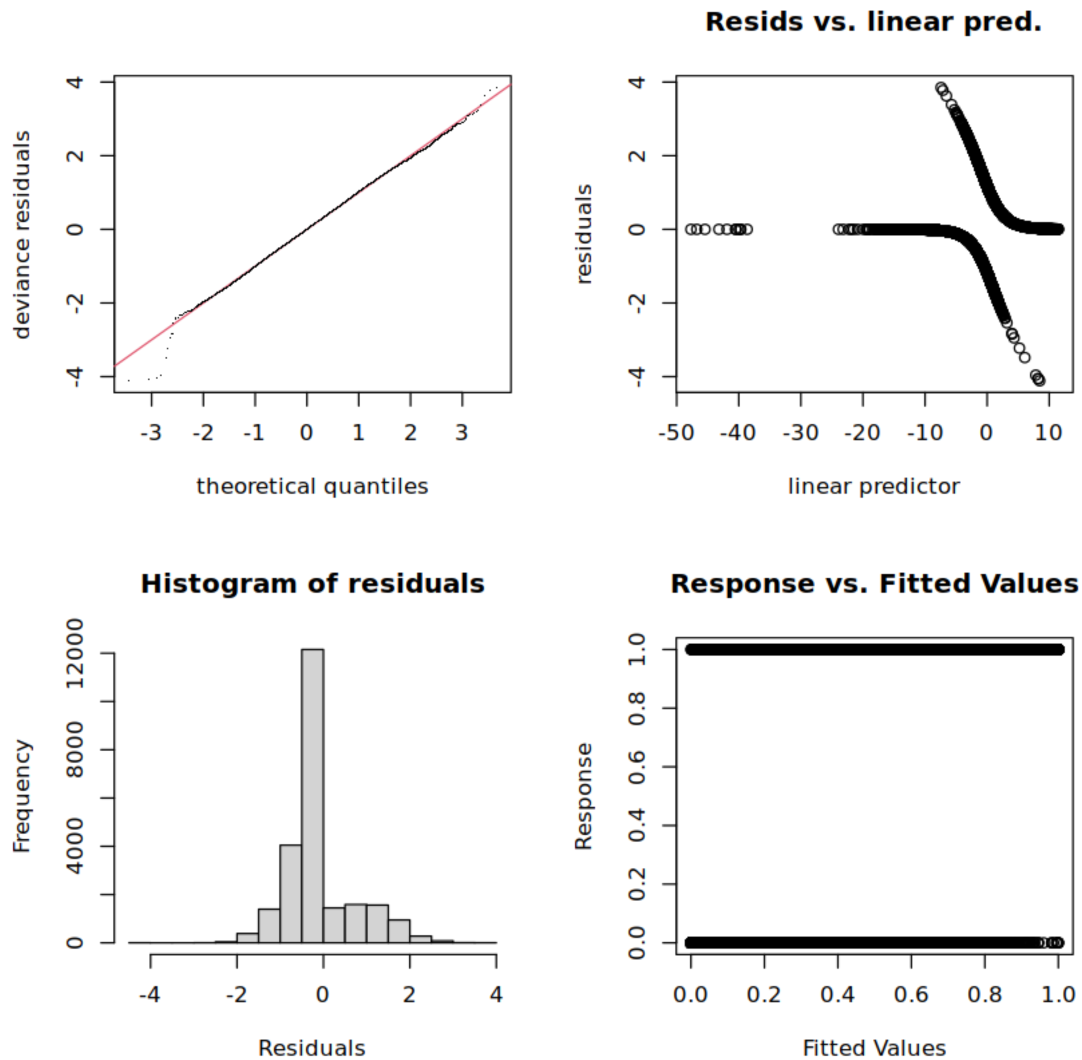
A matrix: 3×6 of type dbl

	para	s(age)	s(education_num)	s(hours_per_week)	s(capital_g
worst	1	0.4493629	1.0000000	0.2823824	0.06770895
observed	1	0.4292099	0.8650929	0.2651937	0.04836111
estimate	1	0.1005919	0.7139160	0.1264677	0.01930184

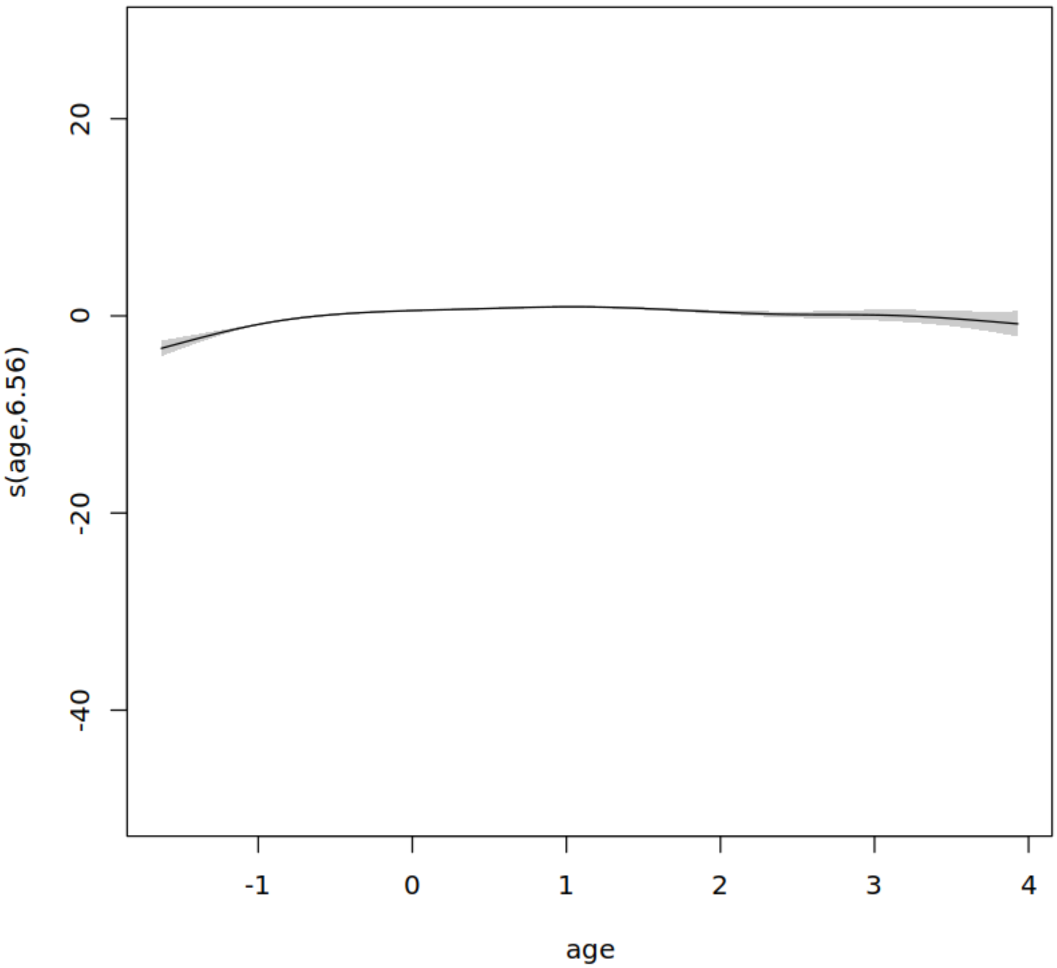
Concurvity test the multicollinearity across the smoothing functions and we can see that education num has high evidence of multicollinearity implying that other smoothing functions are enough to explain the effects of smooth(education_num). Due to this we will remove the smoothing function for this predictor and fit it normally.

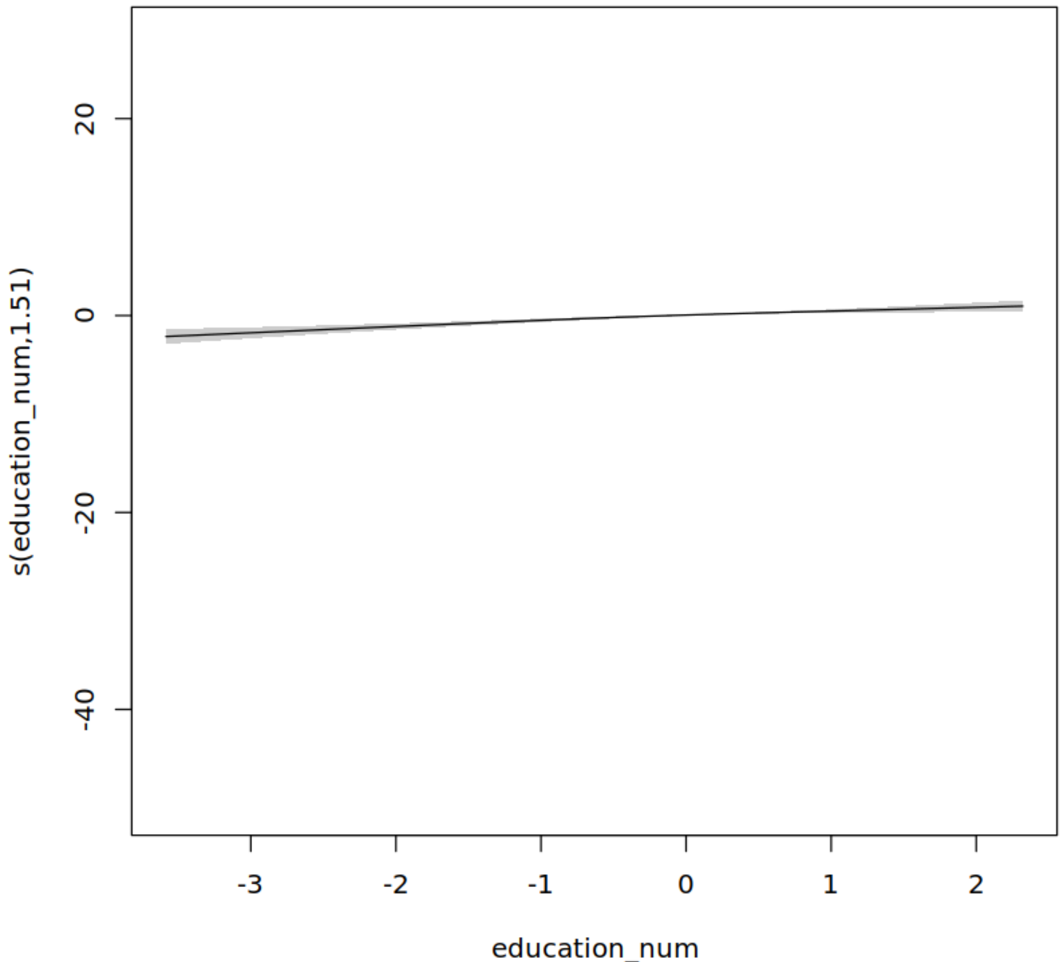
[729]:
```
gam.check(gam_fit)
```

```
Method: ML   Optimizer: outer newton
full convergence after 5 iterations.
Gradient range [-0.0002438782,0.0002070823]
(score 7491.922 & scale 1).
Hessian positive definite, eigenvalue range [0.1354166,4.427607].
Model rank =  71 / 71

Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

                     k'   edf k-index p-value
s(age)             8.00  6.56    0.99    0.57
s(education_num)   8.00  1.51    1.00    0.85
s(hours_per_week)  6.00  5.14    0.99    0.62
s(capital_gain)   10.00  7.95    0.95    0.01 **
s(capital_loss)   18.00 16.30    0.99    0.56
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

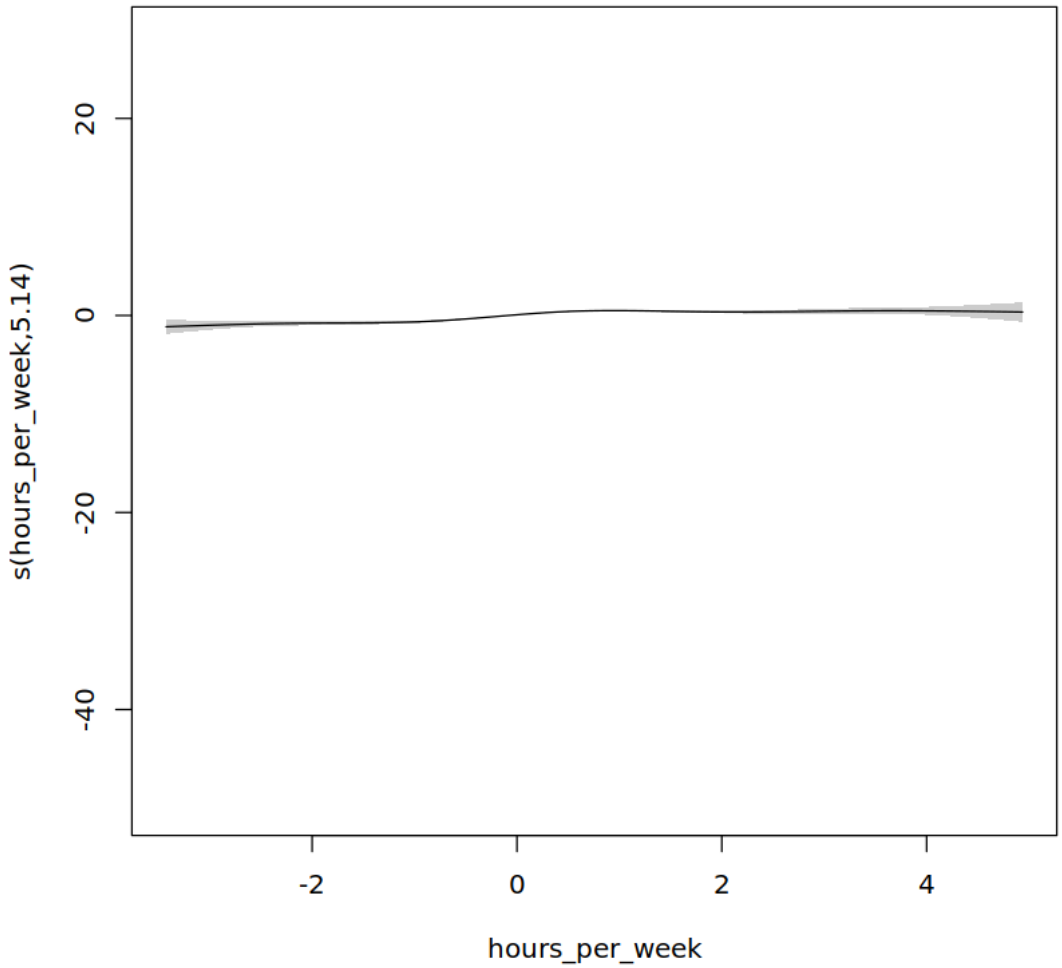


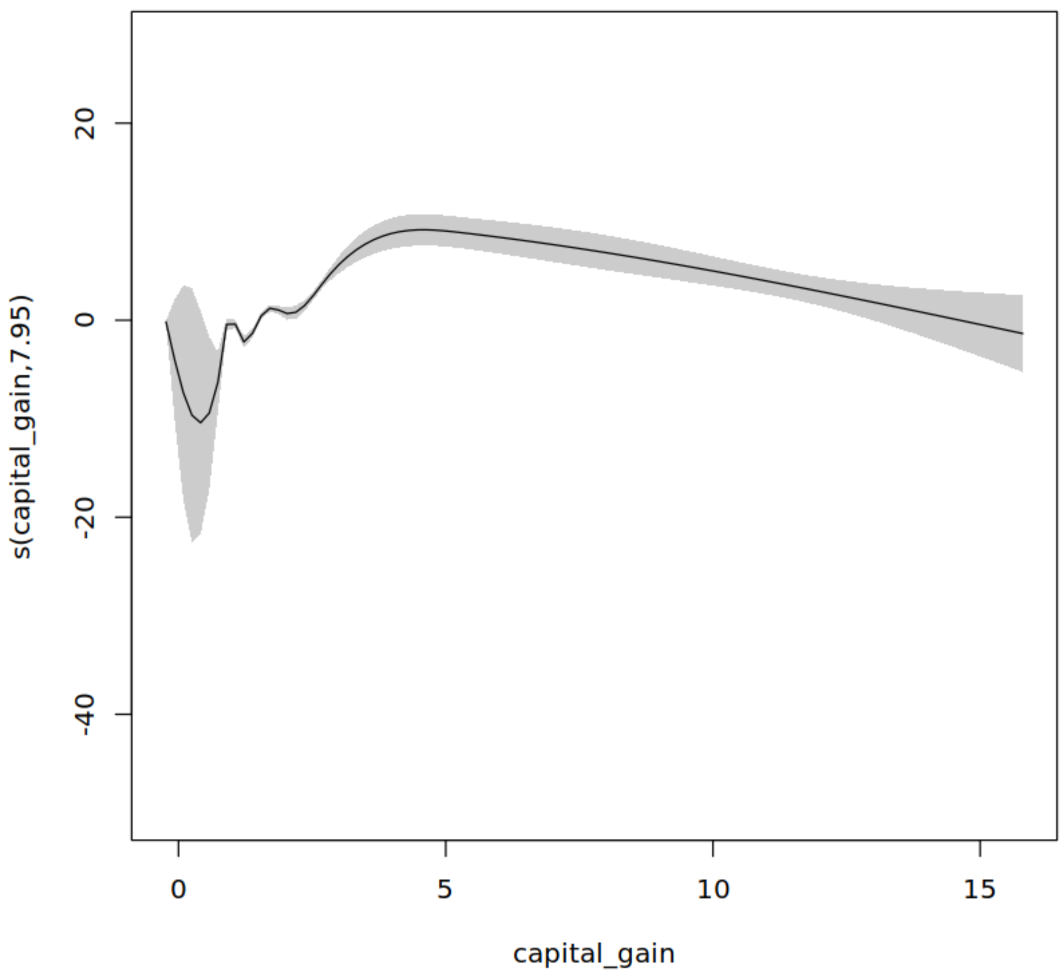
[740]:
```
plot.gam(gam_fit, shade = TRUE)
```


20
0
-20
-40
s(age,6.56)
-1
0
1
2
3
4
age

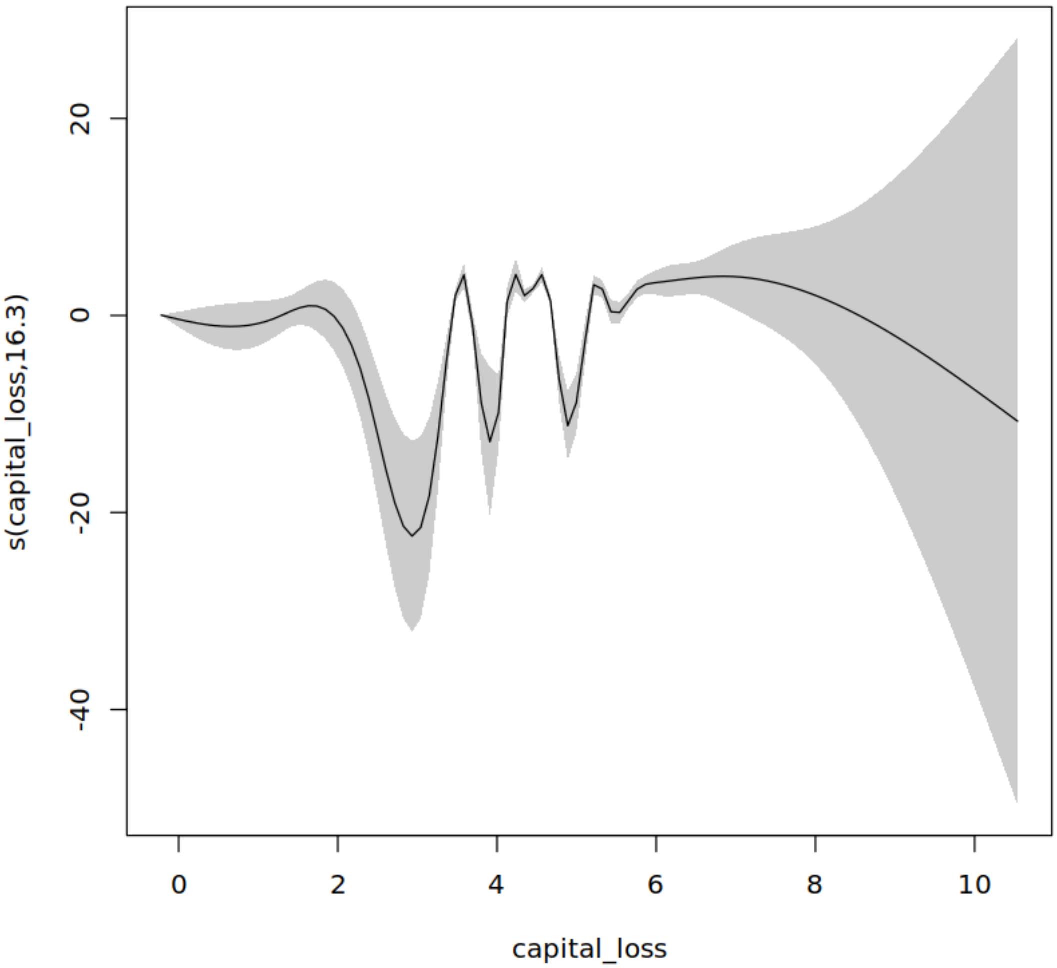


20
0
-20
-40
s(education_num,1.51)
-3
-2
-1
0
1
2
education_num


20
0
-20
-40
s(hours_per_week,5.14)
-2
0
2
4
hours_per_week



s(capital_gain,7.95)
20
0
-20
-40
0
5
10
15
capital_gain



Age, education num, and hours per week appears to be linearlly associated with log odds, however edf of them was way higher than cutoff of 1.5 where at less than or equal to 1.5 we may decide that linear relationship fits better for them. Still education number had lowest edf and since it is having higher multicollinearity too we are fine with removing the smoothing term for this model.

Besides them Capital Gain and Capital Loss indeed have a non linear relationships which is being modeled fine overall.

[778]:
```
#we will use non weighted model for the final analysis because weighted model␣
↪was not wroking well with gam analysis and model comparison overall
#this could have been rectified if we have full dataset with proper variation

gam_final <- gam(data = train_gam, formula = income_group ~ s(age, k = 9, bs =␣
↪"cr")+
```

```
                              s(hours_per_week, k = 7, bs = "cr") +
                              s(capital_gain, k = 15, bs = "cr") +
                              s(capital_loss, k = 15, bs = "cr") +
                              education_num +
                              workclass + education +
                              marital_status + relationship +
                              occupation + sex + race,
                              family = binomial, method = "ML")
```

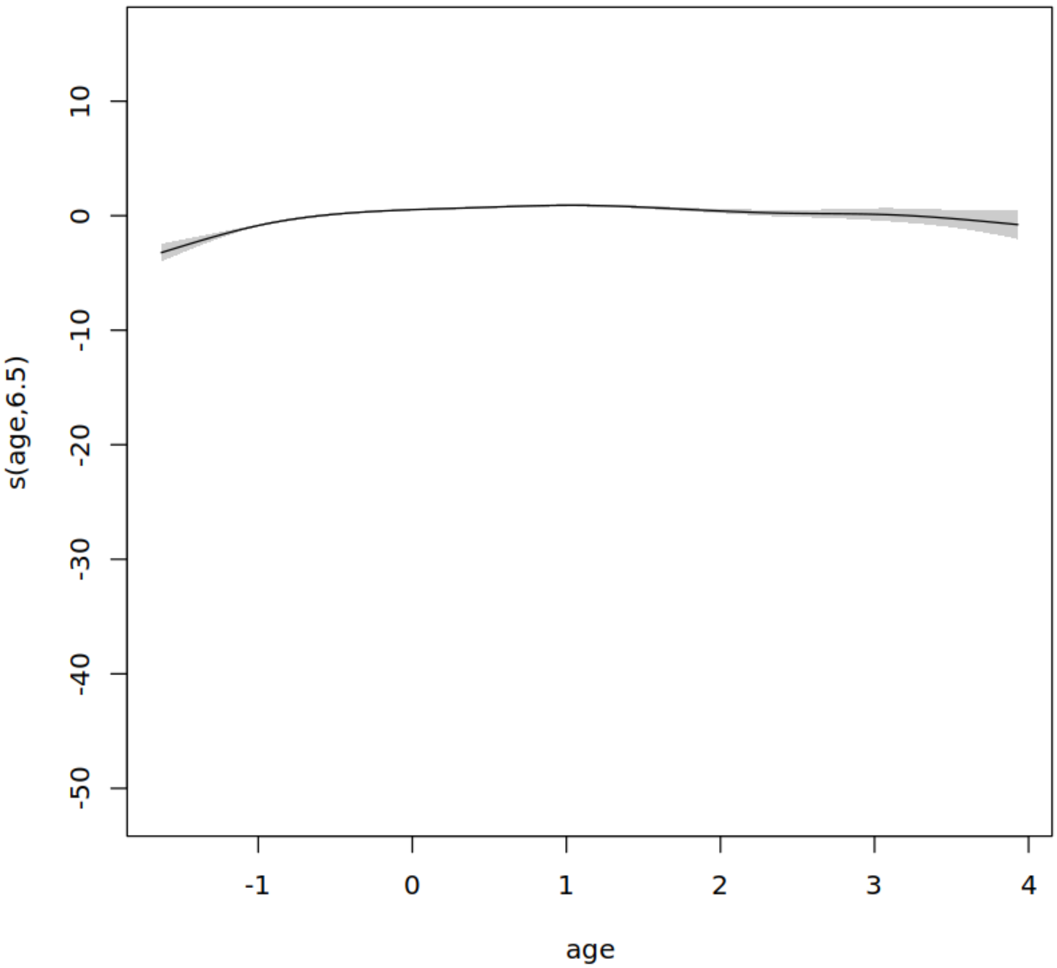
[755]:
```
concurvity(gam_final, full = TRUE)
```

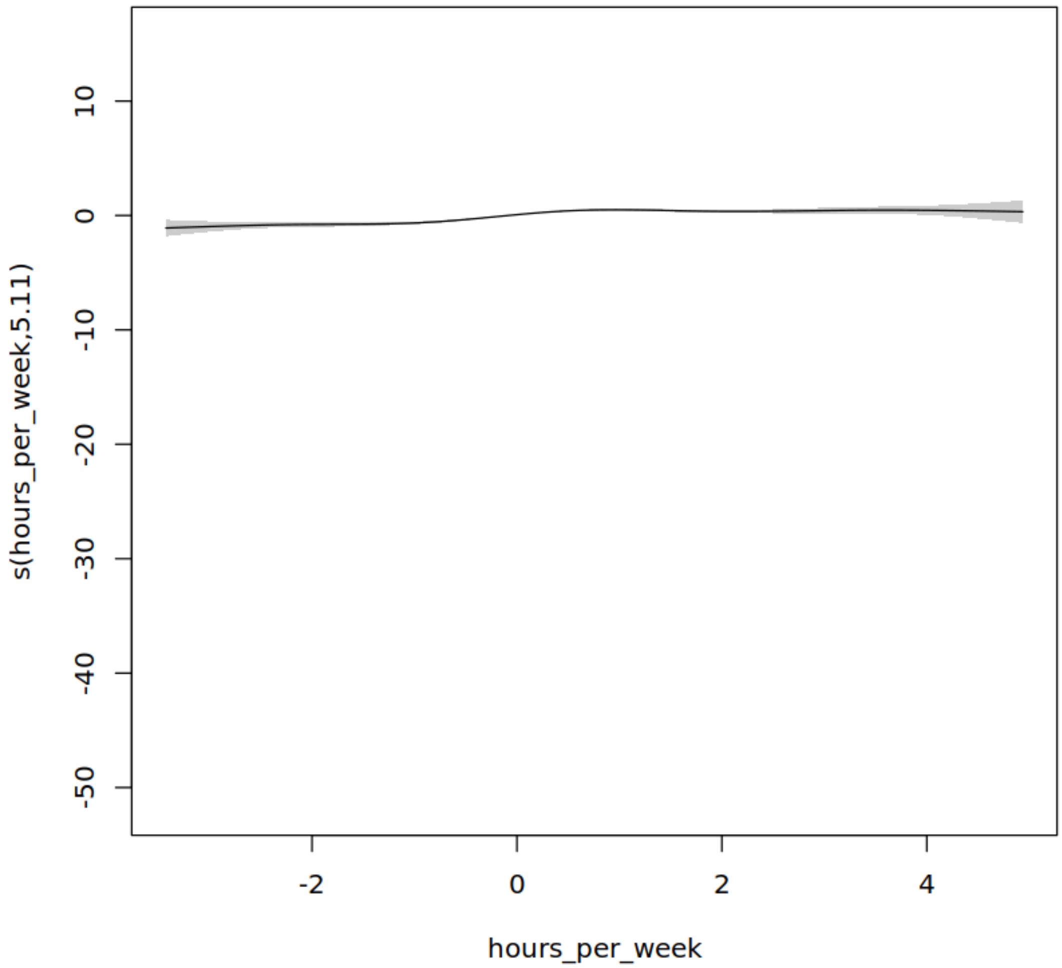
A matrix: 3×5 of type dbl

	para	s(age)	s(hours_per_week)	s(capital_gain)	s(capita
worst	0.9943058	0.43200054	0.2794156	0.06820396	0.04090
observed	0.9943058	0.41905725	0.2630703	0.03022055	0.00667
estimate	0.9943058	0.09652853	0.1252004	0.01729114	0.01263

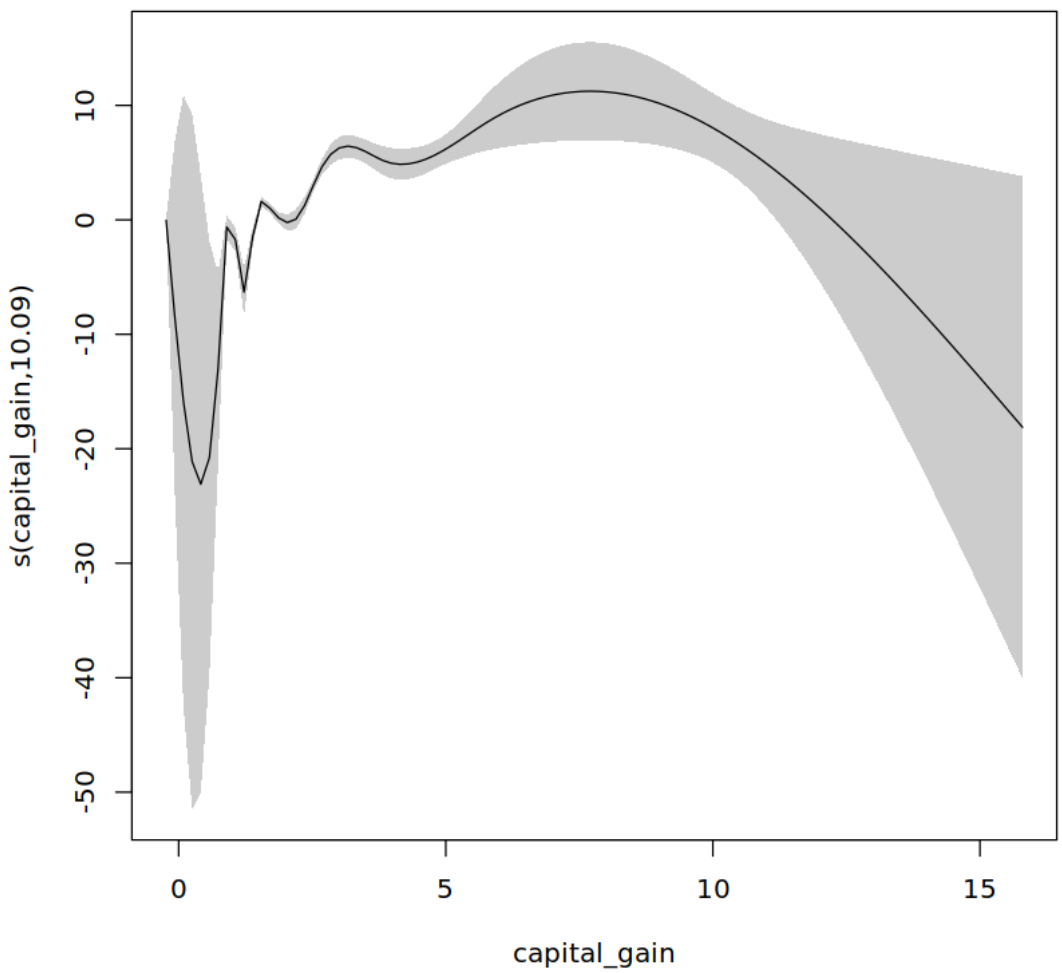
Concurvity analysis is fine now.

[760]:
```
plot.gam(gam_final, shade = TRUE)
```

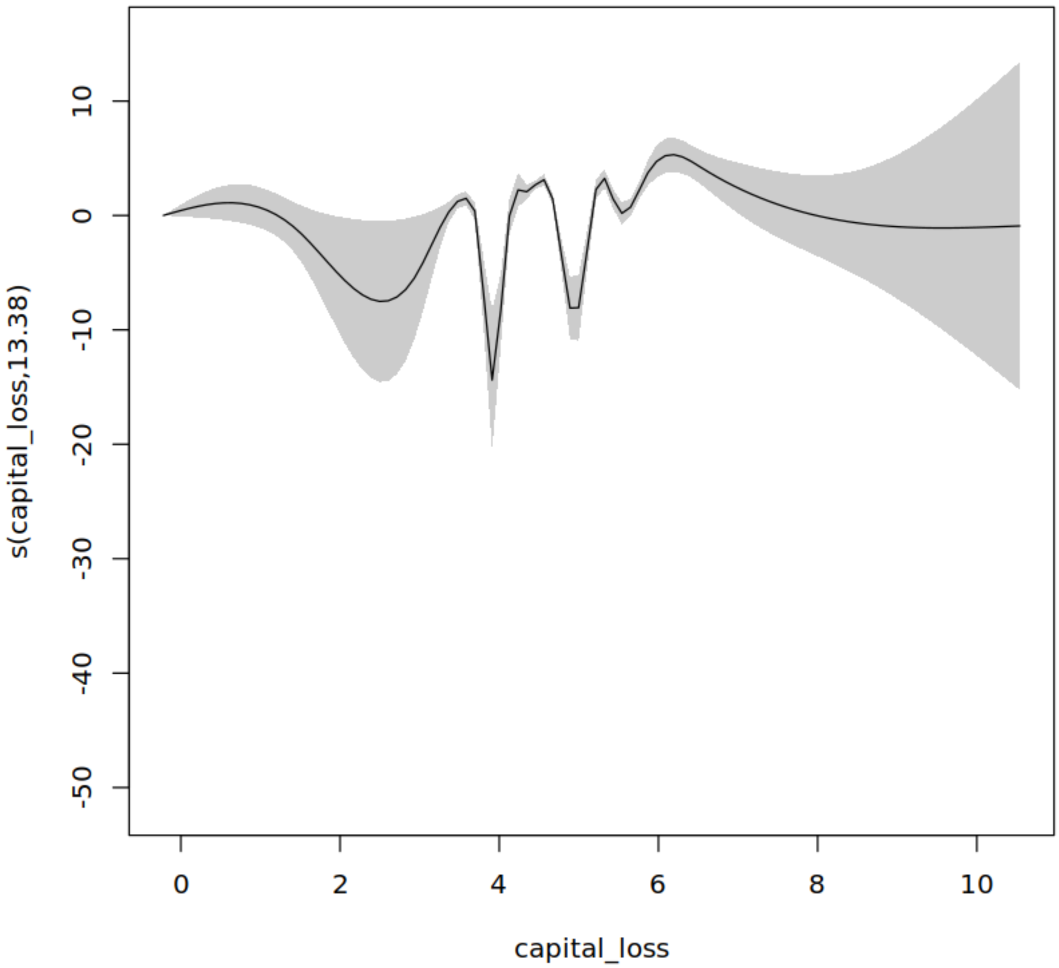

10
0
-10
-20
-30
-40
-50
s(age,6.5)
-1
0
1
2
3
4
age



s(hours_per_week,5.11)
10
0
-10
-20
-30
-40
-50
-2
0
2
4
hours_per_week



s(capital_gain,10.09)
10
0
-10
-20
-30
-40
-50
0
5
10
15
capital_gain

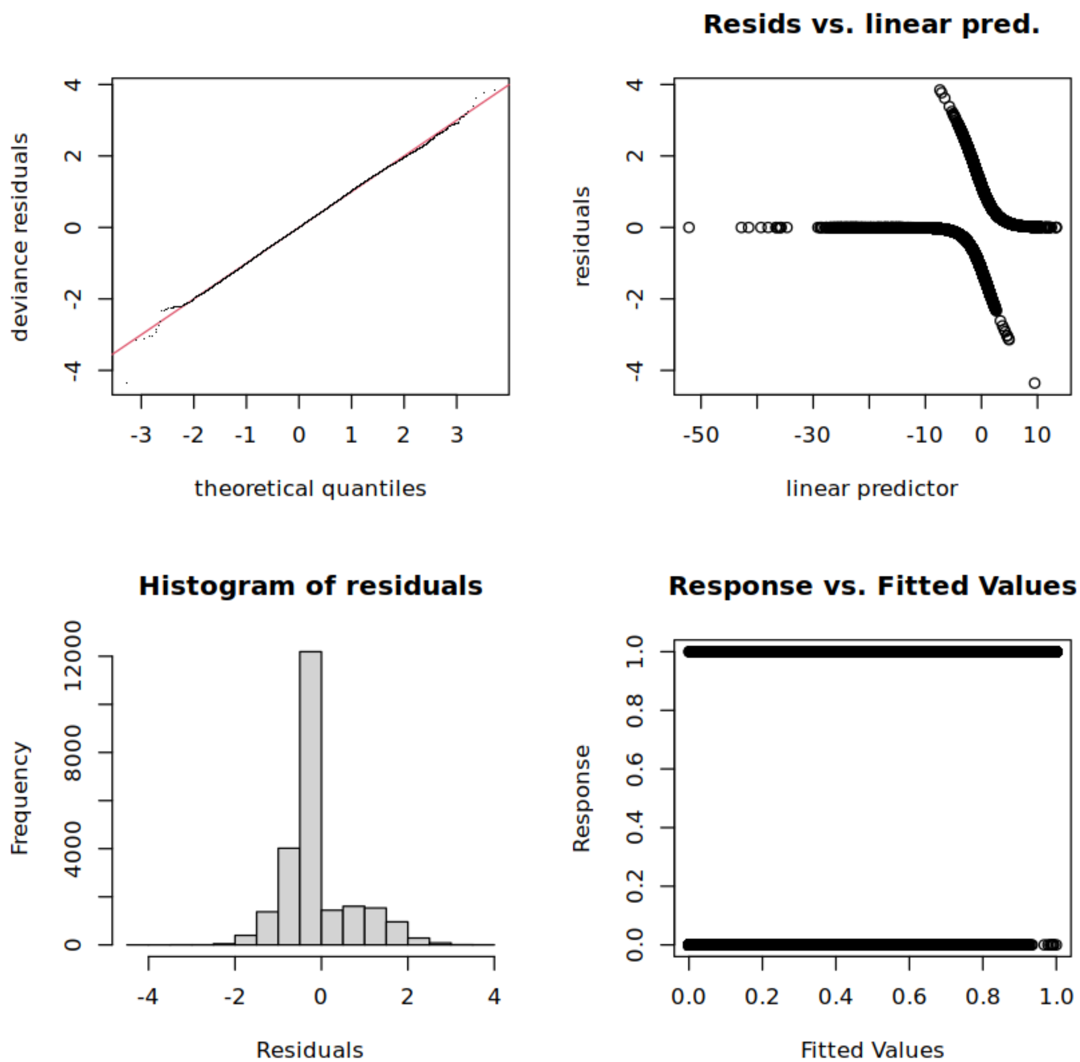


We reduced the k value for the capital loss because in previous plot analysis the variation across the confidence bands for higher values was way too high for our liking. This may lead to only slight reduction in explained deviance which we can accept for our model.

[777]:
```
gam.check(gam_final)
```

```
Method: ML   Optimizer: outer newton
full convergence after 5 iterations.
Gradient range [-0.0005510738,3.358223e-06]
(score 7466.743 & scale 1).
Hessian positive definite, eigenvalue range [0.7245655,1.949829].
Model rank =  64 / 64
```

```
Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

                     k'   edf k-index p-value
s(age)             8.00  6.50    1.01   0.920
s(hours_per_week)  6.00  5.11    1.00   0.735
s(capital_gain)   14.00 10.09    0.97   0.055 .
s(capital_loss)   14.00 13.38    0.97   0.040 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```



Overall plots seems to be fine. P value for smoothing K values seems to be fine with capital loss having potential for having higher k value than chosen 15. Residuals are cenetered around 0 and deviance residual plot is appearing good. And by observing the response vs fitted plot we can see that we have highly confident model overall.

[779]:
```
summary(gam_final)
```

```
Family: binomial
Link function: logit

Formula:
income_group ~ s(age, k = 9, bs = "cr") + s(hours_per_week, k = 7,
    bs = "cr") + s(capital_gain, k = 15, bs = "cr") + s(capital_loss,
    k = 15, bs = "cr") + education_num + workclass + education +
    marital_status + relationship + occupation + sex + race

Parametric coefficients:
                                    Estimate Std. Error z value Pr(>|z|)
(Intercept)                       -3.378e+00  3.799e-01  -8.891  < 2e-16 ***
education_num                      5.374e-01  7.766e-02   6.920 4.52e-12 ***
workclassPrivate                   6.423e-03  5.875e-02   0.109 0.912943
workclassSelf-employed            -2.718e-01  7.711e-02  -3.525 0.000424 ***
workclassUnpaid/Other             -3.474e+01  2.023e+07   0.000 0.999999
educationDoctorate                 3.223e-01  1.979e-01   1.628 0.103422
educationHigh School              -1.242e-01  1.440e-01  -0.863 0.388352
educationLess than High School    -3.394e-01  3.414e-01  -0.994 0.320188
educationMasters                   2.854e-02  9.472e-02   0.301 0.763191
educationOther                     5.922e-01  1.690e-01   3.503 0.000460 ***
educationSome College/Associate   -7.182e-02  9.746e-02  -0.737 0.461163
marital_status Not Married        -8.678e-01  1.832e-01  -4.737 2.17e-06 ***
relationshipOther/Nonfamily        6.465e-01  1.754e-01   3.686 0.000228 ***
relationshipSpouse/Partner         2.070e+00  2.377e-01   8.709  < 2e-16 ***
occupationManual/Service          -9.495e-01  5.705e-02 -16.644  < 2e-16 ***
occupationOffice/Clerical & Sales -4.301e-01  5.828e-02  -7.380 1.59e-13 ***
occupationOther                   -8.544e-01  9.710e-02  -8.800  < 2e-16 ***
sex Male                           2.347e-01  6.160e-02   3.810 0.000139 ***
race Asian-Pac-Islander            5.895e-01  2.900e-01   2.033 0.042071 *
race Black                         5.526e-01  2.767e-01   1.997 0.045811 *
race Other                         1.731e-01  4.244e-01   0.408 0.683418
race White                         7.708e-01  2.652e-01   2.906 0.003657 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Approximate significance of smooth terms:
                     edf Ref.df Chi.sq p-value
s(age)             6.501  7.281  458.9  <2e-16 ***
s(hours_per_week)  5.114  5.704  207.5  <2e-16 ***
s(capital_gain)   10.094 10.411  457.5  <2e-16 ***
s(capital_loss)   13.385 13.709  367.2  <2e-16 ***
---
```

```
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

R-sq.(adj) =   0.47   Deviance explained = 45.2%
-ML = 7466.7  Scale est. = 1         n = 23948
```

We fit the model with all the predictors with age, hpw, capital gain, and capital loss allowed to model non linear relationship with log odds, for which we chose cubic splines and optimized the bandwidth K during hyperparameter tuning. All the categories have some levels with p value less than 0.05, workclass other have small number of observations because of which model is not able to model it properly.

Education, marital status, occupation, and sex are strong predictors of high income in our model and non linear smoothing functions are highly significant, they are not just simply straight lines and explain the relationship to some level. We will consider only those predictors/group levels having proper t tests for some coefficient observations: 1. Race white has largest positive coefficient at 0.8 followed by non family relationship at 0.6. Adding up the sex male positive coefficeints, we can say considering everything else constant un married white men have higher chances to fall in high income group. 2. For negative coefficients we have Manual/service occupation at -0.8 which is somewhat unexpected but understandable. Following that we have other occupation and clerical/sales. Negative coefficients for these sub groups show that while keeping every other thing constant falling under groups with negative coefficients can reduce the chances a person has high income. 3. Base intercept is -3.3 and its for female sex having Amer-Indian-Eskimo race having Bachelors education working in a Managerial/Professional position while being married and having child relationship. Probability of such an individual having high income in 1994 is 0.033, which is quite low. 4. Numerical variables besides education numbers have non linear relationships, we can fit them with a linear relationship too for better explainability but that will impact the model performance. Regardless of that 1 sd increase for education_num has positive benefit for the log odds or overall probability, for example if we add 1 sd education_num we are getting probability as 0.055 which is quite an improvement.

[802]:
```r
#AIC Comparison GAM vs GLM Model

print(AIC(gam_final))
print(AIC(glm_final))
```

```
[1] 14801.19
[1] 16247
```

AIC is lower for Generalized Additive Model final fit compared to final GLM and its also lower than the previous best fit for Additive models.

[800]:
```r
# Predict probabilities for the test set
pred_probs <- predict(gam_final, newdata = test_gam, type = "response")
pred_class <- ifelse(pred_probs >= 0.5, 1, 0)

pred_probs2 <- predict(glm_final, newdata = test_gam, type = "response")
pred_class2 <- ifelse(pred_probs2 >= 0.5, 1, 0)

actual <- as.numeric(as.character(test_gam$income_group))
```

```r
accuracy_gam <- mean(pred_class == actual)
cat("Accuracy of the final GAM model:", round(accuracy, 4), "\n")
accuracy_glm <- mean(pred_class2 == actual)
cat("Accuracy of the final GLM model:", round(accuracy, 4), "\n")
```

```
Accuracy of the final GAM model: 0.8656
Accuracy of the final GLM model: 0.8656
```

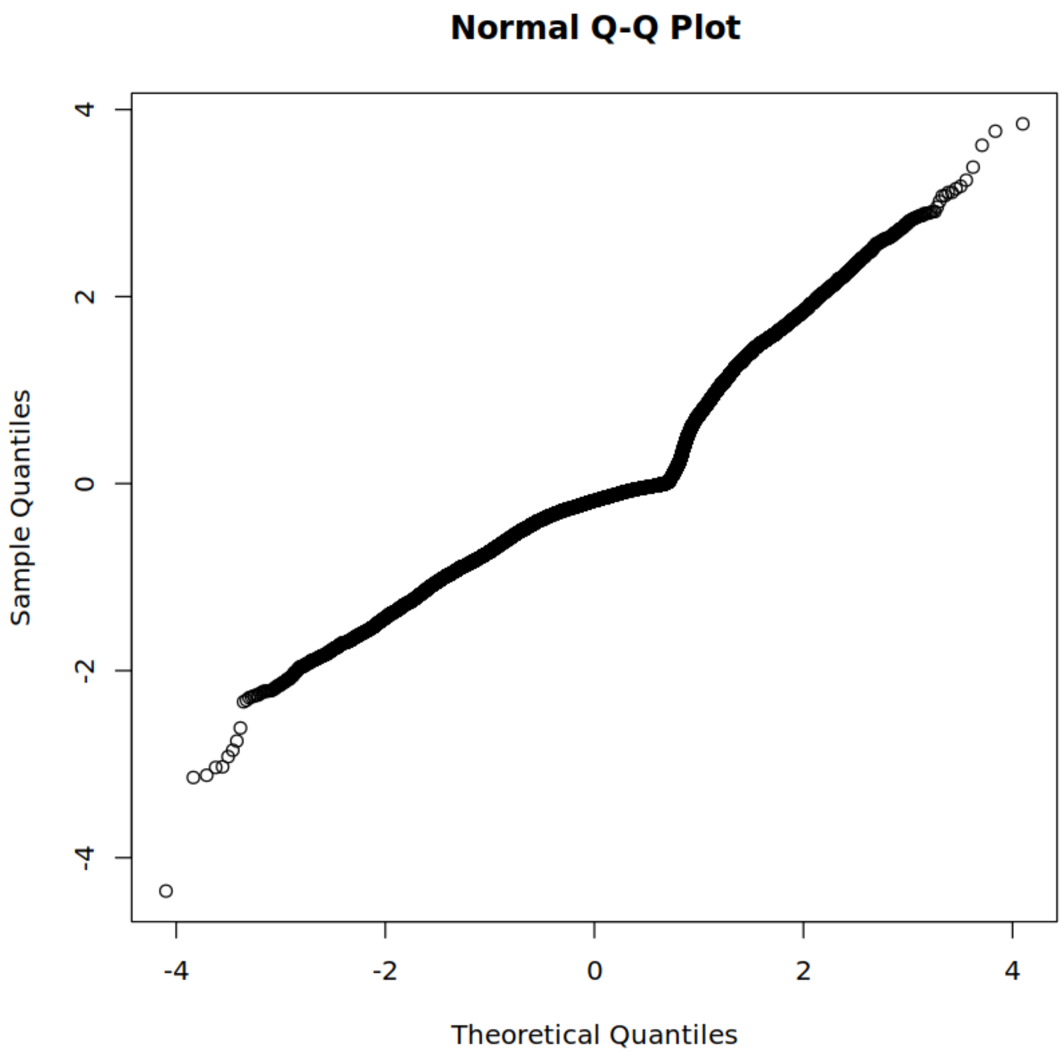
[803]:
```r
table(Predicted = pred_class, Actual = actual)
```

```
         Actual
Predicted    0    1
        0 4333  567
        1  238  850
```

[809]:
```r
f1_gam <- F1_Score(y_pred = pred_class, y_true = actual, positive = "1")

f1_glm <- F1_Score(y_pred = pred_class2, y_true = actual, positive = "1")

cat("F1 Score GAM:", round(f1_gam, 4), "\n")
cat("F1 Score GLM:", round(f1_glm, 4), "\n")
```

```
F1 Score GAM: 0.6786
F1 Score GLM: 0.6498
```

[875]:
```r
qqnorm(resid(gam_final))
```



Generalized Additive model we picked as our final linear model has a good amount of variance explainability, the fact corroborated by mupltiple anova tests and AIC comparisons across this section. Besides the inferential potential this model has, we also got better predictive model as even if Accuracy is similar at 0.87 F1 Score for GAM model is indeed higher.

It is difficult to get a better linear model than this with this dataset as most of the predictors are categorial variables with problems in few numerical variables we have as most of them have non linear relationship with log odds of the ratio of a person earining more than 50K compared to not earning such income per annum. But if we have more predictors we can indeed improve the model, we can also perform Machine Learning methods suited for classification only if we are focused on only predictions, but considering all that we got a good additive model which can explain the relationship between predictors and response while having good enough predictive power.

1.3 Bayesian Model

Most of the predictors are categorical the number of unknown group variability must be quite less overall, implying that because of so many factor levels likelihood of having more latent groups seems unlikely due to which we can not use mixture models, so we will use bayesian hierarchical models for this dataset.

We used rjags models first as it has better visual representation of the hierarchical model however even 1 single iteration for MCMC gibbs sampling was taking us 30 minutes or so even with competent CPU/RAM combination making it near impossible to test and fit multiple models with MCMC giibs sampling algorithm. Due to this we used stan based model which uses more efficient Hamiltonian Monte Carlo algorithm.

In Hierarchical Models besides allowing the predictors to have their own prior distribution we can also allow different prior distributions for the factor levels or groups withing the predictors. Suppose we have chosen prior for Sex as a t distribution then with hierarchical model we can assign different group level prior for male and female categories. Following this we can allow multiple variables having group level priors but this could increase complexity so much that model itself would not only be time consuming to fit but also have divergent transition errors which can lead to not trustworthy Monte Carlo Chains.

For rjags model we will not test or run the model in this project but we used the following model, observation of which can help in explaining the overall hierarchical model we are interested in:

model{ for (i in 1:N) { income_group[i] ~ dbern(p[i]) logit(p[i]) <- alpha[sex_idx[i]] + inprod(beta[], X[i,]) }

```
for (j in 1:J){
    alpha[j] ~ dnorm(mu_alpha, tau_alpha)
}

mu_alpha ~ dnorm(0, 0.01)
tau_alpha ~ dgamma(0.01, 0.01)

for (k in 1:K){
    beta[k] ~ ddexp(0.0, sqrt(2.0))
}
```

}'

[ ]:

```

```

[876]:

```
head(data)
```

A data.frame: 6 × 15

	age	workclass	fnlwgt	education	education_num	marital_status
	<int>	<chr>	<int>	<chr>	<int>	<chr>
1	39	State-gov	77516	Bachelors	13	Never-married
2	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse
3	38	Private	215646	HS-grad	9	Divorced
4	53	Private	234721	11th	7	Married-civ-spouse
5	28	Private	338409	Bachelors	13	Married-civ-spouse
6	37	Private	284582	Masters	14	Married-civ-spouse

[94]:
```
df_bm <- subset(data, select = -c(fnlwgt, native_country)) #removing weights␣
↪for the bayesian model
```

[95]:
```
unique(df_bm$income_group)
```

1. ' <=50K' 2. ' >50K'

[96]:
```
df_bm$income_group <- ifelse(df_bm$income_group == " >50K", 1, 0)

summary(df_bm)
```

```
      age          workclass           education          education_num  
 Min.   :17.00   Length:29936       Length:29936       Min.   : 1.00  
 1st Qu.:28.00   Class :character   Class :character   1st Qu.: 9.00  
 Median :37.00   Mode  :character   Mode  :character   Median :10.00  
 Mean   :38.39                                         Mean   :10.11  
 3rd Qu.:47.00                                         3rd Qu.:12.00  
 Max.   :90.00                                         Max.   :16.00  
 marital_status      occupation        relationship           race          
 Length:29936       Length:29936       Length:29936       Length:29936      
 Class :character   Class :character   Class :character   Class :character  
 Mode  :character   Mode  :character   Mode  :character   Mode  :character  



     sex             capital_gain      capital_loss     hours_per_week 
 Length:29936       Min.   :    0.0   Min.   :   0.00   Min.   : 1.00  
 Class :character   1st Qu.:    0.0   1st Qu.:   0.00   1st Qu.:40.00  
 Mode  :character   Median :    0.0   Median :   0.00   Median :40.00  
                    Mean   :  603.6   Mean   :  88.68   Mean   :40.73  
                    3rd Qu.:    0.0   3rd Qu.:   0.00   3rd Qu.:45.00  
                    Max.   :41310.0   Max.   :4356.00   Max.   :98.00  
  income_group  
 Min.   :0.000  
 1st Qu.:0.000  
 Median :0.000  
 Mean   :0.245  
 3rd Qu.:0.000  
 Max.   :1.000  
```

[97]:
```
str(df_bm)
```

```
'data.frame':   29936 obs. of  13 variables:
 $ age           : int  39 50 38 53 28 37 49 52 31 42 …
 $ workclass     : chr  " State-gov" " Self-emp-not-inc" " Private" " Private"
…
 $ education     : chr  " Bachelors" " Bachelors" " HS-grad" " 11th" …
 $ education_num : int  13 13 9 7 13 14 5 9 14 13 …
```

```
 $ marital_status: chr  " Never-married" " Married-civ-spouse" " Divorced" "
Married-civ-spouse" …
 $ occupation    : chr  " Adm-clerical" " Exec-managerial" " Handlers-cleaners"
" Handlers-cleaners" …
 $ relationship  : chr  " Not-in-family" " Husband" " Not-in-family" " Husband"
…
 $ race          : chr  " White" " White" " White" " Black" …
 $ sex           : chr  " Male" " Male" " Male" " Male" …
 $ capital_gain  : int  2174 0 0 0 0 0 0 0 14084 5178 …
 $ capital_loss  : int  0 0 0 0 0 0 0 0 0 0 …
 $ hours_per_week: int  40 13 40 40 40 40 16 45 50 40 …
 $ income_group  : num  0 0 0 0 0 0 0 1 1 1 …
```

Since most of the predictors are already categorial variables for this dataset we can be quite confident that we don't need to find any latent variable as these grouped variables are more than enough to find the variance across the population. Hence we will not use a mixture model but a bayesian hierarchical model.

We can allow variability across different levels for categorial variables with a bayesian hierarchical model, however if we use all the categorial variables and allow each of them have their population and group based priors then it may lead to overfitting. For this project we will start with one categorial variable, sex. Then we will fit a model allowing intercepts to vary by workclass and sex.

[98]:
```
# Make sex a factor (for random intercepts)
df_bm$sex <- as.factor(df_bm$sex)

# Scale numeric predictors
df_bm$age <- scale(df_bm$age)
df_bm$education_num <- scale(df_bm$education_num)
df_bm$capital_gain <- scale(df_bm$capital_gain)
df_bm$capital_loss <- scale(df_bm$capital_loss)
df_bm$hours_per_week <- scale(df_bm$hours_per_week)
```

[99]:
```
set.seed(10)
idx <- sample(seq_len(nrow(df_bm)), 2000)
df_small <- df_bm[idx, ]
# Fit your model on df_small
```

[33]:
```
# Formula: random intercept for sex, all other predictors as fixed effects
# Fitting a smaller dataset first

fit_sex <- stan_glmer(
  income_group ~ age + education_num + capital_gain + capital_loss +
↪hours_per_week +
    workclass + education + marital_status + occupation + relationship + race +
    (1 | sex),
  data = df_small,
  family = binomial(link = "logit"),
```

```
  prior = laplace(location = 0, scale = 1),           # Laplace prior for betas␣
↪(LASSO-like)
  prior_intercept = normal(location = 0, scale = 10), # Normal prior for␣
↪intercept
  prior_covariance = decov(regularization = 2),       # Prior for group-level␣
↪SDs
  chains = 4, iter = 2000, cores = parallel::detectCores(), seed = 123
)
```

```
fixed-effect model matrix is rank deficient so dropping 1 column / coefficient

Warning message:
"There were 41 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them."
Warning message:
"Examine the pairs() plot to diagnose sampling problems
"
```

[41]:
```
summary(fit_sex)
```

```
Model Info:
 function:     stan_glmer
 family:       binomial [logit]
 formula:      income_group ~ age + education_num + capital_gain + capital_loss +
           hours_per_week + workclass + education + marital_status +
           occupation + relationship + race + (1 | sex)
 algorithm:    sampling
 sample:       4000 (posterior sample size)
 priors:       see help('prior_summary')
 observations: 2000
 groups:       sex (2)

Estimates:
                                       mean   sd   10%   50%   90%
(Intercept)                          -2.8    0.9 -3.9  -2.8  -1.7
age                                   0.3    0.1  0.2   0.3   0.4
education_num                         0.7    0.2  0.5   0.7   1.0
capital_gain                          0.9    0.1  0.7   0.9   1.0
capital_loss                          0.2    0.1  0.2   0.2   0.3
hours_per_week                        0.3    0.1  0.2   0.3   0.4
workclass Local-gov                  -0.2    0.3 -0.6  -0.2   0.2
workclass Private                     0.0    0.2 -0.3   0.0   0.3
workclass Self-emp-inc               -0.1    0.4 -0.6  -0.1   0.3
workclass Self-emp-not-inc           -0.1    0.3 -0.5  -0.1   0.3
workclass State-gov                  -0.7    0.4 -1.2  -0.7  -0.2
```

```
education 11th                          0.0    0.5 -0.5   0.0   0.6
education 12th                         -0.1    0.6 -0.8   0.0   0.6
education 1st-4th                      -0.4    1.2 -1.9  -0.2   0.8
education 5th-6th                       0.5    0.8 -0.4   0.5   1.6
education 7th-8th                      -0.2    0.7 -1.1  -0.2   0.5
education 9th                          -0.4    0.8 -1.5  -0.3   0.5
education Assoc-acdm                   -0.8    0.4 -1.3  -0.8  -0.2
education Assoc-voc                    -0.2    0.3 -0.6  -0.2   0.2
education Bachelors                     0.0    0.3 -0.3   0.0   0.4
education Doctorate                     0.1    0.6 -0.6   0.1   1.0
education HS-grad                       0.0    0.2 -0.2   0.0   0.2
education Masters                       0.5    0.4  0.0   0.5   1.0
education Preschool                     0.0    1.4 -1.6   0.0   1.7
education Prof-school                   0.3    0.6 -0.3   0.3   1.1
marital_status Married-AF-spouse        1.8    1.5  0.0   1.6   3.8
marital_status Married-civ-spouse       2.0    0.5  1.3   2.0   2.6
marital_status Married-spouse-absent    0.9    0.7  0.0   0.8   1.8
marital_status Never-married           -0.1    0.3 -0.5  -0.1   0.2
marital_status Separated               -0.1    0.5 -0.8  -0.1   0.5
marital_status Widowed                  0.4    0.5 -0.2   0.4   1.1
occupation Craft-repair                 0.5    0.3  0.1   0.5   0.8
occupation Exec-managerial              0.7    0.3  0.4   0.7   1.1
occupation Farming-fishing             -1.0    0.5 -1.7  -1.0  -0.3
occupation Handlers-cleaners           -1.2    0.8 -2.3  -1.1  -0.2
occupation Machine-op-inspct           -0.5    0.4 -1.0  -0.5   0.0
occupation Other-service               -0.5    0.4 -1.0  -0.5   0.0
occupation Priv-house-serv             -0.3    1.3 -1.8  -0.1   1.1
occupation Prof-specialty               0.8    0.3  0.4   0.8   1.1
occupation Protective-serv              0.8    0.4  0.3   0.8   1.4
occupation Sales                        0.2    0.3 -0.1   0.2   0.6
occupation Tech-support                 0.1    0.4 -0.3   0.1   0.6
occupation Transport-moving             0.1    0.3 -0.4   0.1   0.5
relationship Not-in-family              0.0    0.5 -0.6   0.0   0.6
relationship Other-relative            -0.8    0.8 -1.9  -0.7   0.1
relationship Own-child                 -1.1    0.6 -1.9  -1.1  -0.3
relationship Unmarried                 -0.2    0.5 -0.8  -0.1   0.4
relationship Wife                       1.0    0.4  0.5   1.0   1.5
race Asian-Pac-Islander                -0.2    0.5 -0.8  -0.1   0.4
race Black                              0.1    0.5 -0.4   0.1   0.7
race Other                             -0.3    0.8 -1.2  -0.2   0.6
race White                             -0.2    0.4 -0.7  -0.2   0.3
b[(Intercept) sex:_Female]             -0.1    0.5 -0.7  -0.1   0.4
b[(Intercept) sex:_Male]                0.2    0.6 -0.3   0.1   0.8
Sigma[sex:(Intercept),(Intercept)]      0.6    1.3  0.0   0.2   1.7

Fit Diagnostics:
           mean   sd   10%   50%   90%
mean_PPD 0.2    0.0  0.2   0.2   0.3
```

```
The mean_ppd is the sample average posterior predictive distribution of the␣
↪outcome variable (for details see help('summary.stanreg')).

MCMC diagnostics
                                    mcse Rhat n_eff
(Intercept)                         0.0  1.0   910
age                                 0.0  1.0  4269
education_num                       0.0  1.0  2040
capital_gain                        0.0  1.0  3925
capital_loss                        0.0  1.0  3732
hours_per_week                      0.0  1.0  4179
workclass Local-gov                 0.0  1.0  2614
workclass Private                   0.0  1.0  2631
workclass Self-emp-inc              0.0  1.0  3026
workclass Self-emp-not-inc          0.0  1.0  2939
workclass State-gov                 0.0  1.0  3247
education 11th                      0.0  1.0  4236
education 12th                      0.0  1.0  4715
education 1st-4th                   0.0  1.0  3330
education 5th-6th                   0.0  1.0  3187
education 7th-8th                   0.0  1.0  3304
education 9th                       0.0  1.0  4193
education Assoc-acdm                0.0  1.0  3043
education Assoc-voc                 0.0  1.0  3476
education Bachelors                 0.0  1.0  1883
education Doctorate                 0.0  1.0  3059
education HS-grad                   0.0  1.0  3178
education Masters                   0.0  1.0  2081
education Preschool                 0.0  1.0  3382
education Prof-school               0.0  1.0  2425
marital_status Married-AF-spouse    0.0  1.0  3320
marital_status Married-civ-spouse   0.0  1.0  1926
marital_status Married-spouse-absent 0.0  1.0  3050
marital_status Never-married        0.0  1.0  3804
marital_status Separated            0.0  1.0  4541
marital_status Widowed              0.0  1.0  4226
occupation Craft-repair             0.0  1.0  1863
occupation Exec-managerial          0.0  1.0  1846
occupation Farming-fishing          0.0  1.0  3103
occupation Handlers-cleaners        0.0  1.0  3420
occupation Machine-op-inspct        0.0  1.0  2329
occupation Other-service            0.0  1.0  2544
occupation Priv-house-serv          0.0  1.0  3216
occupation Prof-specialty           0.0  1.0  2183
occupation Protective-serv          0.0  1.0  2770
occupation Sales                    0.0  1.0  1662
```

```
occupation Tech-support                 0.0  1.0  3033
occupation Transport-moving             0.0  1.0  2586
relationship Not-in-family              0.0  1.0  1868
relationship Other-relative             0.0  1.0  3654
relationship Own-child                  0.0  1.0  2591
relationship Unmarried                  0.0  1.0  1857
relationship Wife                       0.0  1.0  2909
race Asian-Pac-Islander                 0.0  1.0  2557
race Black                              0.0  1.0  2538
race Other                              0.0  1.0  2526
race White                              0.0  1.0  2304
b[(Intercept) sex:_Female]              0.0  1.0   478
b[(Intercept) sex:_Male]                0.0  1.0   481
Sigma[sex:(Intercept),(Intercept)]      0.0  1.0  1468
mean_PPD                                0.0  1.0  4259
log-posterior                           0.3  1.0  1103

For each parameter, mcse is Monte Carlo standard error, n_eff is a crude measure
 ↪of effective sample size, and Rhat is the potential scale reduction factor on
 ↪split chains (at convergence Rhat=1).
```

Multiple coefficients are close to 0, we will use the gam fitted model where we removed a lot of levels for further analysis, and these levels with little to no impact must have caused 41 divergent transitions, implying that there a lot of multicollinearity in the dataset. We will also remove the race variable while using more informative priors for the predictors to reduce the divergency errors.

Besides that model seems good to go. Sex based estimates are indeed different, hours per week, capital gain, and others have positive estimates correctly and governemnt based employment has negative estimates overall which is not surprising. We will test a new model with smaller dataset again before fitting on a training dataset and analyzing it further.

[100]:
```r
df_bm2 <- subset(df_gm, select = -c(fnlwgt))
df_bm2$income_group <- ifelse(df_bm2$income_group == " >50K", 1, 0)
```

[101]:
```r
# Make sex a factor (for random intercepts)
df_bm2$sex <- as.factor(df_bm2$sex)
df_bm2$occupation <- as.factor(df_bm2$occupation)

# Scale numeric predictors
df_bm2$age <- scale(df_bm2$age)
df_bm2$education_num <- scale(df_bm2$education_num)
df_bm2$capital_gain <- scale(df_bm2$capital_gain)
df_bm2$capital_loss <- scale(df_bm2$capital_loss)
df_bm2$hours_per_week <- scale(df_bm2$hours_per_week)
```

[102]:
```r
set.seed(10)
idx <- sample(seq_len(nrow(df_bm2)), 2000)
df_small <- df_bm2[idx, ]
```

```
# Fit your model on df_small
```

[103]:
```
unique(df_small$education)
```

1. 'Some College/Associate' 2. 'High School' 3. 'Bachelors' 4. 'Masters' 5. 'Less than High School'
6. 'Other' 7. 'Doctorate'

[68]:
```
#Now taaking occupation as group random estimate or group level effect

fit_2 <- stan_glmer(formula = income_group ~ age + education_num + capital_gain␣
 ↪+ hours_per_week +
            workclass + education + marital_status + relationship + (1 |␣
 ↪occupation) +
            (1 | sex), data = df_small, family = binomial(link = 'logit'),
            prior = student_t(df = 7, 0, 2.5),    # using more informative␣
 ↪student t prior
            prior_intercept = normal(location = 0, scale = 10),
            chains = 3, iter = 2000, cores = parallel::detectCores(), seed = 123
)
summary(fit_2)
```

```
Warning message:
"There were 18 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them."
Warning message:
"Examine the pairs() plot to diagnose sampling problems
"


Model Info:
 function:     stan_glmer
 family:       binomial [logit]
 formula:      income_group ~ age + education_num + capital_gain +␣
 ↪hours_per_week +
           workclass + education + marital_status + relationship + (1 |
           occupation) + (1 | sex)
 algorithm:    sampling
 sample:       3000 (posterior sample size)
 priors:       see help('prior_summary')
 observations: 2000
 groups:       occupation (4), sex (2)

Estimates:
                                                     mean   sd   10%   50%   90%
(Intercept)                                         -2.3    0.8 -3.4  -2.3  -1.3
age                                                  0.3    0.1  0.2   0.3   0.4
education_num                                        0.6    0.2  0.3   0.6   0.9
```

```
capital_gain                                       0.9    0.1  0.7   0.9   1.0
hours_per_week                                     0.3    0.1  0.2   0.3   0.4
workclassPrivate                                   0.2    0.2 -0.1   0.2   0.4
workclassSelf-employed                             0.0    0.2 -0.3   0.0   0.3
educationDoctorate                                 0.5    0.7 -0.4   0.5   1.4
educationHigh School                              -0.4    0.4 -0.9  -0.4   0.2
educationLess than High School                    -0.6    1.0 -1.9  -0.6   0.6
educationMasters                                   0.6    0.3  0.2   0.6   1.0
educationOther                                     0.6    0.6 -0.2   0.6   1.3
educationSome College/Associate                   -0.3    0.3 -0.7  -0.3   0.1
marital_status Not Married                        -1.6    0.5 -2.3  -1.6  -0.9
relationshipOther/Nonfamily                        1.1    0.5  0.5   1.1   1.8
relationshipSpouse/Partner                         1.9    0.6  1.1   1.9   2.8
b[(Intercept) occupation:Management/Professional]  0.4    0.3  0.1   0.4   0.8
b[(Intercept) occupation:Manual/Service]          -0.2    0.3 -0.6  -0.2   0.1
b[(Intercept) occupation:Office/Clerical_&_Sales] -0.1    0.3 -0.4  -0.1   0.3
b[(Intercept) occupation:Other]                   -0.1    0.3 -0.5  -0.1   0.3
b[(Intercept) sex:_Female]                         0.0    0.3 -0.3   0.0   0.3
b[(Intercept) sex:_Male]                           0.0    0.3 -0.4   0.0   0.3
Sigma[occupation:(Intercept),(Intercept)]          0.3    0.5  0.0   0.2   0.7
Sigma[sex:(Intercept),(Intercept)]                 0.4    1.4  0.0   0.0   0.8

Fit Diagnostics:
           mean   sd   10%   50%   90%
mean_PPD 0.2    0.0  0.2   0.2   0.3

The mean_ppd is the sample average posterior predictive distribution of the
 ↪outcome variable (for details see help('summary.stanreg')).

MCMC diagnostics
                                                  mcse Rhat n_eff
(Intercept)                                       0.0  1.0  1425
age                                               0.0  1.0  4041
education_num                                     0.0  1.0  1032
capital_gain                                      0.0  1.0  3472
hours_per_week                                    0.0  1.0  3684
workclassPrivate                                  0.0  1.0  2311
workclassSelf-employed                            0.0  1.0  2381
educationDoctorate                                0.0  1.0  2505
educationHigh School                              0.0  1.0  1042
educationLess than High School                    0.0  1.0  1163
educationMasters                                  0.0  1.0  3147
educationOther                                    0.0  1.0  1592
educationSome College/Associate                   0.0  1.0  1214
marital_status Not Married                        0.0  1.0  2530
relationshipOther/Nonfamily                       0.0  1.0  2411
relationshipSpouse/Partner                        0.0  1.0  2066
```

```
b[(Intercept) occupation:Management/Professional] 0.0  1.0  1359
b[(Intercept) occupation:Manual/Service]          0.0  1.0  1240
b[(Intercept) occupation:Office/Clerical_&_Sales] 0.0  1.0  1409
b[(Intercept) occupation:Other]                   0.0  1.0  1641
b[(Intercept) sex:_Female]                        0.0  1.0  1374
b[(Intercept) sex:_Male]                          0.0  1.0  1187
Sigma[occupation:(Intercept),(Intercept)]         0.0  1.0  1281
Sigma[sex:(Intercept),(Intercept)]                0.1  1.0   218
mean_PPD                                          0.0  1.0  2980
log-posterior                                     0.1  1.0  1134

For each parameter, mcse is Monte Carlo standard error, n_eff is a crude measure
 ↪of effective sample size, and Rhat is the potential scale reduction factor on
 ↪split chains (at convergence Rhat=1).
```

This is a big improvement from before as only 18 divergent transitions errors are left and overall rhat, mcse, and effective sample size seems reasonable. But we will try to tune the model further to reduce divergent transitions and also because here group level effects for sex are near 0.

[71]:
```
#group level effects for sex variable were 0 in previous fit most likely
 ↪because its variability has already been explained across different
#predictors and their level, we will remove it and only use occupation as group
 ↪level prior in third fit

fit_3 <- stan_glmer(formula = income_group ~ age + education_num + capital_gain
 ↪+ hours_per_week +
           workclass + education + marital_status + relationship + (1 |
 ↪occupation) +
           sex, data = df_small, family = binomial(link = 'logit'),
           prior = student_t(df = 7, 0, 2.5),    # using more informative
 ↪student t prior
           prior_intercept = normal(location = 0, scale = 10),
           chains = 3, iter = 2000, cores = parallel::detectCores(), seed = 123
)
summary(fit_3)
```

```
Model Info:
 function:     stan_glmer
 family:       binomial [logit]
 formula:      income_group ~ age + education_num + capital_gain +
 ↪hours_per_week +
           workclass + education + marital_status + relationship + (1 |
           occupation) + sex
 algorithm:    sampling
 sample:       1500 (posterior sample size)
 priors:       see help('prior_summary')
 observations: 2000
```

```
 groups:       occupation (4)

Estimates:
                                                   mean   sd   10%   50%   90%
(Intercept)                                       -2.2    0.8 -3.3  -2.2  -1.2
age                                                0.3    0.1  0.2   0.3   0.4
education_num                                      0.5    0.2  0.3   0.5   0.8
capital_gain                                       0.9    0.1  0.7   0.9   1.0
hours_per_week                                     0.3    0.1  0.2   0.3   0.4
workclassPrivate                                   0.2    0.2 -0.1   0.2   0.4
workclassSelf-employed                             0.0    0.3 -0.3   0.0   0.3
educationDoctorate                                 0.5    0.7 -0.4   0.5   1.4
educationHigh School                              -0.4    0.4 -1.0  -0.4   0.1
educationLess than High School                    -0.7    1.0 -2.0  -0.7   0.5
educationMasters                                   0.6    0.3  0.2   0.6   1.0
educationOther                                     0.6    0.6 -0.1   0.6   1.4
educationSome College/Associate                   -0.4    0.3 -0.7  -0.4   0.0
marital_status Not Married                        -1.6    0.5 -2.2  -1.6  -0.9
relationshipOther/Nonfamily                        1.1    0.5  0.6   1.1   1.8
relationshipSpouse/Partner                         1.9    0.7  1.1   1.9   2.8
sex Male                                          -0.1    0.2 -0.3  -0.1   0.2
b[(Intercept) occupation:Management/Professional]  0.4    0.3  0.1   0.4   0.8
b[(Intercept) occupation:Manual/Service]          -0.2    0.3 -0.6  -0.2   0.1
b[(Intercept) occupation:Office/Clerical_&_Sales] -0.1    0.3 -0.4  -0.1   0.3
b[(Intercept) occupation:Other]                   -0.1    0.4 -0.5  -0.1   0.3
Sigma[occupation:(Intercept),(Intercept)]          0.3    0.6  0.0   0.2   0.7

Fit Diagnostics:
           mean   sd   10%   50%   90%
mean_PPD 0.2    0.0  0.2   0.2   0.3

The mean_ppd is the sample average posterior predictive distribution of the
 ↪outcome variable (for details see help('summary.stanreg')).

MCMC diagnostics
                                                  mcse Rhat n_eff
(Intercept)                                       0.0  1.0   939
age                                               0.0  1.0  1669
education_num                                     0.0  1.0   603
capital_gain                                      0.0  1.0  1974
hours_per_week                                    0.0  1.0  1908
workclassPrivate                                  0.0  1.0  1580
workclassSelf-employed                            0.0  1.0  1401
educationDoctorate                                0.0  1.0  1597
educationHigh School                              0.0  1.0   672
educationLess than High School                    0.0  1.0   742
educationMasters                                  0.0  1.0  1266
```

```
educationOther                                   0.0  1.0  1483
educationSome College/Associate                  0.0  1.0   683
marital_status Not Married                       0.0  1.0  1354
relationshipOther/Nonfamily                      0.0  1.0  1276
relationshipSpouse/Partner                       0.0  1.0  1156
sex Male                                         0.0  1.0  1829
b[(Intercept) occupation:Management/Professional] 0.0  1.0   559
b[(Intercept) occupation:Manual/Service]         0.0  1.0   511
b[(Intercept) occupation:Office/Clerical_&_Sales] 0.0  1.0   507
b[(Intercept) occupation:Other]                  0.0  1.0   564
Sigma[occupation:(Intercept),(Intercept)]        0.0  1.0   651
mean_PPD                                         0.0  1.0  1311
log-posterior                                    0.1  1.0   556

For each parameter, mcse is Monte Carlo standard error, n_eff is a crude measure
 ↪of effective sample size, and Rhat is the potential scale reduction factor on
 ↪split chains (at convergence Rhat=1).
```

We fixed divergent transitions error in this model so we will use it. Besides that model seems to be indeed a better fit overall from previous ones.

[105]:
```
set.seed(123)
#training the model with 70% of the data

train_index <- sample(1:nrow(df_bm2), size = 0.7*nrow(df_bm2))
df_bm_train <- df_bm2[train_index,]
df_bm_test <- df_bm2[-train_index,]
```

[106]:
```
#checking the size for datasets

dim(df_bm_train)
dim(df_bm_test)
```

1. 20955 2. 13

1. 8981 2. 13

[133]:
```
#fitting final bayesian hierarchical model

fit_bm <- stan_glmer(formula = income_group ~ age + education_num +
 ↪capital_gain + hours_per_week +
          workclass + education + marital_status + relationship + sex + (1 |
 ↪occupation)
          , data = df_bm_train, family = binomial(link = 'logit'),
          prior = student_t(df = 6, 0, 2.5),   # using slightly more
 ↪informative prior than before
          prior_intercept = normal(location = 0, scale = 10),
          chains = 3, iter = 2000, cores = parallel::detectCores(), seed = 123
 ↪
```

```
          # weights = df_gm$fnlwgt[train_index] #we also tested the model with␣
↪weights, it did indeed improve our analysis but it takes a lot of time
          #to fit them, 3+ hours, and it also adds an extra layer of␣
↪complexity which we could avoid
)
```

```
Warning message:
"There were 3 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them."
Warning message:
"Examine the pairs() plot to diagnose sampling problems
"
```

There is only 3 divergent error appearing, we have fitted the model multiple times and most of the time it got 0 divergent transition like we showed in previous fit so its a random effect at this level. We can definitely remove some predictors and get 0 divergent errors all the time but we will lose some model capabilities as we saw from GAM model all the variables do contribute to the relationship to some level. Moreover, the posterior predictions for all the models were same along with accuracy/f1 score regardless of the random 3 divergent transitions we are getting in this kernel run randomly.

Besides from the summary below we have proper rhat, n_eff, and other metrics so this model is fine for further analysis.

[134]:
```
summary(fit_bm)
```

```
Model Info:
 function:     stan_glmer
 family:       binomial [logit]
 formula:      income_group ~ age + education_num + capital_gain +␣
↪hours_per_week +
           workclass + education + marital_status + relationship + sex +
           (1 | occupation)
 algorithm:    sampling
 sample:       3000 (posterior sample size)
 priors:       see help('prior_summary')
 observations: 20955
 groups:       occupation (4)

Estimates:
                                                    mean   sd   10%   50%   90%
(Intercept)                                        -3.4    0.5 -3.9  -3.4  -2.8
age                                                 0.4    0.0  0.4   0.4   0.4
education_num                                       0.6    0.1  0.5   0.6   0.7
capital_gain                                        0.8    0.0  0.7   0.8   0.8
hours_per_week                                      0.4    0.0  0.3   0.4   0.4
```

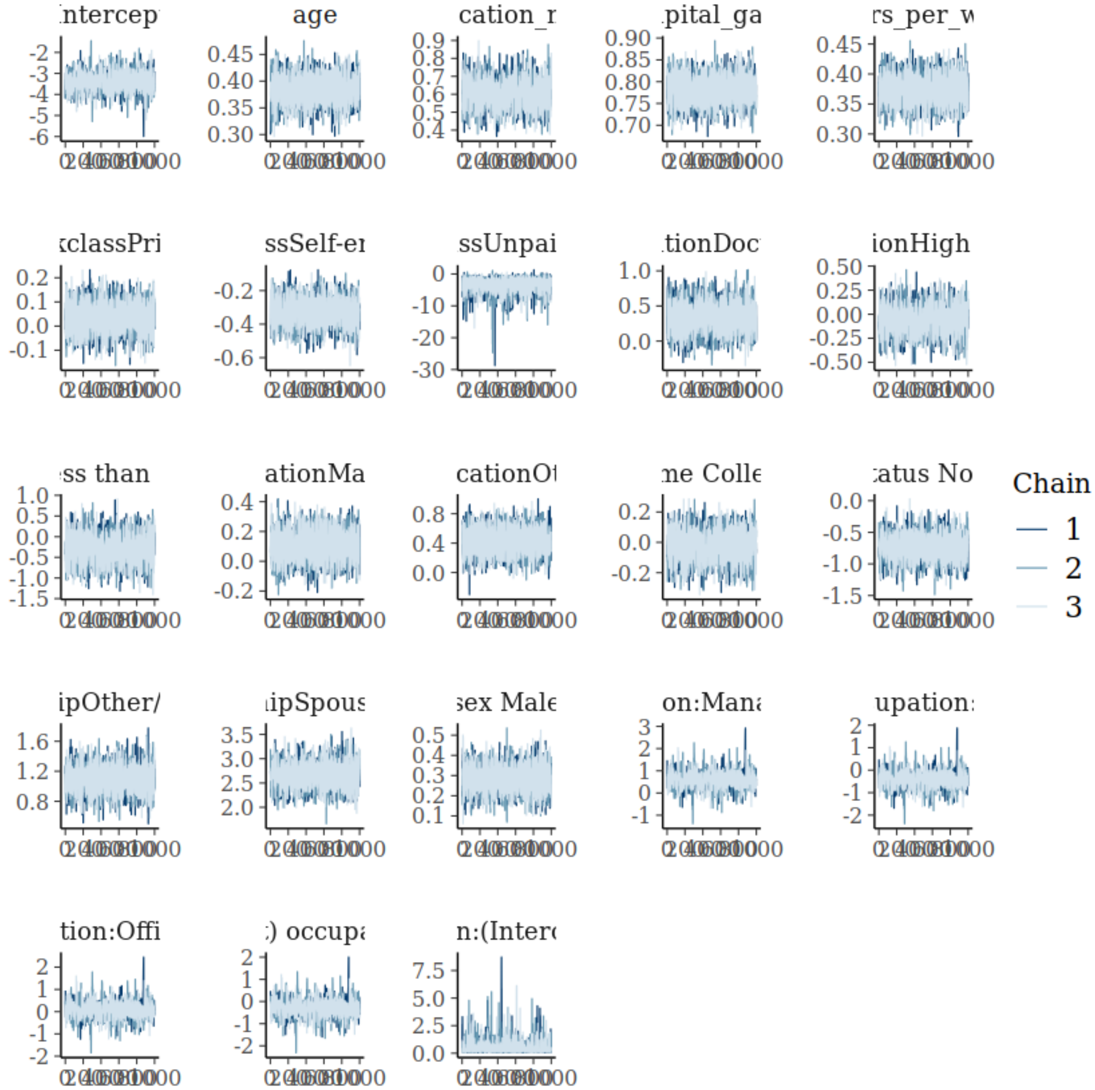
```
workclassPrivate                                0.0    0.1  0.0   0.0   0.1
workclassSelf-employed                         -0.3    0.1 -0.4  -0.3  -0.2
workclassUnpaid/Other                          -3.4    2.3 -6.2  -3.0  -1.2
educationDoctorate                              0.3    0.2  0.1   0.3   0.6
educationHigh School                           -0.1    0.1 -0.3  -0.1   0.1
educationLess than High School                 -0.3    0.3 -0.7  -0.3   0.1
educationMasters                                0.1    0.1  0.0   0.1   0.2
educationOther                                  0.4    0.2  0.2   0.4   0.7
educationSome College/Associate                 0.0    0.1 -0.2   0.0   0.1
marital_status Not Married                     -0.7    0.2 -1.0  -0.7  -0.5
relationshipOther/Nonfamily                     1.1    0.2  0.9   1.1   1.3
relationshipSpouse/Partner                      2.7    0.3  2.3   2.7   3.0
sex Male                                        0.3    0.1  0.2   0.3   0.4
b[(Intercept) occupation:Management/Professional]  0.6    0.3  0.2   0.6   1.0
b[(Intercept) occupation:Manual/Service]       -0.4    0.4 -0.8  -0.4   0.0
b[(Intercept) occupation:Office/Clerical_&_Sales]  0.1    0.3 -0.3   0.1   0.5
b[(Intercept) occupation:Other]                -0.3    0.4 -0.7  -0.3   0.1
Sigma[occupation:(Intercept),(Intercept)]       0.5    0.6  0.1   0.3   1.1

Fit Diagnostics:
           mean   sd   10%   50%   90%
mean_PPD 0.2    0.0  0.2   0.2   0.3

The mean_ppd is the sample average posterior predictive distribution of the
 outcome variable (for details see help('summary.stanreg')).

MCMC diagnostics
                                                  mcse Rhat n_eff
(Intercept)                                       0.0  1.0   901
age                                               0.0  1.0  4176
education_num                                     0.0  1.0  1050
capital_gain                                      0.0  1.0  4654
hours_per_week                                    0.0  1.0  4057
workclassPrivate                                  0.0  1.0  2858
workclassSelf-employed                            0.0  1.0  2892
workclassUnpaid/Other                             0.1  1.0  2051
educationDoctorate                                0.0  1.0  2558
educationHigh School                              0.0  1.0  1097
educationLess than High School                    0.0  1.0  1230
educationMasters                                  0.0  1.0  3025
educationOther                                    0.0  1.0  2868
educationSome College/Associate                   0.0  1.0  1242
marital_status Not Married                        0.0  1.0  2353
relationshipOther/Nonfamily                       0.0  1.0  2462
relationshipSpouse/Partner                        0.0  1.0  1820
sex Male                                          0.0  1.0  3935
b[(Intercept) occupation:Management/Professional] 0.0  1.0   797
```

```
b[(Intercept) occupation:Manual/Service]         0.0  1.0   779
b[(Intercept) occupation:Office/Clerical_&_Sales] 0.0  1.0   785
b[(Intercept) occupation:Other]                  0.0  1.0   794
Sigma[occupation:(Intercept),(Intercept)]        0.0  1.0  1068
mean_PPD                                         0.0  1.0  2942
log-posterior                                    0.1  1.0   980

For each parameter, mcse is Monte Carlo standard error, n_eff is a crude measure
 ↪of effective sample size, and Rhat is the potential scale reduction factor on
 ↪split chains (at convergence Rhat=1).
```

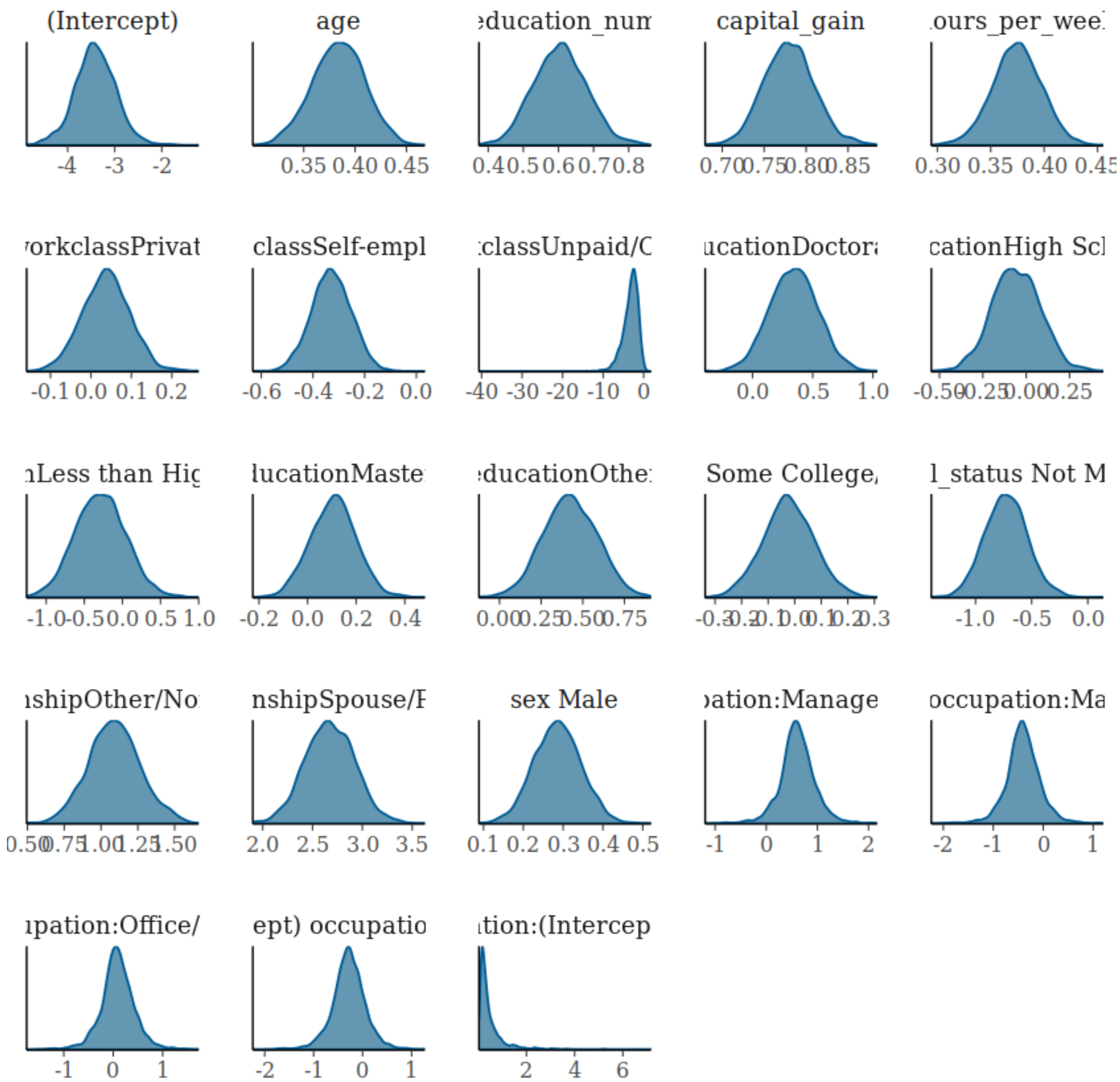
We fitted a full model besides race and we alloed occupation have the hierarchical priors. We have 0 to little effect divergent errors in this model so its good model compared to various others we tested.

In bayesian model we have posterior predictive distributions for all the predictors and even if we can't form population estimate specficially its still useful in explaining relationship across the factors and resposne: 1. Generally model seems to be a good fit with controllable or minimal sd for all the predictors. MCSE or Monte Carlo Standard Errors are also 0 for most of the predictors with 0.1 being maximum for worclass other/unpaid which is understandable as it has low number of observations. 2. Intercept here is log odds for observation quite similar to additive model with similar probability of 0.032 attributing to a married female with bachelors education. But since we allowed hierarchical priors for the occuption we can use this information, and indeed as per bayesian model for same set of predictors we got probability of 0.05 for the person falling in high income category. 3. There are some sub groups or levels with no effect on the baseline estimate like Working Privately, in additive model too it was not impactful as per the model and this is an interesting finding overall, that compared to government private employment does not increase in likelhood that a person can have higher income during that time in 1994. 4. Education categories seems to be adding to the odds properly with bachelors as baseline. As more education is added log odds are also increasing. 5. Numerical variables are fitted properly here with good effect on mean overall. 6. We allowed baseline odds to vary across the Occupation levels and indeed we can see impact of this as professional and clerical/sales occupation add to the log odds overall.

[135]:
```
#Now we will perform convergence tests

plot(fit_bm, plotfun = "trace")
```



```
[126]: plot(fit_bm, plotfun = "dens")
```



Traceplots shows clear convergence and density plotss are fine too with most of the predictors centered around 0 with a small variance.

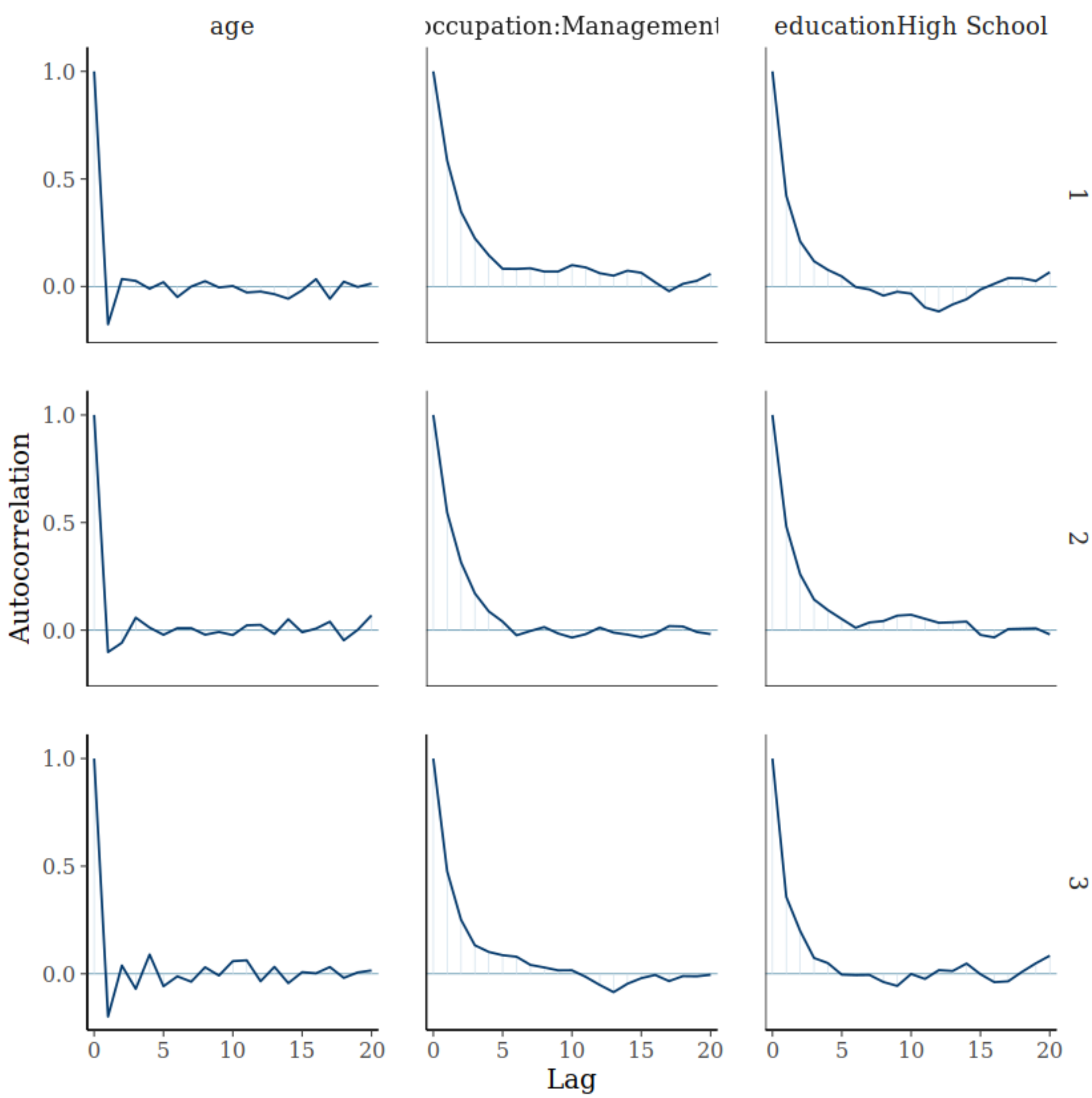
[136]:
```
rhat(fit_bm)
```

(Intercept) 1.00084443217634 **age** 0.999809181050913 **education__num** 1.00179433189152 **capital__gain** 0.99987860288949 **hours__per__week** 0.999682848289446 **workclassPrivate** 0.999281040928417 **workclassSelf-employed** 0.999799670694952 **workclassUnpaid/Other** 1.00276377237268 **educationDoctorate** 1.00046476552193 **educationHigh School** 1.00247524386339 **educationLess than High School** 1.00196742630757 **educationMasters** 1.00095290746393 **educationOther** 1.00041559557237 **educationSome College/Associate** 1.00155706199084 **marital__status Not Married** 0.999677233861531 **relationshipOther/Nonfamily** 0.999058566947009 **relationshipSpouse/Partner** 0.999370783533312 **sex Male** 0.999700534967139 **b{[}(Intercept)**

occupation:Management/Professional{]} 1.00169874766041 **b{[}(Intercept) occupation:Manual/Service{]}** 1.00157235985017 **b{[}(Intercept) occupation:Office/Clerical_\&_Sales{]}** 1.00197417889505 **b{[}(Intercept) occupation:Other{]}** 1.0016997531775 **Sigma{[}occupation:(Intercept),(Intercept){]}** 0.999840871526285

R Hat value is around 1 for all the predictors implying that all the chains have converged properly. Overall even with above few metrics we can be quite sure that our bayesian model have created good chains for our analysis.

[137]:
```
mcmc_acf(as.array(fit_bm), pars = c("age", "b[(Intercept) occupation:Management/
↪Professional]", 'educationHigh School'), lags = 20)
```



[140]:
```
autocorr.diag(as.mcmc(as.matrix(fit_bm)))
```

A matrix: 5 × 23 of type dbl

	(Intercept)	age	education_num	capital_gain	hours_per_
Lag 0	1.000000000	1.00000000	1.0000000000	1.00000000	1.000000000
Lag 1	0.430024848	-0.15816733	0.4473283244	-0.21085583	-0.16263189
Lag 5	0.069269521	-0.01859894	0.0363415104	-0.01439364	0.026479401
Lag 10	0.028792131	0.01390960	0.0109456361	0.02143026	-0.00630439
Lag 50	0.001449844	-0.03192735	0.0004870541	0.01957865	0.008035997

Above are auto correlation plots and table for some predictors as we can't plot all of them, but generally all of them have low autocorrelation and at around lag of 5-10 all of the predictors have near zero autocorrelation with 0th value. This implies that bayesian model is producing chains with good information overall.

[138]:
```
#HPD for the fitted model
HPDinterval(as.mcmc(as.matrix(fit_bm)))
```

A matrix: 23 × 2 of type dbl

	lower	upper
(Intercept)	-4.27411373	-2.50629723
age	0.33692633	0.43285796
education_num	0.44531071	0.75201459
capital_gain	0.72382593	0.84408102
hours_per_week	0.32754487	0.41771275
workclassPrivate	-0.08006447	0.15052487
workclassSelf-employed	-0.47464443	-0.17253378
workclassUnpaid/Other	-7.76769435	-0.09870139
educationDoctorate	-0.06117520	0.73915139
educationHigh School	-0.34416069	0.23839471
educationLess than High School	-0.95110282	0.35415501
educationMasters	-0.07855185	0.29267394
educationOther	0.08640692	0.76376469
educationSome College/Associate	-0.22686118	0.16298631
marital_status Not Married	-1.09455574	-0.32522887
relationshipOther/Nonfamily	0.75526555	1.43396421
relationshipSpouse/Partner	2.17092193	3.16132920
sex Male	0.16017974	0.40589707
b[(Intercept) occupation:Management/Professional]	-0.12787326	1.28051684
b[(Intercept) occupation:Manual/Service]	-1.10128126	0.31213377
b[(Intercept) occupation:Office/Clerical_&_Sales]	-0.61911013	0.77136572
b[(Intercept) occupation:Other]	-0.97573533	0.43591261
Sigma[occupation:(Intercept),(Intercept)]	0.04244746	1.64038093

[139]:
```
effectiveSize(as.mcmc(as.matrix(fit_bm)))
```

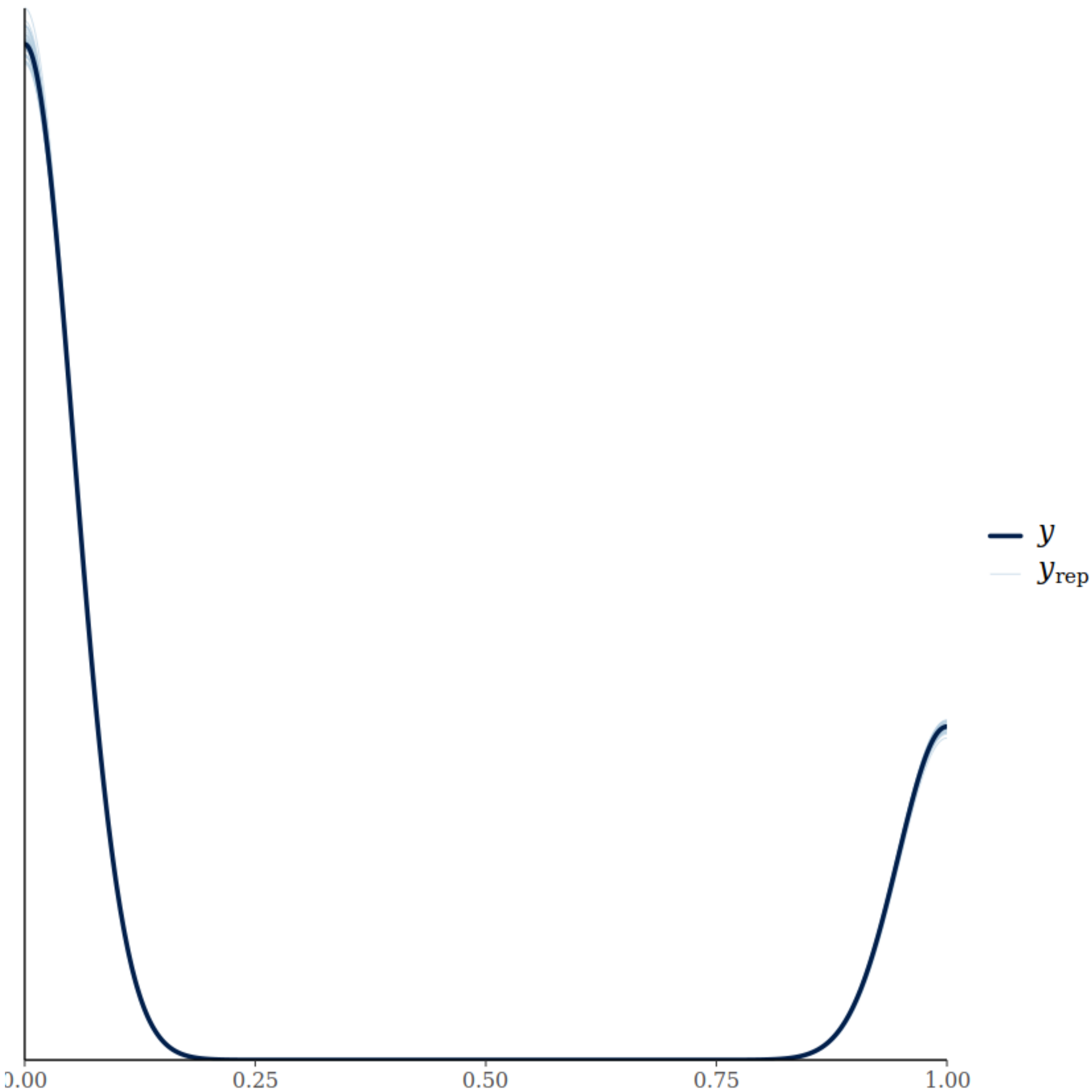
(Intercept) 1043.432007222 **age** 4125.93089541038 **education_num** 1046.36184559946 **capital_gain** 4601.63851627629 **hours_per_week** 4163.91864250986 **workclassPrivate** 3001.86626012807 **workclassSelf-employed** 2950.22928485037 **workclassUnpaid/Other** 2034.85122316487 **educationDoctorate** 2861.11047826482 **educationHigh School** 1094.39516318017 **educationLess than High School** 1227.31752729083 **educationMasters** 3045.90238638773 **educationOther** 2945.40604635185 **educationSome College/Associate** 1233.68365653286 **marital_status Not Married** 2454.67983328488

relationshipOther/Nonfamily 2438.548899392 **relationshipSpouse/Partner** 1902.99140308948 **sex Male** 4027.54062208811 **b{[}(Intercept) occupation:Management/Professional{]}** 899.541550534613 **b{[}(Intercept) occupation:Manual/Service{]}** 886.693660700589 **b{[}(Intercept) occupation:Office/Clerical_\&_Sales{]}** 849.019067541313 **b{[}(Intercept) occupation:Other{]}** 868.471037787417 **Sigma{[}occupation:(Intercept),(Intercept){]}** 1080.66659854549

Effective sample sizes are good overall. We took 3 chains with 1000 iterations for burnin and 1000 iterations for recording, most of the variables have effective sample size across 1000-5000 where 1000 is considered good size generally in practice. For some we have less than 1000 but they are close to 1000 nevertheless. Benefit of all this is that we have high number of independent draws across all the variables, and this was expected because of low autocorrelation.

Overall from above tests, plots, and summary observations we can be confirmed that the model has converged properly with 3 chains at 2000 iterations. We can run smaller chains too but since stan_glmer by default use 1000 for burn it we are left with 1000 for our analysis so atleast 2000 iterations should be minimum for this dataset.

[141]:
```
pp_check(fit_bm)
```



[ ]:

[142]:
```
colMeans(as.mcmc(as.matrix(fit_bm)))
```

(Intercept) -3.38251143728929 **age** 0.383443598706606 **education__num** 0.603480911891468 **capital__gain** 0.77935838684509 **hours__per__week** 0.373972470415425 **workclassPrivate** 0.036639926745194 **workclassSelf-employed** -0.330312185102112 **workclassUnpaid/Other** -3.44709222171144 **educationDoctorate** 0.345486839399312 **educationHigh School** -0.0606884735148477 **educationLess than High School** -0.29746580396587 **educationMasters** 0.102807648993801 **educationOther** 0.424631668522416 **educationSome College/Associate** -0.0266316978134757 **marital__status Not Married** -0.736036325718673 **relationshipOther/Nonfamily** 1.08922517011035 **relationshipSpouse/Partner** 2.66343359987772 **sex Male** 0.286791761418874 **b{[}(Intercept)**

occupation:Management/Professional{]} 0.597469148493296 **b{[}(Intercept)**
occupation:Manual/Service{]} -0.413234929166908 **b{[}(Intercept)**
occupation:Office/Clerical_\&_Sales{]} 0.0792504772399275 **b{[}(Intercept)**
occupation:Other{]} -0.2910269141275 **Sigma{[}occupation:(Intercept),(Intercept){]}**
0.504309377902313

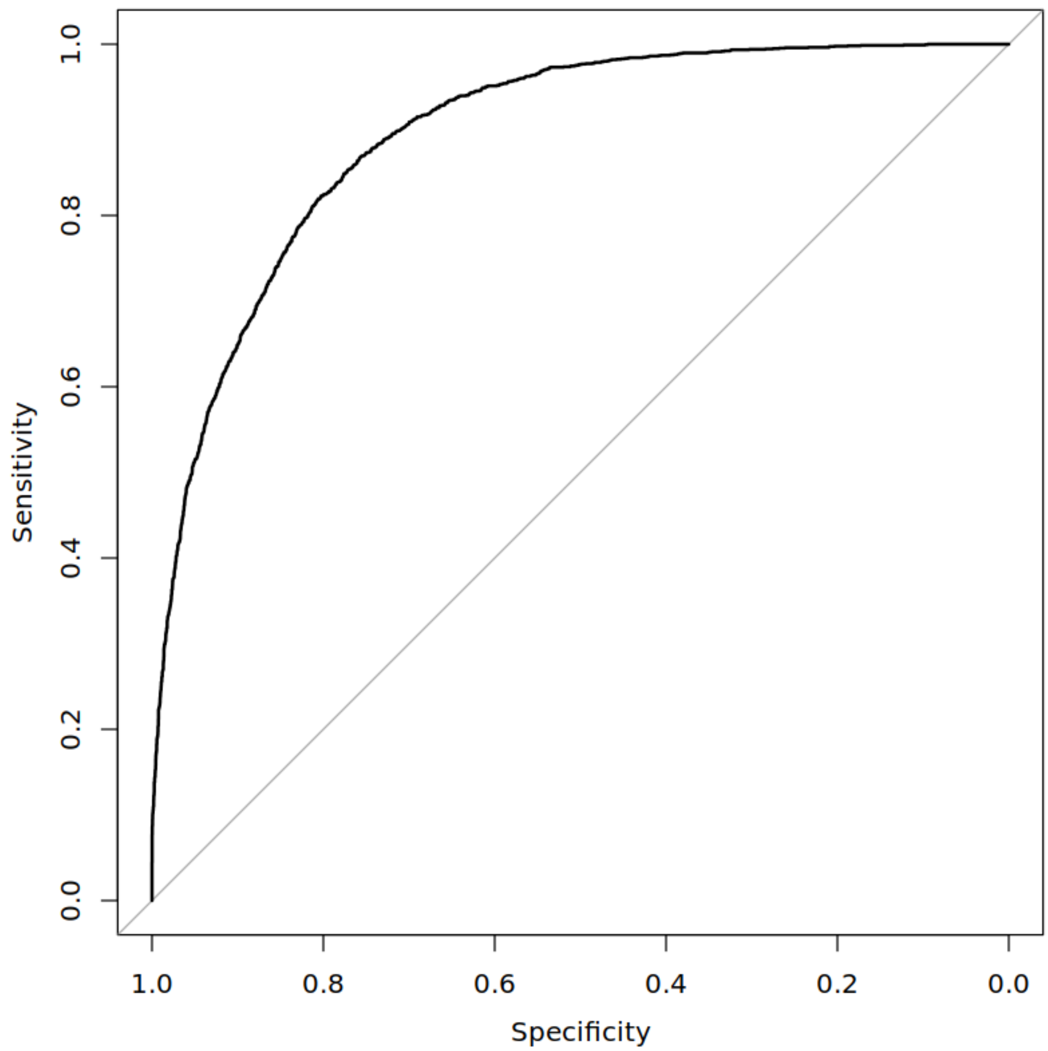
[181]:
```
# Predict on train set
train_probs <- posterior_linpred(fit_bm, newdata = df_bm_train, transform =␣
↪TRUE)
train_pred <- ifelse(colMeans(train_probs) > 0.5, 1, 0)
train_acc <- mean(train_pred == df_bm_train$income_group)

# Predict on test set
test_probs <- posterior_linpred(fit_bm, newdata = df_bm_test, transform = TRUE)
test_pred <- ifelse(colMeans(test_probs) > 0.5, 1, 0)
test_acc <- mean(test_pred == df_bm_test$income_group)

cat("Train Accuracy:", train_acc, "\n")
cat("Test Accuracy:", test_acc, "\n")
```

```
Instead of posterior_linpred(…, transform=TRUE) please call posterior_epred(),
which provides equivalent functionality.

Instead of posterior_linpred(…, transform=TRUE) please call posterior_epred(),
which provides equivalent functionality.


Train Accuracy: 0.8384634
Test Accuracy: 0.8467877
```

[185]:
```
pred_probs <- posterior_linpred(fit_bm, newdata = df_bm_test, transform = TRUE)
pred_mean <- colMeans(pred_probs)

roc_obj <- roc(df_bm_test$income_group, pred_mean)
auc(roc_obj)
plot(roc_obj)
```

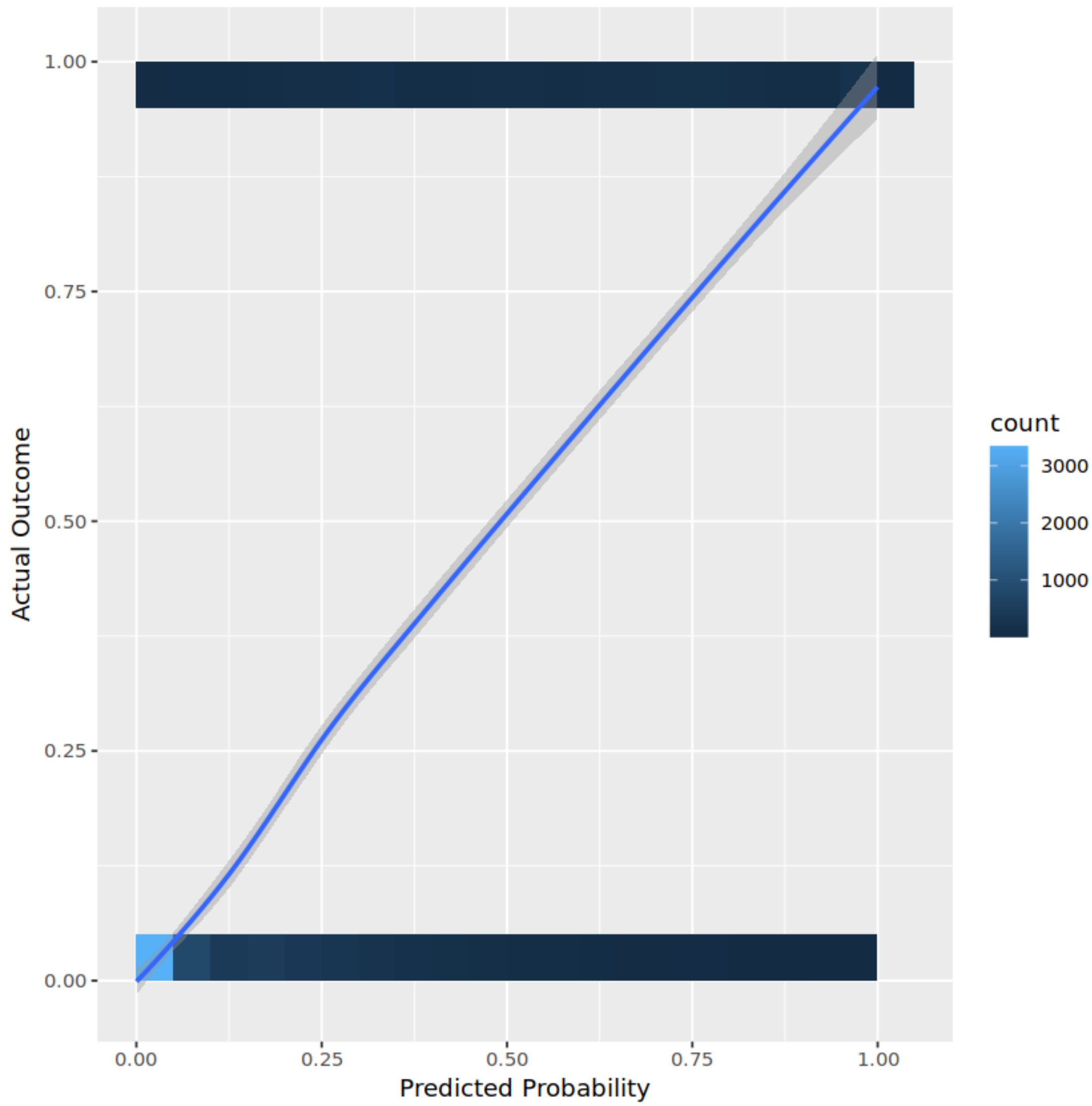
```
Instead of posterior_linpred(…, transform=TRUE) please call posterior_epred(),
which provides equivalent functionality.

Setting levels: control = 0, case = 1

Setting direction: controls < cases
```

0.894996646595153

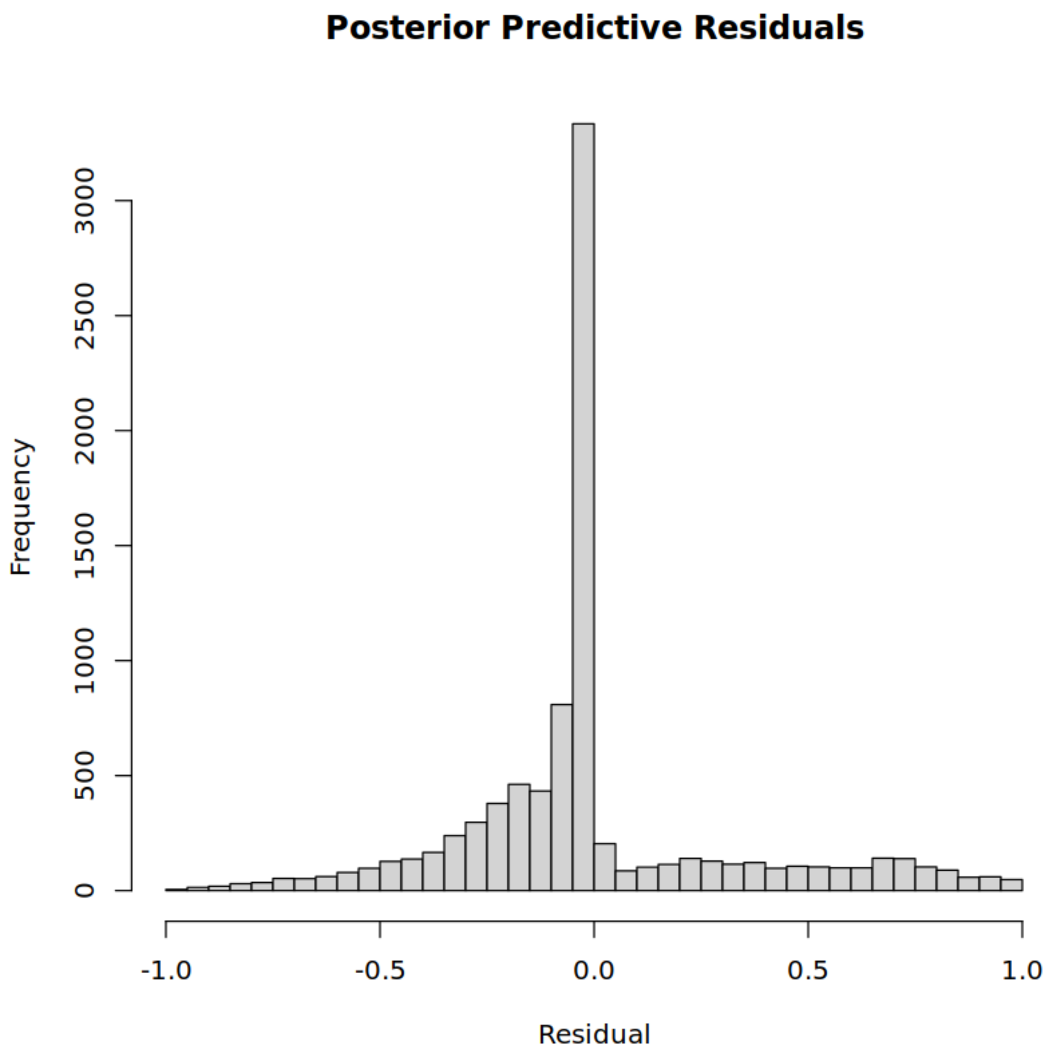


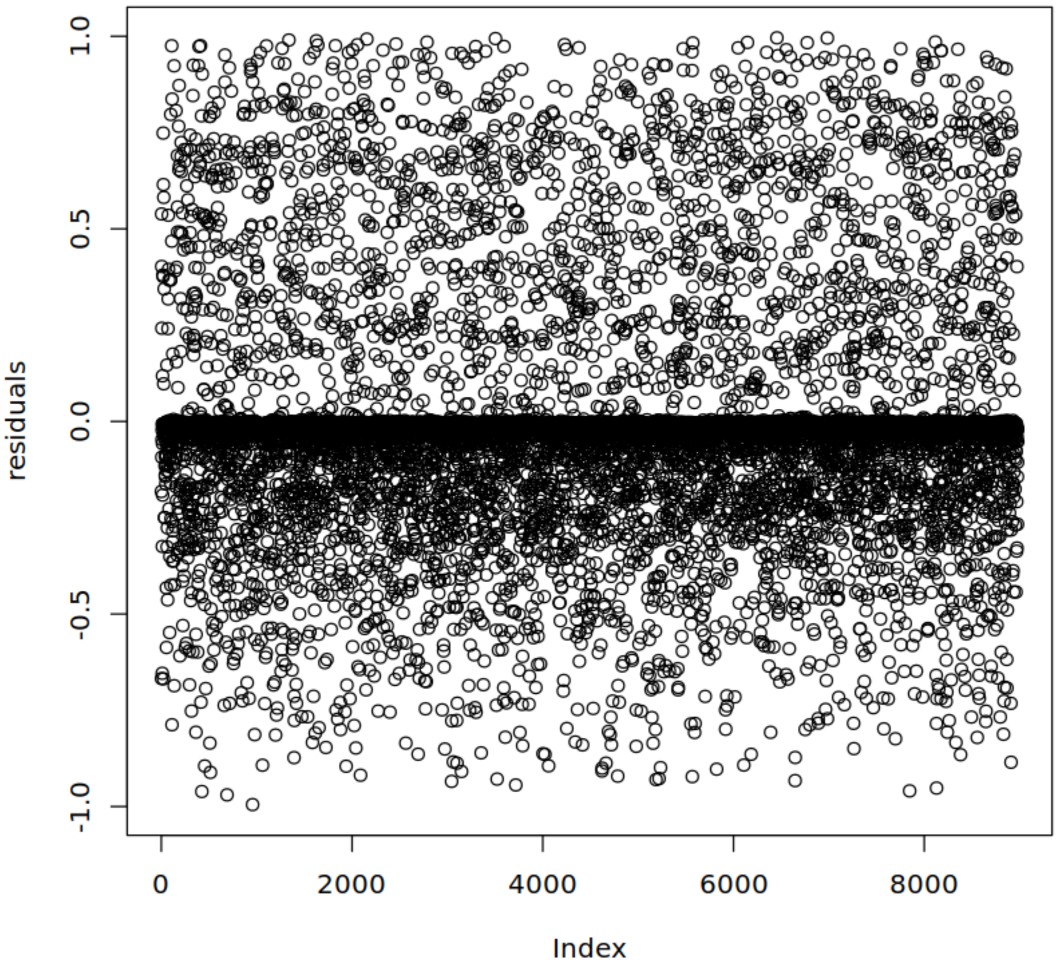
[186]:
```
#predictive probabilites are reliable
df_calib <- data.frame(
  pred = pred_mean,
  actual = df_bm_test$income_group
)
ggplot(df_calib, aes(x = pred, y = actual)) +
  geom_bin2d(bins = 20) +
  geom_smooth(method = "loess") +
  labs(x = "Predicted Probability", y = "Actual Outcome")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



[187]:
```
# Get posterior predictive draws for the test set
yrep <- posterior_predict(fit_bm, newdata = df_bm_test)
# Calculate mean predicted value for each observation
yrep_mean <- colMeans(yrep)
# Calculate residuals
residuals <- df_bm_test$income_group - yrep_mean

# Plot residuals
hist(residuals, breaks = 30, main = "Posterior Predictive Residuals", xlab =␣
 ↪"Residual")
plot(residuals)
```


Posterior Predictive Residuals
Frequency
0
500
1000
1500
2000
2500
3000
-1.0
-0.5
0.0
0.5
1.0
Residual



Residuals are cenetered around 0 with a spread. Spread does not have any specific pattern and no major outliers exist outside the spread. Overall, residuals look healthy.

[192]:
```
# True/False Positives/Negatives
TP <- sum(test_pred == 1 & df_bm_test$income_group == 1)
FP <- sum(test_pred == 1 & df_bm_test$income_group == 0)
FN <- sum(test_pred == 0 & df_bm_test$income_group == 1)

precision <- TP / (TP + FP)
recall <- TP / (TP + FN)
F1 <- 2 * (precision * recall) / (precision + recall)

cat("F1 Score:", F1, "\n")
```

```
F1 Score: 0.6396019
```

[189]:
```
fit_matrix <- as.matrix(fit_bm)
cat('Proportion of population having income greater than 50 K:
↪',round(mean(train_pred),2),'\n')
```

```
Proportion of population having income greater than 50 K: 0.2
```

[190]:
```
# For a new person
newdata <- data.frame(
  age = 0,  # mean if scaled
  education_num = 0,  # mean if scaled
  capital_gain = 0,
  hours_per_week = 0,
  workclass = "Private",
  education = "Bachelors",
  marital_status = " Married",
  relationship = "Spouse/Partner",
  sex = " Female",
  occupation = "Manual/Service"
)

yrep_new <- posterior_predict(fit_bm, newdata = newdata, draws = 1000)
mean_prob <- mean(yrep_new)
# 95% credible interval:
cat("Posterior mean probability:", mean_prob, "\n")
```

```
Posterior mean probability: 0.254
```

Now we will check other proportions across the MCMC sample distribution which we got.

[191]:
```
# Subset test data
df_female <- subset(df_bm_test, sex == " Female")
df_male <- subset(df_bm_test, sex == " Male")

# Get posterior predictive draws for all females
yrep_female <- posterior_predict(fit_bm, newdata = df_female)
yrep_male <- posterior_predict(fit_bm, newdata = df_male)

# For each posterior draw, calculate the proportion of high income
prop_high_income_draws_female <- rowMeans(yrep_female)
prop_high_income_draws_male <- rowMeans(yrep_male)

prob_gt_20 <- mean(prop_high_income_draws_female >= 0.441)
cat("Posterior probability that proportion of women having high income >= 44.1%:
↪", prob_gt_20, "\n")
cat('Posterior Mean for Women having high income:',␣
↪mean(prop_high_income_draws_female),"\n")
```

```
cat('Posterior Mean for Men having high income:',␣
 ↪mean(prop_high_income_draws_male),"\n")
cat('Posterior Mean for Men having high income compare to women:',␣
 ↪mean(prop_high_income_draws_male > prop_high_income_draws_female))
```

```
Posterior probability that proportion of women having high income >= 44.1%: 0
Posterior Mean for Women having high income: 0.1076074
Posterior Mean for Men having high income: 0.3073083
Posterior Mean for Men having high income compare to women: 1
```

[145]:
```
#checking privately employed employees proportion
df_private <- subset(df_bm_test, workclass == "Private")
df_govt <- subset(df_bm_test, workclass == "Government")

yrep_private <- posterior_predict(fit_bm, newdata = df_private)
yrep_govt <- posterior_predict(fit_bm, newdata = df_govt)

# For each posterior draw, calculate the proportion of high income
prop_high_income_draws_private <- rowMeans(yrep_private)
prop_high_income_draws_govt <- rowMeans(yrep_govt)

cat('Posterior Mean for People employed in Private Sector having high income:',␣
 ↪mean(prop_high_income_draws_private), '\n')
cat('Posterior Mean for People employed in Government Sector having high income:
 ↪', mean(prop_high_income_draws_govt))
```

```
Posterior Mean for People employed in Private Sector having high income:
0.210934
Posterior Mean for People employed in Government Sector having high income:
0.3085451
```

[171]:
```
#checking posterior mean for people with age equal to or greater than 38 (mean␣
 ↪age) and education_num greater than 10(mean education year)
#having income_group greater than 50K USD per annum

df_age <- df_bm_test[df_bm_test$age >= 0 & df_bm_test$education_num >= 0,]

cat('Posterior Mean for People having age greater than 38 and education years␣
 ↪greater than 10 having high income:',␣
 ↪mean(colMeans(posterior_predict(fit_bm, new_data = df_age))))
```

```
Posterior Mean for People having age greater than 38 and education years greater
than 10 having high income: 0.2470878
```

In Bayesian Model here we got the stationary distribution with which we can have even more indepth analysis than shown above. But it is to be noted that since this model is not including weights into considerations sub groups having low number of observations in the sample will not be having good posterior prediction with this model, but for all other cases there should not be any problem as we saw before in weight analysis in EDA and Linear Model section that while weights can improve the model besides the issues with subgroups, all other proportions and other findings like Accuracy, F Scores, and others are same for both weight and not weighted models.

1.4 Neural Network

In this section we will git a basic neurel network. For more complex neurel network Tensorflow/Pytorch seems to be a better fit while running them on python. Still, a basic neurel network will be enough for us to compare the models in this project.

For the linear model we will cover all the variables besides fmlwgt as linear model does not have issue with more predictors like Linear Model and Bayesian Models have

[864]:
```
df <-  subset(data, select = -fnlwgt)
df$income_group <- ifelse(df$income_group == " >50K", 1, 0)
dummies <- dummyVars(income_group ~ ., data = df)
df_nn <- as.data.frame(predict(dummies, newdata = df))
num_cols <- sapply(df_nn, is.numeric)
df_nn[num_cols] <- scale(df_nn[num_cols])
df_nn$income_group <- as.factor(df$income_group)

head(df_nn)
```

A data.frame: 6 × 104

	age <dbl>	workclass Federal-gov <dbl>	workclass Local-gov <dbl>	workclass Private <dbl>	workc <dbl
1	0.04659375	-0.1801466	-0.2716969	-1.6888387	-0.188
2	0.88419263	-0.1801466	-0.2716969	-1.6888387	-0.188
3	-0.02955160	-0.1801466	-0.2716969	0.5921031	-0.188
4	1.11262868	-0.1801466	-0.2716969	0.5921031	-0.188
5	-0.79100512	-0.1801466	-0.2716969	0.5921031	-0.188
6	-0.10569695	-0.1801466	-0.2716969	0.5921031	-0.188

[865]:
```
set.seed(123)
train_idx <- sample(1:dim(df_nn)[1], size = 0.8*dim(df_nn)[1])
train_nn <- df_nn[train_idx, ]
test_nn <- df_nn[-train_idx, ]
```

[855]:
```
head(train_nn)
```

A data.frame: 6 × 104

	age <dbl>	workclass Federal-gov <dbl>	workclass Local-gov <dbl>	workclass Private <dbl>	wo <
20485	1.4933554	-0.1801466	-0.2716969	-1.6888387	-0
20537	0.3511752	-0.1801466	3.6804495	-1.6888387	-0
29145	1.3410647	-0.1801466	-0.2716969	0.5921031	-0
27293	-0.1818423	-0.1801466	-0.2716969	0.5921031	-0
31390	-0.2579877	-0.1801466	-0.2716969	0.5921031	-0
3275	1.0364833	-0.1801466	3.6804495	-1.6888387	-0

[869]:
```
#fitting a simple neural network with 8 neurons
nn_model <- nnet(income_group ~ ., data = train_nn, size = 8, decay = 0.001,␣
↪trace = FALSE)
```

[871]:
```
pred_probs <- predict(nn_model, newdata = test_nn, type = "raw")
pred_class <- ifelse(pred_probs >= 0.5, 1, 0)
actual <- as.numeric(as.character(test_nn$income_group))
accuracy <- mean(pred_class == actual)
cat("Neural Network Accuracy:", round(accuracy, 4), "\n")
cat('F1 Score for Neural Network:', F1_Score(y_pred = pred_class, y_true =␣
↪actual, positive = "1"))
```

```
Neural Network Accuracy: 0.8562
F1 Score for Neural Network: 0.6732448
```

Neural Network accuracy and F1 score is slightly lower than Generalized Additive Model. Of course we can improve the neural network by using more complex packages such as keras and tensorflow which would be best trained by using a GPU. For this project a simple neurel network is enough and even a simple neurel network of 8 hidden units and 1 layer is giving us comparable predictive results as GAM.

However it is to be noted that while GAM is using smoothing function for some of the variables it still has more explainability power than the Neurel Network here and depending upon the task, reader may prefer using GAM over neural network.

1.5 Conclusion

In this project we covered Census Income data based on US Census 1994 available at UC Irvine ML repo. We first observed the datset and then formed the hypothesis based on the studies and facts known across multiple studies during the 1990-1993 period. After that we performed data cleaning, explantory data analysis, and hypothesis tests.

For the hypothesis tests we used weights to properly form population estimates and test them. However, we rejected all our hypothesis, average number of education year as per census 1994 data was not equal to 11.5 as media claimed, its around 10 years with low variance. Similarly proportion of women having high income was not equal to proportion of women actually in the workforce that is 44.1%, we got an estimate of 14% only, a very big difference. And once again we rejected the hypothesis that proportion of people in US falling in high income group of earning more than 50K USD in 1994 was 15%, we got the estimate at 24%. Considering that our hypothesis was based on 1990-93 data but even then in one year inflation can not be so big that people earning more than 50K USD a year, more than 100K USD per year in 2025 inflation adjusted, increased by more than 50% in a year.

Besides hypothesis tests we also performed various analysis on data itself and found multiple relationships across them like sex, married status, occupation, and age having major influence in deciding if a person would be in high income bracket. After cleaning the data and preparing it for the models we fitted Linear Model specifically to test these relationships, and for some levels we indeed got appropriate positive and negative coefficients which told us the relationships like more education years helping in more income. But we also got some concerning evidence that some predictors provided unfair advatage during that period, for example being a white male in manageriaal position while being married in 1994 would ensure that a person have higher probability of having higher income even if all the factors besides gender or race or both are different, and with huge difference at hat.

After that we formed bayesian hiearchical model with which we were able to MCMC chains and good posterior distributions for multiple predictors while allowing group level priors for occupation vary as it was chosen as Hierarchical prior. Our MCMC Model converged properly and we were able to form multiple posterior predictions like government employees having 31% of having high income compared to 21% for private sector, which is quite different from today. Overall probability for high income group was 20% which is actually lower than 24% we got from frequentist approach and is actually near to our assumed proportion for same period. We could analyze the model further for more interesting finding. Overall Bayesian Model allowed us to obtain Stationary Distribution for our dataset and we got a good usable model with which we can form probabilistic statements about parameters or predictions with credible intervals.

And in the end we fitted a small nueral network and observed that even with a relatively basic and small netwrok neurel network had great predictive capability as cen be observed below.

Model	Accuracy	F1 Score
Generalized Linear Model	0.8656	0.6498
Generalized Additive Model	0.8656	0.6786
Bayesian Hierarchical Model	0.8468	0.6396
Neural Network	0.8562	0.6732

While Neural Network would not be good for inference like additive model or for making probablity statements with credible intervals in bayesial model it worked great for accuracy and F1 scores with both scores quite near our best model for prediction, Generalized Additive Model. We can improve Neural Network a lot by by using more complex neural networks in tensorflow or pytorch.

Overall, in the project we performed a proper statistical analysis for the Census Income Data starting while also focusing on planned hypothesis with proper power analysis. For understanding the relationships across the variables Generalized Additive Model helped us a lot, and by using the additive model we were able to form a linear model which outperformed all other models too. Bayesian Model also helped us to understand the relationships across the data while helping us form direct credible estimates with the stationary distribution captured by Monte Carlo Chains while allowing for hierarchical priors. And in the end we formed a basic neural network to find that even with basic neural network with least amount of work and hyperparameter tuning could provide us predictive capability approximately equal to our best fit model.

1.6 Bibliography

These are external sources we used to collect the data and form the hypothesis.
1. Census Income - https://archive.ics.uci.edu/dataset/20/census+income 2. NCES Digest of educational studies - https://nces.ed.gov/pubs91/91660.pdf 3. World Bank US labor data-https://data.worldbank.org/indicator/SL.TLF.TOTL.FE.ZS?locations=US 4. Wage stagnation chart US - https://www.epi.org/publication/charting-wage-stagnation/